Numerical analysis of superconducting phases in the extended Hubbard model with non-local pairing

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Thesis for the Master's degree in Physics

Abstract

[To be continued...]

- Appendix on s-wave SC ghost simulations on HM
- Appendix on triplet pairing in MFT self-consistent scheme
- Set ComputerModern font in all CairoMakie figures

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List of symbols and abbreviations

AF	Antiferromagnetic
BCS	Bardeen-Cooper-Schrieffer (theory)
BZ	Brillouin zone
β	Inverse temperature
$\Delta^{(\gamma)}$	γ-wave component of the gap function
DoF	Degree of freedom
EHM	Extended Hubbard model
EV(s)	Expectation value(s)
FS	Fermi surface
HF	Hartree-Fock
HFP(s)	Hartree-Fock parameter(s)
HM	Hubbard model
L_ℓ	Number of lattice sites in direction ℓ
iHF	Iterative Hartree-Fock
LRT	Linear Response Theory
MBZ	Magnetic Brillouin zone
MFT	Mean-Field Theory
NBZ	Nematic Brillouin zone
NN(s)	Nearest neighbor(s)
PH	Particle-hole
PHS	Particle-hole symmetry
RWC(s)	Relevant Wick's contraction(s)
RMP(s)	Renormalized model parameter(s)
SC	Superconductor
SDW	Spin-density wave
SSB	Spontaneous symmetry breaking
T_c	Critical temperature

Introduction

This thesis project is about my favorite ice cream flavor. [To be continued...]

Chapter 1

Phase transitions and SSB in EHM

The general idea of this project is to apply HF methods to the EHM, derive analytically the approximated quadratic hamiltonian of Eq. (**??**) and then apply the iHF procedure sketched in Sec. **??** to extract the self-consistent values of the various HFPs. One of the advantages of using iHF methods is that we can impose rather simply the symmetries of the HF approximated wavefunction.

This is most useful when dealing with competing phases. A phase transition is basically described by some spontaneous symmetry breaking (SSB) and the rising to a non-zero value of an order parameter. We will deal with three phases:

1. the normal phase, where no symmetry is broken;
2. the antiferromagnetic phase, where we allow for spin density to be non-uniform across sites;
3. the superconducting phase, where we do not conserve particle number.

Each phase is discussed in a separate chapter. Many other "phases" are possible, based on which symmetries are respected and which are not.

For each phase, we will extract the best approximation to the true ground state wavefunction. In computational terms, we will find three solutions to the general self-consistency problem of Eq. (**??**). In the end we want to compare the free energy of these three solutions, assessing which one is the most stable. Any HF approximation, being by construction not the true ground state, overestimates energy. Thus, comparing energies of our solutions we get a hint of the features of the true ground state.

1.1 The EHM symmetries

In this section we present the symmetries of the extended Hubbard model introduced in Eq. (**??**)

$$\hat{H}_{\mathrm{EHM}} = \underbrace{-t\sum_{\langle ij\rangle}\sum_{\sigma}\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}}_{\hat{H}_t}\underbrace{+U\sum_i \hat{n}_{i\uparrow}\hat{n}_{i\downarrow}}_{\hat{H}_U}\underbrace{-V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}\hat{n}_{i\sigma}\hat{n}_{j\sigma'}}_{\hat{H}_V}$$

The symmetries of the model can be divided in two groups: the ones related to the geometry of the square lattice, and those related to the algebraic properties of $\hat{H}_{\mathrm{EHM}}$. We start by the lattice symmetries.

1.1.1 Lattice symmetries

We study the EHM on the planar square lattice, with one orbital per site. We indicate with L_x the number of sites in direction x, and L_y in direction y. The hamiltonian of Eq. (**??**) respects three fundamental lattice symmetries:

1. Discrete translations symmetry in both the x and y directions. We refer to the groups of one-site translations as as $\mathrm{T}^{(x)}_1$, $\mathrm{T}^{(x)}_1$. It holds:

$$\mathrm{T}^{(\ell)}_n \supset \mathrm{T}^{(\ell)}_{kn} \qquad k\in\mathbb{N} \tag{1.1}$$

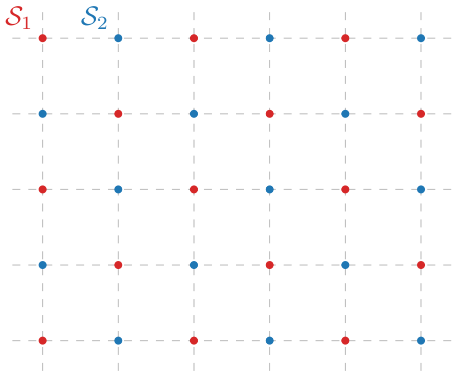


Figure 1.1 | Square lattice $\mathcal{L}$ bipartition in two sublattices $\mathcal{S}_1$ (red), $\mathcal{S}_2$ (blue).

A shift in direction ℓ by kn sites can be decomposed in k shift by n sites. Invariance by n-sites shifts, thus, implies invariance also for kn-sites shifts.

2. Discrete $\pi/2$ rotations symmetry. This symmetry is described by the four-fold cyclic point group C_4, containing rotations by angle $\pi/2, \pi, 3\pi/2$ and identity. A similar property as for the translations holds for cyclic point groups,

$$\mathrm{C}_n \supset \mathrm{C}_{n/k} \qquad k, n/k \in \mathbb{N} \tag{1.2}$$

Invariance for rotations of angle $2\pi/n$ implies invariance for rotations of angle $k \times 2\pi/n$.

3. Inversion symmetry. We define I_2 as the inversion group in two-dimensional space, containing the reflection operator and identity. This point may appear redundant, because

$$\mathrm{I}_2 \equiv \mathrm{C}_2 \tag{1.3}$$

In 2D space, to rotate by π around the origin and to mirror the position of a point are the same operation.

The planar square lattice is bipartite. Let us give a definition. Consider a graph $\mathcal{L}$ and the set of its vertices $V(\mathcal{L}) = \{v_i\}$. Let $E(\mathcal{L})$ be the set of graph edges,

$$E(\mathcal{L}) = \{(v, w) \mid v, w \in V(\mathcal{L}) \wedge v,\ w \text{ are connected}\}$$

The graph is said to be bipartite if two sub-graphs $\mathcal{S}_1$, $\mathcal{S}_2$ exist such that:

- The vertices of $\mathcal{L}$ split in the two sub-graphs,

$$V(\mathcal{L}) = V(\mathcal{S}_1) \cup V(\mathcal{S}_2)$$

- Vertices of one sub-graph are linked only to vertices of the other sub-graph. In other words

$$v, w \in \mathcal{S}_i \qquad \Longrightarrow \qquad (v, w) \notin E(\mathcal{L})$$

The planar square lattice is bipartite: it can be divided in two square lattices, as represented in Fig. 1.1. Two sites are "connected" if linked by a non-zero hopping operator. Which is, if they are NNs. Note that one-site translations leave the hamiltonian invariant but exchange the two sublattices, while two-sites translations don't.

1.1.2 Spin and charge symmetries

The conventional HM shows a global U(2) symmetry, extended to $\mathrm{SU}(2) \times \mathrm{SU}(2)/\mathbb{Z}_2 = \mathrm{SO}(4)$ symmetry at half-filling on a bipartite lattice [12], such as the square lattice. For the EHM, we limit to prove that we can identify a $\mathrm{U}(2) \subset \mathrm{SO}(4)$ global symmetry.

Gauge symmetry. The EHM is symmetric under the global transformation

$$\hat{c}_{i\sigma} \to e^{-i\eta}\hat{c}_{i\sigma} \qquad \hat{c}^\dagger_{i\sigma} \to e^{i\eta}\hat{c}^\dagger_{i\sigma} \qquad \eta \in \mathbb{R} \tag{1.4}$$

To prove this, it suffices to note that

$$\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma'} \quad \to \quad e^{i\eta}\hat{c}^\dagger_{i\sigma}e^{-i\eta}\hat{c}_{j\sigma'} = \hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma'} \tag{1.5}$$

As a consequence, any product of fermionic operators with an equal number of annihilations and creations is symmetric. Hence the kinetic part $\hat{H}_t$ and the interactions $\hat{H}_U$, $\hat{H}_V$ are invariant. As a consequence, the total number of particles (charge)

$$\hat{N} = \sum_{i\sigma} \hat{c}^\dagger_{i\sigma}\hat{c}_{i\sigma}$$

is a good quantum number. The symmetry is described by the transformation of Eq. (1.4), which is a pure phase shift by η. The relative group is $\mathrm{U}^{\mathrm{c}}(1)$, with a superscript "c" indicating "charge".

Global spin symmetry The EHM is described by three operators: NN hopping, independent of σ; local repulsion between opposite-spin densities; non-local attraction, independent of σ. If we rotate all spins by an equal amount, the hamiltonian must remain identical. An identical observation holds for the number operator, $\hat{N}$. Thus $\hat{H}_{\mathrm{EHM}} - \mu\hat{N}$ must be exhibit a global SU(2) symmetry, related to the rotation in 3D space of all spins altogether.

Let us prove this rigorously. Let the on-site spinor:

$$\hat{\Psi}_i \equiv \begin{bmatrix} \hat{c}_{i\uparrow} \\ \hat{c}_{i\downarrow} \end{bmatrix} \tag{1.6}$$

and the Pauli matrices

$$\tau^x \equiv \begin{bmatrix} & 1 \\ 1 & \end{bmatrix} \qquad \tau^y \equiv \begin{bmatrix} & -i \\ i & \end{bmatrix} \qquad \tau^z \equiv \begin{bmatrix} 1 & \\ & -1 \end{bmatrix}$$

We define the local spin operator as:

$$\hat{S}^\alpha_i \equiv \frac{1}{2}\hat{\Psi}^\dagger_i \tau^\alpha \hat{\Psi}_i \qquad \left[\hat{S}^\alpha_j, \hat{S}^\beta_j\right] = i\sum_\gamma \epsilon_{\alpha\beta\gamma}\hat{S}^\gamma_j \tag{1.7}$$

with $\epsilon_{\alpha\beta\gamma}$ the Levi-Civita tensor. It can be checked by explicit calculation that the $\mathfrak{su}(2)$ spin algebra is actually respected. For example, consider $\alpha = x$, $\beta = y$:

$$\begin{aligned}
\left[\hat{S}^x_j, \hat{S}^y_j\right] &= \frac{1}{4}\left[\hat{\Psi}^\dagger_j \begin{bmatrix} & 1 \\ 1 & \end{bmatrix} \hat{\Psi}_j, \hat{\Psi}^\dagger_j \begin{bmatrix} & -i \\ i & \end{bmatrix} \hat{\Psi}_j\right] \\
&= \frac{1}{4}\left[\hat{c}^\dagger_{j\uparrow}\hat{c}_{j\downarrow} + \hat{c}^\dagger_{j\downarrow}\hat{c}_{j\uparrow}, -i\hat{c}^\dagger_{j\uparrow}\hat{c}_{j\downarrow} + i\hat{c}^\dagger_{j\downarrow}\hat{c}_{j\uparrow}\right] \\
&= \frac{i}{4}\left[\hat{c}^\dagger_{j\uparrow}\hat{c}_{j\downarrow}, \hat{c}^\dagger_{j\downarrow}\hat{c}_{j\uparrow}\right] - \frac{i}{4}\left[\hat{c}^\dagger_{j\downarrow}\hat{c}_{j\uparrow}, \hat{c}^\dagger_{j\uparrow}\hat{c}_{j\downarrow}\right] \\
&= \frac{i}{2}\left[\hat{c}^\dagger_{j\uparrow}\hat{c}_{j\downarrow}, \hat{c}^\dagger_{j\downarrow}\hat{c}_{j\uparrow}\right] \\
&= \frac{i}{2}\left(\hat{c}^\dagger_{j\uparrow}\hat{c}_{j\downarrow}\hat{c}^\dagger_{j\downarrow}\hat{c}_{j\uparrow} - \hat{c}^\dagger_{j\downarrow}\hat{c}_{j\uparrow}\hat{c}^\dagger_{j\uparrow}\hat{c}_{j\downarrow}\right) \\
&= \frac{i}{2}\left(-\hat{c}^\dagger_{j\uparrow}\hat{c}^\dagger_{j\downarrow}\hat{c}_{j\downarrow}\hat{c}_{j\uparrow} + \hat{c}^\dagger_{j\uparrow}\hat{c}_{j\uparrow} + \hat{c}^\dagger_{j\downarrow}\hat{c}^\dagger_{j\uparrow}\hat{c}_{j\uparrow}\hat{c}_{j\downarrow} - \hat{c}^\dagger_{j\downarrow}\hat{c}_{j\downarrow}\right) \\
&= \frac{i}{2}\left(\hat{c}^\dagger_{j\uparrow}\hat{c}_{j\uparrow} - \hat{c}^\dagger_{j\downarrow}\hat{c}_{j\downarrow}\right) \\
&= i\hat{S}^z_j
\end{aligned}$$

We used Fermi anticommutation rules, $\{\hat{c}_{i\sigma}, \hat{c}^\dagger_{j\sigma'}\} = \delta_{ij}\delta_{\sigma\sigma'}$. Identical derivations can be carried out for all other components.

Let the raising and lowering operators:

$$\hat{S}_j^+ \equiv \frac{\hat{S}_j^x + i\hat{S}_j^y}{2} \qquad \hat{S}_j^- \equiv \frac{\hat{S}_j^x - i\hat{S}_j^y}{2}$$

and the global spin operators

$$\hat{S}^\alpha \equiv \sum_{i\in\mathcal{L}} \hat{S}_i^\alpha \qquad \left[\hat{S}^\alpha, \hat{S}^\beta\right] = i\sum_\gamma \epsilon_{\alpha\beta\gamma}\hat{S}^\gamma$$

being $\mathcal{L}$ the lattice. Explicitly,

$$\hat{S}^z \equiv \frac{1}{2}\sum_{i\in\mathcal{L}} (\hat{n}_{i\uparrow} - \hat{n}_{i\downarrow}) \qquad \hat{S}^+ \equiv \frac{1}{2}\sum_{i\in\mathcal{L}} \hat{c}_{i\uparrow}^\dagger \hat{c}_{i\downarrow} \qquad \hat{S}^- \equiv \left(\hat{S}^+\right)^\dagger$$

The EHM hamiltonian and the number operator commutate with the global spin operator. For the kinetic part $\hat{H}_t$ and the local repulsion $\hat{H}_U$ this is a well known result [12, 27],

$$\left[\hat{H}_t + \hat{H}_U, \hat{\mathbf{S}}\right] = 0 \qquad \left[\hat{N}, \hat{\mathbf{S}}\right] = 0$$

For the non-local attraction, since it depends solely on total densities on neighbouring sites, spin-rotational invariance must hold. Anyway, let us give a short formal proof following this chain of results:

1. The non-local attraction can be written as:

$$\begin{aligned}\hat{H}_V &= V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'} \hat{n}_{i\sigma}\hat{n}_{j\sigma'} \\ &= V\sum_{\langle ij\rangle} \hat{\Psi}_i^\dagger\hat{\Psi}_i\hat{\Psi}_j^\dagger\hat{\Psi}_j\end{aligned} \tag{1.8}$$

having we used $\hat{\Psi}_i^\dagger\hat{\Psi}_i = \hat{c}_{i\uparrow}^\dagger\hat{c}_{i\uparrow} + \hat{c}_{i\downarrow}^\dagger\hat{c}_{i\downarrow}$.

2. Similarly, the number operator is given by:

$$\begin{aligned}\hat{N} &= \sum_{i\in\mathcal{L}}\sum_\sigma \hat{n}_{i\sigma} \\ &= \sum_{i\in\mathcal{L}} \hat{\Psi}_i^\dagger\hat{\Psi}_i\end{aligned} \tag{1.9}$$

3. Let now $\hat{R}(\boldsymbol{\theta})$ be any global spin rotation,

$$\hat{R}(\boldsymbol{\theta}) \equiv \exp\left(-i\boldsymbol{\theta}\cdot\hat{\mathbf{S}}\right) \quad \text{where} \quad \boldsymbol{\theta} = (\theta_x, \theta_y, \theta_z)$$

Spinors transform as:

$$\hat{R}(\boldsymbol{\theta})\hat{\Psi}_i\hat{R}(\boldsymbol{\theta})^\dagger = \mathcal{R}(\boldsymbol{\theta})\hat{\Psi}_i$$

with $\mathcal{R}(\boldsymbol{\theta}) \in \mathrm{SU}(2)$ the unitary matrix associated to said rotation,

$$\mathcal{R}(\boldsymbol{\theta}) = \exp\left(-\frac{i}{2}\boldsymbol{\theta}\cdot\boldsymbol{\tau}\right) \quad \text{where} \quad \boldsymbol{\tau} = (\tau_x, \tau_y, \tau_z) \tag{1.10}$$

Let us prove this. Let $\hat{A} = i\boldsymbol{\theta}\cdot\hat{\mathbf{S}}$ e $\hat{B} = \hat{\Psi}_i$. The following identity, a consequence of Baker-Campbell-Housdorff formula, holds:

$$e^{\hat{A}}\hat{B}e^{-\hat{A}} = \hat{B} + \left[\hat{A}, \hat{B}\right] + \frac{1}{2!}\left[\hat{A}, \left[\hat{A}, \hat{B}\right]\right] + \frac{1}{3!}\left[\hat{A}, \left[\hat{A}, \left[\hat{A}, \hat{B}\right]\right]\right] + \cdots$$

The first order of the expansion gives:

$$
\begin{aligned}
[i\boldsymbol{\theta}\cdot\hat{\mathbf{S}},\hat{\Psi}_i] &= i\sum_\alpha \theta_\alpha[\hat{S}^\alpha,\hat{\Psi}_i] \\
&= -\frac{i}{2}\sum_\alpha \theta_\alpha\tau^\alpha\hat{\Psi}_i \\
&= -\frac{i}{2}\boldsymbol{\theta}\cdot\boldsymbol{\tau}\hat{\Psi}_i
\end{aligned}
$$

We omitted some purely algebraic passages. The second order:

$$
\begin{aligned}
\Big[i\boldsymbol{\theta}\cdot\hat{\mathbf{S}},\Big[i\boldsymbol{\theta}\cdot\hat{\mathbf{S}},\hat{\Psi}_i\Big]\Big] &= \Big[i\boldsymbol{\theta}\cdot\hat{\mathbf{S}},-\frac{i}{2}(\boldsymbol{\theta}\cdot\boldsymbol{\tau})\hat{\Psi}_i\Big] \\
&= -\frac{i}{2}\boldsymbol{\theta}\cdot\boldsymbol{\tau}\Big[i\boldsymbol{\theta}\cdot\hat{\mathbf{S}},\Psi_i\Big] \\
&= \frac{1}{4}(\boldsymbol{\theta}\cdot\boldsymbol{\tau})^2\hat{\Psi}_i
\end{aligned}
$$

and so on. Collecting all terms,

$$
\begin{aligned}
\hat{R}(\boldsymbol{\theta})\hat{\Psi}_i\hat{R}(\boldsymbol{\theta})^\dagger &= \hat{\Psi}_i - \frac{i}{2}\boldsymbol{\theta}\cdot\boldsymbol{\tau}\hat{\Psi}_i + \frac{1}{2!\times 4}(\boldsymbol{\theta}\cdot\boldsymbol{\tau})^2\hat{\Psi}_i + \cdots \\
&= \sum_{k=1}^{\infty}\frac{1}{k!}\left(-\frac{i}{2}\boldsymbol{\theta}\cdot\boldsymbol{\tau}\right)^k\hat{\Psi}_i \\
&= \mathcal{R}(\boldsymbol{\theta})\hat{\Psi}_i
\end{aligned}
$$

In last passage, we identified the series as the Taylor expansion of the exponential, recognising the expression for $\mathcal{R}(\boldsymbol{\theta})$ given in Eq. (1.10).

4. Then, the group acts on the product $\hat{\Psi}_i^\dagger\hat{\Psi}_i$ as

$$
\begin{aligned}
\hat{R}\hat{\Psi}_i^\dagger\hat{\Psi}_i\hat{R}^\dagger &= \hat{R}\hat{\Psi}_i^\dagger\hat{R}^\dagger\hat{R}\hat{\Psi}_i\hat{R}^\dagger \\
&= \hat{\Psi}_i^\dagger\mathcal{R}^\dagger\mathcal{R}\hat{\Psi}_i \\
&= \hat{\Psi}_i^\dagger\hat{\Psi}_i
\end{aligned}
$$

being $\mathcal{R}\in\mathrm{SU}(2)$. Recalling Eqns. (1.8), (1.9), both $\hat{H}_V$ and $\hat{N}$ are SU(2) symmetric.

The EHM is symmetric under global spin rotations. Note that $\hat{H}_V$ is actually symmetric under local spin rotations, meaning that we could rotate the spin at each site by a different amount and get the same interaction. The same holds for $\hat{H}_U$. It is the kinetic operator that makes the EHM globally (and not locally) SU(2) symmetric.

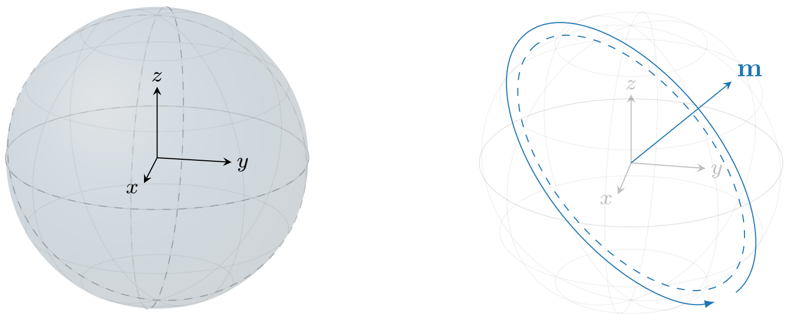
Note that the $U^{\mathrm{c}}(1)$ gauge symmetry and the SU(2) spin rotations symmetry can be combined,

$$
\mathrm{U}^{\mathrm{c}}(1)\otimes\mathrm{SU}(2)=\mathrm{U}(2)
$$

We will refer to U(2) symmetry to indicate full charge-spin conservation. Finally, note the group of 3D rotations contains the subgroup of 2D rotations around a given axis z,

$$
\mathrm{U}^z(1)\subset\mathrm{SU}(2)
$$

where the z superscript serves to distinguish from the gauge group. The z axis does not need to be identified with the "real" z axis, perpendicular with the lattice plane. Any given direction works. Inverting our logic, we can build the SU(2) generators, starting from $\hat{S}^z$, after we have chosen the direction z. In terms of commutation relations, a given operator can respect $\mathrm{U}^z(1)$ but not SU(2): in that case, it will commutate to zero with $\hat{S}^z$, and not the other spin operators. A graphic rendition of this SSB is given in Fig. 1.2, representing for a given vector $\mathbf{m}$ the passage from full SU(2) rotational symmetry around any direction to reduced U(1) rotational symmetry around just one direction.



a | SU(2) rotations domain before SSB.

b | U(1) rotations domain after SSB.

Figure 1.2 | Graphic representation of the concept of $\mathrm{SU}(2) \to \mathrm{U}^z(1)$ SSB. "Before" SSB, we have full symmetry under any possible rotation (solid color sphere, left panel). "After" SSB we have symmetry only for rotations around some vector $\mathbf{m}$ (dashed tilted circumference, right panel).

1.1.3 Particle-hole symmetry at half-filling

The half-filled EHM on the square lattice exhibits particle-hole symmetry (PHS). The presence of this additional symmetry is a consequence of the fact that the square lattice is bipartite. Let us prove this. Consider the generic unitary particle-hole (PH) transformation

$$\mathrm{P} \quad : \quad \hat{c}_{i\sigma} \quad \to \quad e^{i\eta_{i\sigma}}\hat{h}^{\dagger}_{i\sigma} \tag{1.11}$$

being in general η_i a phase dependent on the fermionic quantum state $(i\sigma)$. Inserting the above transformation in the EHM, we get

$$\mathrm{P}\{\hat{H}\} = -t\sum_{\langle ij\rangle}\sum_{\sigma} e^{i(\eta_{i\sigma}+\eta_{j\sigma})}\hat{h}_{i\sigma}\hat{h}^{\dagger}_{j\sigma} + U\sum_{i}\hat{h}_{i\uparrow}\hat{h}_{i\downarrow}\hat{h}^{\dagger}_{i\downarrow}\hat{h}^{\dagger}_{i\uparrow} - V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}\hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^{\dagger}_{j\sigma'}\hat{h}^{\dagger}_{i\sigma}$$

For the two interactions $\hat{H}_U$, $\hat{H}_V$ the phase factors cancelled out. We aim to find the conditions under which the above hamiltonian is mapped onto the EHM of (**??**),

$$\mathrm{P}\{\hat{H}\} \stackrel{?}{=} -t\sum_{\langle ij\rangle}\sum_{\sigma}\hat{h}^{\dagger}_{i\sigma}\hat{h}_{j\sigma} + U\sum_{i\in\mathcal{L}}\hat{h}^{\dagger}_{i\uparrow}\hat{h}^{\dagger}_{i\downarrow}\hat{h}_{i\downarrow}\hat{h}_{i\uparrow} - V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}\hat{h}^{\dagger}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}_{j\sigma'}\hat{h}_{i\sigma}$$

Before starting the derivation, which is an extension of the argument carried out for the conventional HM [12], let us discuss some properties of the hole operators. Due to the presence of the phase factors in the PHS operative definition, the Fermi anticommutation rules hold regardlessly of the phase factor:

$$\begin{aligned}\{\hat{h}_{i\sigma}, \hat{h}^{\dagger}_{j\sigma'}\} &= e^{i(\eta_{i\sigma}-\eta_{j\sigma'})}\{\hat{c}^{\dagger}_{i\sigma}, \hat{c}_{j\sigma'}\} \\ &= e^{i(\eta_{i\sigma}-\eta_{j\sigma'})}\delta_{ij}\delta_{\sigma\sigma'} \\ &= \delta_{ij}\delta_{\sigma\sigma'}\end{aligned}$$

The on-site density operator is given by:

$$\hat{h}_{i\sigma}\hat{h}^{\dagger}_{i\sigma} = \hat{n}_{i\sigma}$$

being $\hat{n}_{i\sigma}$ the fermionic number operator on site i with spin σ. It follows, for quartic interactions

$$\begin{aligned}
\hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'}\hat{h}^\dagger_{i\sigma} &= -\hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^\dagger_{i\sigma}\hat{h}^\dagger_{j\sigma'} \\
&= \hat{h}_{i\sigma}\hat{h}^\dagger_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'} - \hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'}\delta_{ij}\delta_{\sigma\sigma'} \\
&= -\hat{h}^\dagger_{i\sigma}\hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'} + \hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'} - \hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'}\delta_{ij}\delta_{\sigma\sigma'} \\
&= \hat{h}^\dagger_{i\sigma}\hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'}\hat{h}_{j\sigma'} - \hat{h}^\dagger_{i\sigma}\hat{h}_{i\sigma} + \hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'} - \hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'}\delta_{ij}\delta_{\sigma\sigma'} \\
&= -\hat{h}^\dagger_{i\sigma}\hat{h}^\dagger_{j\sigma'}\hat{h}_{i\sigma}\hat{h}_{j\sigma'} - \hat{h}^\dagger_{i\sigma}\hat{h}_{i\sigma} + \hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'} - \left(\hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'} - \hat{h}^\dagger_{i\sigma}\hat{h}_{j\sigma'}\right)\delta_{ij}\delta_{\sigma\sigma'} \\
&= \hat{h}^\dagger_{i\sigma}\hat{h}^\dagger_{j\sigma'}\hat{h}_{j\sigma'}\hat{h}_{i\sigma} + (\hat{n}_{i\sigma} + \hat{n}_{j\sigma'}) - 1 - \left(\hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'} - \hat{h}^\dagger_{i\sigma}\hat{h}_{j\sigma'}\right)\delta_{ij}\delta_{\sigma\sigma'} \qquad (1.12)
\end{aligned}$$

The first term of the last passage is mirrored with respect to the starting term. Now, let us deal with the kinetic, local and non-local interactions separately.

Kinetic term. Consider the kinetic term,

$$\hat{H}_t = -t\sum_{\langle ij\rangle}\sum_{\sigma}\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}$$

Apply the PH transform of Eq. (1.11)

$$\mathrm{P}\{\hat{H}_t\} = -t\sum_{\langle ij\rangle}\sum_{\sigma} e^{i(\eta_{i\sigma}+\eta_{j\sigma})}\hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma}$$

Anti-commutate the two fermionic operators, and exchange site labels $i \leftrightarrow j$. We get:

$$\mathrm{P}\{\hat{H}_t\} = -t\sum_{\langle ij\rangle}\sum_{\sigma} e^{i(\eta_{i\sigma}+\eta_{j\sigma}+\pi)}\hat{h}^\dagger_{i\sigma}\hat{h}_{j\sigma}$$

A necessary condition on the unitary operation P is the following,

$$\eta_{i\sigma} + \eta_{j\sigma} = (2k-1)\pi \quad \text{with} \quad k \in \mathbb{Z} \qquad (1.13)$$

valid for NN sites.

Local repulsion. Substitute Eq. (1.12) in $\hat{H}_U$,

$$\begin{aligned}
\mathrm{P}\{\hat{H}_U\} &= U\sum_{i\in\mathcal{L}}\hat{h}_{i\uparrow}\hat{h}_{i\downarrow}\hat{h}^\dagger_{i\downarrow}\hat{h}^\dagger_{i\uparrow} \\
&= U\sum_{i\in\mathcal{L}}\hat{h}^\dagger_{i\uparrow}\hat{h}^\dagger_{i\downarrow}\hat{h}_{i\downarrow}\hat{h}_{i\uparrow} + U\sum_{i\in\mathcal{L}}(\hat{n}_{i\uparrow}+\hat{n}_{i\downarrow}) - U\sum_{i\in\mathcal{L}}(1) \\
&= U\sum_{i}\hat{h}^\dagger_{i\uparrow}\hat{h}^\dagger_{i\downarrow}\hat{h}_{i\downarrow}\hat{h}_{i\uparrow} + U(\hat{N} - L_xL_y)
\end{aligned}$$

For the half-filled state $\langle\hat{N}\rangle = L_xL_y$. The unitary map of Eq. (1.11), independently of the condition of Eq. (1.13), leaves the hamiltonian identical at half-filling.

Non-local attraction. Proceed identically for $\hat{H}_V$,

$$\begin{aligned}
\mathrm{P}\{\hat{H}_V\} &= -V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}\hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'}\hat{h}^\dagger_{i\sigma} \\
&= -V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}\hat{h}^\dagger_{i\sigma}\hat{h}^\dagger_{j\sigma'}\hat{h}_{j\sigma'}\hat{h}_{i\sigma} - V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}(\hat{n}_{i\sigma}+\hat{n}_{j\sigma'}) + V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}(1) \\
&= -V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}\hat{h}^\dagger_{i\sigma}\hat{h}^\dagger_{j\sigma'}\hat{h}_{j\sigma'}\hat{h}_{i\sigma} - 2zV(\hat{N} - L_xL_y)
\end{aligned}$$

Symmetry	Operations
$\mathrm{T}_1^{(x)} \otimes \mathrm{T}_1^{(y)}$	One-site discrete translations separately in both directions
C_4	Four-fold point group (real $\pi/2$ angle rotations)
$\mathrm{U}^{\mathrm{c}}(1)$	Charge-conserving global phase shifts
$\mathrm{SU}(2)$	Three-dimensional rotations in global spin space

Table 1.1 Model symmetries and relative group operations. Note that $\mathrm{U}^{\mathrm{c}}(1)$ represents the gauge symmetry linked to charge conservation.

with $z = 4$ the lattice coordination number. In the last passage, we used that on the square lattice

$$\begin{aligned}\sum_{\langle ij\rangle}(1) &= \sum_{i\in\mathcal{L}/2} z \\ &= \frac{L_x L_y}{2} \times z \end{aligned} \tag{1.14}$$

As before, the non-local attraction is particle-hole symmetric at half-filling. As for the local repulsion, PHS is given just the operation of Eq. (1.11). The condition of Eq. (1.13) is irrelevant.

Recollecting everything, if P is defined by the unitary operation

$$\mathrm{P} \quad : \quad \begin{cases} \hat{c}_{i\sigma} \to \hat{h}^{\dagger}_{i\sigma} & \text{for} \quad i \in \mathcal{S}_1 \\ \hat{c}_{i\sigma} \to -\hat{h}^{\dagger}_{i\sigma} & \text{for} \quad i \in \mathcal{S}_2 \end{cases}$$

and the system is half-filled, PHS is respected. Any other phase, as long as it alternates by π on the two sublattices, is good enough. As for the conventional HM [12], this result can be generalised to any bipartite lattice. In turn, if a lattice is not bipartite, PHS is not present. To summarise, we report in Tab. 1.1 all identified symmetries.

1.2 Symmetry breaking channels in real space

In many-body Physics, a phase transition is described by some order parameter rising from zero to a non-zero value. This is associated with some spontaneous symmetry breaking (SSB). In order to simulate different phases, we need a way to distinguish them based on which symmetries they break (or not). We also need a way to work computationally in a given symmetry sector. HF methods, by reducing the hamiltonian to a non-interacting approximation, offer a simple framework to these ends. Indeed, once we identified all symmetries of a given phase, we just impose them on our approximation of the full hamiltonian. The found ground-state approximation will entail these symmetries by construction.

Recall now the discussion of the HF method of Sec. **??**. The ending point were Eqns. (**??**), (**??**), which identify the self-consistent single-particle fields which approximate the original quartic hamiltonian. In order to get there, we applied Wick's theorem to the quartic part of the hamiltonian and derived the most general expressions of $\mathbf{h}$. In next sections here apply Wick's theorem to the quartic operators of $\hat{H}_{\mathrm{EHM}}$ and discuss the symmetry properties of the resulting decomposition.

Before starting, let us define precisely the three phases we will investigate in terms of symmetries:

- Normal phase: no broken symmetry.
- Antiferromagnetic phase: simultaneous $\mathrm{SU}(2) \overset{\mathrm{SSB}}{\to} \mathrm{U}^z(1)$ and $\mathrm{T}_1^{(x)} \otimes \mathrm{T}_1^{(y)} \overset{\mathrm{SSB}}{\to} \mathrm{T}_2^{(x)} \otimes \mathrm{T}_2^{(y)}$. Among all possible realisations of antiferromagnetism, we realise a SDW with alternating distribution of spin densities;
- Superconducting phase: simultaneous $\mathrm{U}^c(1) \overset{\mathrm{SSB}}{\to} 0$ and $\mathrm{C}_4 \overset{\mathrm{SSB}}{\to} \mathrm{C}_2$. We investigate anisotropic superconductivity, allowing for the particle number to be not conserved and for nematic effects.

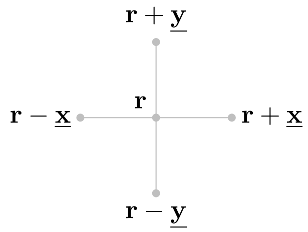


Figure 1.3 | Schematic representation of the four NNs of a given site i for a planar square lattice.

Nematic effects are observed in unconventional superconductors [22], and this is the reason of our interest in the possibility of rotational SSBs. We will theoretically discuss the nematic SSB $C_4 \to C_2$ also for normal and AF phases, as if discrete rotational symmetry was broken. As we will see, while it not trivial to exclude it *a priori*, it will actually turn out that the HF ground state preserves C_4 in both cases.

We now move to a more detailed discussion of each hamiltonian contribution. In next sections, we switch from the site index i to the vector notation $\mathbf{r}$, indicating the site 2D position on the lattice

$$i \quad \to \quad \mathbf{r} = (x, y)$$

with $x, y \in \mathbb{N}$. The lattice versors are:

$$\underline{\mathbf{x}} \equiv (1, 0) \qquad \underline{\mathbf{y}} \equiv (0, 1)$$

as in Fig. 1.3.

1.2.1 Local repulsion

The local repulsion $\hat{H}_U$ is the quartic operator of the conventional HM,

$$\begin{aligned}\hat{H}_U &= U \sum_{\mathbf{r}\in\mathcal{L}} \hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}\downarrow} \\ &= U \sum_{\mathbf{r}\in\mathcal{L}} \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\end{aligned}$$

In order to obtain the self-consistent values of $h^{+-}_{\alpha\beta}$, $h^{++}_{\alpha\beta}$ we need to compute an expectation value of the full hamiltonian. Wick's theorem guarantees that, for the wavefunctions we are using, the expectation value of a quartic fermionic operator is the sum of all fully contracted terms. Recalling the nomenclature of Eq. (??),

$$\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle = \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\uparrow}\rangle\langle \hat{c}^\dagger_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\downarrow}\rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\downarrow}\rangle\langle \hat{c}^\dagger_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}\rangle\langle \hat{c}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle}_{\text{Cooper}} \tag{1.15}$$

where the equal sign holds as long as we perform the averaging operation on the subspace of **gaussian states**. Now we discuss the above EVs separately, identifying if any can be neglected based on selection rules in a given phase.

Hartree term. The Hartree part of the above decomposition is a product of density EVs,

$$\langle \hat{n}_{\mathbf{r}\uparrow}\rangle\langle \hat{n}_{\mathbf{r}\downarrow}\rangle$$

Recalling the groups of Tab. 1.1, we see that the density operator preserves all symmetries. Single-site density operators commutate with spin operators and are gauge-invariant. Moreover, shifting or rotating the lattice leaves density operators unchanged. Thus no selection rule applies, and the above EVs are in principle non-zero.

Fock terms. For the Fock part, notice that

$$\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\downarrow} = \hat{S}^+_{\mathbf{r}}$$

being $S^+_{\mathbf{r}}$ the local raising operator. From SU(2) algebra we have:

$$\left[\hat{S}^+_{\mathbf{r}}, \hat{S}^\alpha\right] \neq 0 \qquad \alpha \in \{x, y, z\}$$

If the ground-state is invariant under global spin rotations, these EVs must be zero. Gauge symmetry gives no selection rules, due to Eq. (1.5).

Cooper terms. The Cooper part can be non-zero only if charge is not conserved. In terms of the gauge symmetry, it suffices to apply the map of Eq. (1.4) to $\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}$

$$\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow} \quad \rightarrow \quad e^{2i\eta}\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}$$

to conclude that this operator is in general not gauge invariant. It is instead SU(2)-symmetric. To prove this, it suffices to compute commutators of $\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}$ with $\hat{S}^z$, $\hat{S}^+$:

$$\begin{aligned}
\left[\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}, \hat{S}^z\right] &= \left[\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}, \frac{1}{2}\sum_{\mathbf{r}'\in\mathcal{L}}(\hat{n}_{\mathbf{r}'\uparrow} - \hat{n}_{\mathbf{r}'\downarrow})\right] \\
&= \frac{1}{2}\left[\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}, \hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}\right] \\
&= \frac{1}{2}\left[\hat{c}^\dagger_{\mathbf{r}\uparrow}, \hat{n}_{\mathbf{r}\uparrow}\right]\hat{c}^\dagger_{\mathbf{r}\downarrow} - \frac{1}{2}\hat{c}^\dagger_{\mathbf{r}\uparrow}\left[\hat{c}^\dagger_{\mathbf{r}\downarrow}, \hat{n}_{\mathbf{r}\downarrow}\right]
\end{aligned}$$

Since:

$$\begin{aligned}
\left[\hat{c}^\dagger_{\mathbf{r}\sigma}, \hat{n}_{\mathbf{r}\sigma}\right] &= \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}\sigma} - \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}\sigma}\hat{c}^\dagger_{\mathbf{r}\sigma} \\
&= -\hat{c}^\dagger_{\mathbf{r}\sigma}\left(1 - \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}\sigma}\right) \\
&= -\hat{c}^\dagger_{\mathbf{r}\sigma}
\end{aligned} \tag{1.16}$$

we have:

$$\left[\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}, \hat{S}^z\right] = -\frac{1}{2}\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow} + \frac{1}{2}\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow} = 0$$

Moreover, considering the raising operator:

$$\begin{aligned}
\left[\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}, \hat{S}^+\right] &= \left[\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}, \frac{1}{2}\sum_{\mathbf{r}'\in\mathcal{L}}\hat{c}^\dagger_{\mathbf{r}'\uparrow}\hat{c}_{\mathbf{r}'\downarrow}\right] \\
&= \left[\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}, \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\downarrow}\right] \\
&= \left[\hat{c}^\dagger_{\mathbf{r}\uparrow}, \hat{c}^\dagger_{\mathbf{r}\uparrow}\right]\hat{c}^\dagger_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\downarrow} - \left(\hat{c}^\dagger_{\mathbf{r}\uparrow}\right)^2\left[\hat{c}^\dagger_{\mathbf{r}\downarrow}, \hat{c}_{\mathbf{r}\downarrow}\right] \\
&= 0
\end{aligned}$$

where we used the facts that every operator commutates with itself, and a squared creation operator is null. These commutation relations, together, prove that Cooper EVs are legitimately non-zero for a SU(2)-preserving phase.

In Tab. 1.2 we collect the selection rules for the Hartree, Fock and Cooper terms of Eq. (1.15), based on the symmetry specifications given for each phase in above paragraphs. Note that Fock-like EVs are prohibited for all phases.

Term	Normal phase	Antiferromagnet	Superconductor
Hartree			
Fock	Prohibited	Prohibited	Prohibited
Cooper	Prohibited	Prohibited	

Table 1.2 Summary of selection rules for the Wick's decomposition of Eq. (1.15) of the local repulsion $\hat{H}_U$. An empty entry indicates no selection rule.

1.2.2 Non-local attraction

In order to analyse similarly the non-local attraction $\hat{H}_V$, it is useful to decompose the operator in spin channels:

$$\begin{aligned}\hat{H}_V &\equiv \underbrace{\hat{H}_V^{\uparrow\uparrow} + \hat{H}_V^{\downarrow\downarrow}}_{\text{Same spin}} + \underbrace{\hat{H}_V^{\uparrow\downarrow} + \hat{H}_V^{\downarrow\uparrow}}_{\text{Opposite spin}} \\ &= \hat{H}^{(\text{s.s.})} + \hat{H}^{(\text{o.s.})} \end{aligned} \tag{1.17}$$

being

$$\hat{H}_V^{\sigma\sigma'} \equiv -V \sum_{\langle \mathbf{r},\mathbf{r}'\rangle} \hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}'\sigma'}$$

Let us start from the opposite spin sector. On the square lattice (as for any bipartite lattice) NNs belong to different sublattices. This implies that any NNs sum can be rewritten as:

$$\sum_{\langle \mathbf{r}\mathbf{r}'\rangle} f(\mathbf{r},\mathbf{r}') = \sum_{\mathbf{r}\in\mathcal{L}/2}\sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} f(\mathbf{r},\mathbf{r}+\boldsymbol{\delta}) \tag{1.18}$$

with f some function. We are using the notation of Fig. 1.3. The symbol "$\mathcal{L}/2$" indicates one of the two sublattices $\mathcal{S}_1$, $\mathcal{S}_2$ represented in Fig. 1.1. We can rewrite:

$$\hat{H}_V^{\sigma\sigma'} = -V \sum_{\mathbf{r}\in\mathcal{L}/2}\sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} \left(\hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\sigma'} + \hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}-\boldsymbol{\delta}\sigma'}\right)$$

Now, if we choose $\mathcal{L}/2 = \mathcal{S}_1$, this gives:

$$\begin{aligned}\hat{H}_V^{\uparrow\downarrow} &= -V \sum_{\mathbf{r}\in\mathcal{S}_1}\sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} \left(\hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\downarrow} + \hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}-\boldsymbol{\delta}\downarrow}\right) \\ \hat{H}_V^{\downarrow\uparrow} &= -V \sum_{\mathbf{r}\in\mathcal{S}_1}\sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} \left(\hat{n}_{\mathbf{r}\downarrow}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\uparrow} + \hat{n}_{\mathbf{r}\downarrow}\hat{n}_{\mathbf{r}-\boldsymbol{\delta}\uparrow}\right)\end{aligned}$$

These two operators are "complementary": $\sigma =\uparrow$ densities act on $\mathcal{S}_1$ for $\hat{H}_V^{\uparrow\downarrow}$, on $\mathcal{S}_2$ for $\hat{H}_V^{\downarrow\uparrow}$; $\sigma =\downarrow$ densities, the opposite. Then their sum can be expressed as:

$$\begin{aligned}\hat{H}_V^{(\text{o.s.})} &\equiv \hat{H}_V^{\uparrow\downarrow} + \hat{H}_V^{\downarrow\uparrow} \\ &= -V \sum_{\mathbf{r}\in\mathcal{L}}\sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} \left(\hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\downarrow} + \hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}-\boldsymbol{\delta}\downarrow}\right)\end{aligned} \tag{1.19}$$

Wick's theorem applied to this quartic operator gives:

$$\begin{aligned}\langle \hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\rangle &= \langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle \\ &\simeq \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\uparrow}\rangle\langle \hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\rangle\langle \hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\rangle\langle \hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle}_{\text{Cooper}}\end{aligned} \tag{1.20}$$

For the same-spin sector we define

$$\begin{aligned}\hat{H}_V^{(\text{s.s.})} &\equiv \hat{H}_V^{\uparrow\uparrow} + \hat{H}_V^{\downarrow\downarrow} \\ &= -V \sum_{\mathbf{r}\in\mathcal{L}/2}\sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}}\sum_{\sigma\in\{\uparrow,\downarrow\}} \left(\hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\sigma} + \hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}-\boldsymbol{\delta}\sigma}\right)\end{aligned} \tag{1.21}$$

Sector	Term	Normal phase	Antiferromagnet	Superconductor
o.s.	Hartree			
	Fock	Prohibited	Prohibited	Prohibited
	Cooper	Prohibited	Prohibited	
s.s.	Hartree			
	Fock			
	Cooper	Prohibited	Prohibited	

Table 1.3 | Summary of selection rules for the Wick's decomposition of Eqns. (1.20), (1.22) of the non-local attraction $\hat{H}_V$. An empty entry indicates no selection rule.

Note that, with respect to the o.s. sector, we are summing on half the sites two times: one for spin $\uparrow$, one for spin $\downarrow$. Applying Wick's theorem:

$$\begin{aligned}\langle \hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\rangle &= \langle \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\hat{c}_{\mathbf{r}\sigma}\rangle \\ &\simeq \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}\sigma}\rangle\langle \hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\rangle\langle \hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\hat{c}_{\mathbf{r}\sigma}\rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\rangle\langle \hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\hat{c}_{\mathbf{r}\sigma}\rangle}_{\text{Cooper}}\end{aligned} \tag{1.22}$$

As we did for $\hat{H}_U$, we analyse each separately.

Hartree terms. As it was for $\hat{H}_U$, Hartree EVs of Eqns. (1.19), (1.21) are not subject to selection rules for any symmetry of Tab. 1.1. Thus they are allowed in both sectors.

Fock terms. The Fock EVs from the two sectors behave differently. Let us compute the following generic commutation relations:

$$\begin{aligned}\left[\hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'},\hat{S}^z\right] &= \left[\hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'},\frac{1}{2}\sum_{\mathbf{r}'\in\mathcal{L}}(\hat{n}_{\mathbf{r}'\uparrow}-\hat{n}_{\mathbf{r}'\downarrow})\right] \\ &= \frac{(-1)^{\delta_{\sigma=\downarrow}}}{2}\left[\hat{c}^\dagger_{\mathbf{r}\sigma},\hat{n}_{\mathbf{r}\sigma}\right]\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'} + \frac{(-1)^{\delta_{\sigma'=\downarrow}}}{2}\hat{c}^\dagger_{\mathbf{r}\sigma}\left[\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'},\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\sigma}\right] \\ &= \left(-\frac{(-1)^{\delta_{\sigma=\downarrow}}}{2}+\frac{(-1)^{\delta_{\sigma'=\downarrow}}}{2}\right)\hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}\end{aligned} \tag{1.23}$$

where in last passage we used Eq. (1.16) and its conjugate version,

$$[\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'},\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\sigma}] = \hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}$$

Now, if $\sigma = \sigma'$ the last passage of Eq. (1.23) is zero; otherwise is not. This proves that the o.s. Fock EVs are non-zero only for a phase that completely breaks SU(2); instead, in the s.s. sector we can have non-zero EVs for a phase that reduces SU(2) to $\mathrm{U}^z(1)$.

Cooper terms. For the Cooper terms, identical considerations as for $\hat{H}_U$ hold. Gauge symmetry must be broken to allow for non zero EVs, while SU(2) symmetry survives. The proof is analogous to the one given for $\hat{H}_U$.

In Tab. 1.3 we summarise these considerations, giving the selection rules on the EVs of Eqns. (1.20), (1.22), dividing in the two sectors "o.s." and "s.s.". By identifying these selection rules, we set grounds for the detailed derivation of each chapter and heavily simplify application of HF model to the EHM. We now move to the last part of this chapter: reciprocal space description of the EHM.

1.3 The EHM in reciprocal space

The EHM is symmetric under discrete lattice translations. We will investigate phases that do not (completely) break translational invariance. It is thus convenient to move to reciprocal space, by

Fourier-transforming the hamiltonian. We will use the following convention:

$$\hat{c}_{\mathbf{r}\sigma} \overset{\mathcal{F}}{=} \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k}\in\mathrm{BZ}} e^{-i\mathbf{k}\cdot\mathbf{r}} \hat{c}_{\mathbf{k}\sigma} \tag{1.24}$$

Throughout the text, $\overset{\mathcal{F}}{=}$ will indicate a passage where a Fourier transform is used. "BZ" is shorthand notation for the Brillouin Zone.

It is useful to recall that the following equation for Kronecker's delta holds:

$$\frac{1}{L_x L_y} \sum_{\mathbf{r}\in\mathcal{L}} e^{i\mathbf{k}\cdot\mathbf{r}} = \delta_{\mathbf{k}=\mathbf{0}} \tag{1.25}$$

On one sublattice:

$$\frac{1}{L_x L_y} \sum_{\mathbf{r}\in\mathcal{L}/2} e^{i\mathbf{k}\cdot\mathbf{r}} = \frac{1}{2}\delta_{\mathbf{k}=\mathbf{0}} \tag{1.26}$$

The notation $\mathcal{L}/2$ is shorthand to indicate $\mathcal{S}_1$ or $\mathcal{S}_2$.

Kinetic part. Let us transform $\hat{H}_t$:

$$\begin{aligned}
\hat{H}_t &= -t \sum_{\langle\mathbf{r}\mathbf{r}'\rangle} \sum_{\sigma} \left[\hat{c}^{\dagger}_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}'\sigma} + \mathrm{h.c.}\right] \\
&\overset{\mathcal{F}}{=} -t \sum_{\langle\mathbf{r}\mathbf{r}'\rangle} \sum_{\sigma} \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k}\in\mathrm{BZ}} e^{i\mathbf{k}\cdot\mathbf{r}} \hat{c}^{\dagger}_{\mathbf{k}\sigma} \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k}'\in\mathrm{BZ}} e^{-i\mathbf{k}'\cdot\mathbf{r}'} \hat{c}_{\mathbf{k}'\sigma} + \mathrm{h.c.} \\
&= -2t \sum_{\mathbf{k},\mathbf{k}'} \sum_{\sigma} \hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}'\sigma} \times \frac{1}{L_x L_y} \sum_{\mathbf{r}\in\mathcal{L}/2} e^{i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{r}} \times \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} e^{i\mathbf{k}'\cdot\boldsymbol{\delta}} \\
&= -2t \sum_{\mathbf{k},\mathbf{k}'} \sum_{\sigma} \hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} \left[\frac{e^{ik_x}+e^{-ik_x}}{2} + \frac{e^{ik_y}+e^{-ik_y}}{2}\right] &(1.27) \\
&= \sum_{\mathbf{k}\in\mathrm{BZ}} \sum_{\sigma} \epsilon_{\mathbf{k}} \hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} &(1.28)
\end{aligned}$$

where we used Eqns. (1.18), (1.26). We defined the bare bands:

$$\epsilon_{\mathbf{k}} \equiv -2t\left(\cos k_x + \cos k_y\right) \tag{1.29}$$

Local repulsion. The local repulsion transforms as:

$$\begin{aligned}
\hat{H}_U &= U \sum_{\mathbf{r}\in\mathcal{L}} \hat{c}^{\dagger}_{\mathbf{r}\uparrow}\hat{c}^{\dagger}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow} \\
&\overset{\mathcal{F}}{=} U \sum_{\mathbf{r}\in\mathcal{L}} \frac{1}{(L_x L_y)^2} \sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4} e^{i\mathbf{k}_1\cdot\mathbf{r}}\hat{c}^{\dagger}_{\mathbf{k}_1\uparrow} e^{i\mathbf{k}_2\cdot\mathbf{r}}\hat{c}^{\dagger}_{\mathbf{k}_2\downarrow} e^{-i\mathbf{k}_3\cdot\mathbf{r}}\hat{c}_{\mathbf{k}_3\downarrow} e^{-i\mathbf{k}_4\cdot\mathbf{r}}\hat{c}_{\mathbf{k}_4\uparrow} \\
&= \frac{U}{L_x L_y} \sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4} \hat{c}^{\dagger}_{\mathbf{k}_1\uparrow}\hat{c}^{\dagger}_{\mathbf{k}_2\downarrow}\hat{c}_{\mathbf{k}_3\downarrow}\hat{c}_{\mathbf{k}_4\uparrow} \times \frac{1}{L_x L_y} \sum_{\mathbf{r}\in\mathcal{L}} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}} \\
&= \frac{U}{L_x L_y} \sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4} \hat{c}^{\dagger}_{\mathbf{k}_1\uparrow}\hat{c}^{\dagger}_{\mathbf{k}_2\downarrow}\hat{c}_{\mathbf{k}_3\downarrow}\hat{c}_{\mathbf{k}_4\uparrow}\delta_{\mathbf{k}_1+\mathbf{k}_2=\mathbf{k}_3+\mathbf{k}_4}
\end{aligned}$$

where we used Eq. (1.25). Let $\mathbf{k}_1+\mathbf{k}_2=\mathbf{k}_3+\mathbf{k}_4=\mathbf{K}$, and define $\mathbf{k}$, $\mathbf{k}'$ such that

$$\mathbf{k}_1 \equiv \mathbf{K}+\mathbf{k} \qquad \mathbf{k}_2 \equiv \mathbf{K}-\mathbf{k} \qquad \mathbf{k}_3 \equiv \mathbf{K}-\mathbf{k}' \qquad \mathbf{k}_4 \equiv \mathbf{K}+\mathbf{k}'$$

Sums over these variables must be intended as over the Brillouin Zone (BZ). Hence:

$$\hat{H}_U \overset{\mathcal{F}}{=} \frac{U}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\uparrow}\hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\downarrow}\hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow}\hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \tag{1.30}$$

Non-local attraction. To transform the non-local attraction $\hat{H}_V$, it is useful to start by defining:

$$h_{\mathbf{r}\sigma\sigma'} \equiv -V \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} \left(\hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\sigma'} + \hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}-\boldsymbol{\delta}\sigma'}\right)$$

with:

$$\hat{H}_V = \sum_{\mathbf{r}\in\mathcal{L}/2}\sum_{\sigma,\sigma'} h_{\mathbf{r}\sigma\sigma'}$$

The product of densities on neighbouring sites transforms as:

$$\begin{aligned}
\hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma'} &= \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\sigma'}\hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma'}\hat{c}_{\mathbf{r}\sigma} \\
&\overset{\mathcal{F}}{=} \frac{1}{(L_xL_y)^2}\sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}} e^{\pm i(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}}\hat{c}^\dagger_{\mathbf{k}_1\sigma}\hat{c}^\dagger_{\mathbf{k}_2\sigma'}\hat{c}_{\mathbf{k}_3\sigma'}\hat{c}_{\mathbf{k}_4\sigma}
\end{aligned}$$

Then:

$$\begin{aligned}
h_{\mathbf{r}\sigma\sigma'} &\overset{\mathcal{F}}{=} -\frac{V}{(L_xL_y)^2}\sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}}\sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}} \\
&\qquad\qquad \times\left(e^{i(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}} + e^{-i(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}}\right)\hat{c}^\dagger_{\mathbf{k}_1\sigma}\hat{c}^\dagger_{\mathbf{k}_2\sigma'}\hat{c}_{\mathbf{k}_3\sigma'}\hat{c}_{\mathbf{k}_4\sigma} \\
&= -\frac{2V}{(L_xL_y)^2}\sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}}\sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}}\cos\left[(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}\right]\hat{c}^\dagger_{\mathbf{k}_1\sigma}\hat{c}^\dagger_{\mathbf{k}_2\sigma'}\hat{c}_{\mathbf{k}_3\sigma'}\hat{c}_{\mathbf{k}_4\sigma}
\end{aligned}$$

The full non-local interaction is given by summing over all sites of $\mathcal{L}/2$ (which is, one sublattice). Applying Eq. (1.26) gives back momentum conservation

$$\frac{1}{L_xL_y}\sum_{\mathbf{r}\in\mathcal{L}/2} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}} = \frac{1}{2}\delta_{\mathbf{k}_1+\mathbf{k}_2=\mathbf{k}_3+\mathbf{k}_4}$$

where we used Eq. (1.26). Define $\mathbf{k}$, $\mathbf{k}'$ and $\mathbf{K}$ as we did fo the local repulsion, and

$$\delta\mathbf{k} \equiv \mathbf{k}-\mathbf{k}'$$

Hence:

$$\begin{aligned}
\hat{H}_V &= \sum_{\mathbf{r}\in\mathcal{L}/2}\sum_{\sigma,\sigma'} h_{\mathbf{r}\sigma\sigma'} \\
&\overset{\mathcal{F}}{=} -\frac{V}{L_xL_y}\sum_{\sigma\sigma'}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\cos\left(\delta\mathbf{k}\cdot\boldsymbol{\delta}\right)\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma'}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma'}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} \\
&= -\frac{V}{L_xL_y}\sum_{\sigma\sigma'}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\left[\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)\right]\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma'}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma'}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} \qquad (1.31)
\end{aligned}$$

Finally, we can separate in same spin and opposite spin channels. This gives:

$$\begin{aligned}
\hat{H}_V \overset{\mathcal{F}}{=} &\overbrace{-\frac{V}{L_xL_y}\sum_{\sigma}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\left[\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)\right]\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}}^{\text{s.s.}} \\
&\underbrace{-\frac{V}{L_xL_y}\sum_{\sigma}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\left[\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)\right]\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\overline{\sigma}}\hat{c}_{\mathbf{K}-\mathbf{k}'\overline{\sigma}}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}}_{\text{o.s.}}
\end{aligned}$$

The o.s. part can be simplified: since

$$\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\overline{\sigma}}\hat{c}_{\mathbf{K}-\mathbf{k}'\overline{\sigma}}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} = \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\overline{\sigma}}\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}'\overline{\sigma}}$$

due to fermionic anti-commutation rules, and since the composite transform

$$\mathbf{k} \to -\mathbf{k} \qquad \mathbf{k}' \to -\mathbf{k}'$$

leaves the form factor $\cos(\delta k_x) + \cos(\delta k_y)$ invariant, we reduce the spin sum to two times the $\sigma = \uparrow$ part,

$$\hat{H}_V^{(\text{o.s.})} = -\frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \left[\cos(\delta k_x) + \cos(\delta k_y)\right] \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \tag{1.32}$$

The $\mathcal{F}$-transform of the s.s. part is given explicitly by:

$$\begin{aligned}\hat{H}_V^{(\text{s.s.})} = -\frac{V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} &\left[\cos(\delta k_x) + \cos(\delta k_y)\right] \\ &\times \left[\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\uparrow} \hat{c}_{\mathbf{K}-\mathbf{k}'\uparrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} + \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\downarrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\downarrow}\right]\end{aligned} \tag{1.33}$$

All these expressions are going to be used in the analysis of each phase. In reciprocal space, for a quartic operator $\hat{c}^\dagger_{\mathbf{k}_1} \hat{c}^\dagger_{\mathbf{k}_2} \hat{c}_{\mathbf{k}_3} \hat{c}_{\mathbf{k}_4}$ we distinguish contractions as follows:

- Hartree contractions:
  $$\hat{c}^\dagger_{\mathbf{k}_1} \hat{c}^\dagger_{\mathbf{k}_2} \hat{c}_{\mathbf{k}_3} \hat{c}_{\mathbf{k}_4} \qquad \hat{c}^\dagger_{\mathbf{k}_1} \hat{c}^\dagger_{\mathbf{k}_2} \hat{c}_{\mathbf{k}_3} \hat{c}_{\mathbf{k}_4}$$
- Fock contractions:
  $$\hat{c}^\dagger_{\mathbf{k}_1} \hat{c}^\dagger_{\mathbf{k}_2} \hat{c}_{\mathbf{k}_3} \hat{c}_{\mathbf{k}_4} \qquad \hat{c}^\dagger_{\mathbf{k}_1} \hat{c}^\dagger_{\mathbf{k}_2} \hat{c}_{\mathbf{k}_3} \hat{c}_{\mathbf{k}_4}$$
- Cooper contractions:
  $$\hat{c}^\dagger_{\mathbf{k}_1} \hat{c}^\dagger_{\mathbf{k}_2} \hat{c}_{\mathbf{k}_3} \hat{c}_{\mathbf{k}_4} \qquad \hat{c}^\dagger_{\mathbf{k}_1} \hat{c}^\dagger_{\mathbf{k}_2} \hat{c}_{\mathbf{k}_3} \hat{c}_{\mathbf{k}_4}$$

with appropriate sign factors. With the EHM expressed in reciprocal space, there is no general feature left to address. We can move to the detailed discussion of each phase.

1.4 Square lattice spatial symmetry channels

This short conclusive section is apparently rather uncorrelated with all we said in the previous parts. We here introduce a set of orthonormal functions that results particularly useful to represent quantities defined on NNs sites on the square lattice. Let us start from a generic function of two variables $f(\mathbf{r}, \mathbf{r}')$. Suppose that, for a given centre $\mathbf{r}$, the function is non-zero only on the site itself $(\mathbf{r}' = \mathbf{r})$ and its NNs $\mathbf{r}' = \mathbf{r} \pm \boldsymbol{\delta}$, with $\boldsymbol{\delta} \in \{\underline{\mathbf{x}}, \underline{\mathbf{y}}\}$. In other words,

$$f(\mathbf{r}, \mathbf{r}') \neq 0 \qquad \Longleftrightarrow \qquad |\mathbf{r} - \mathbf{r}'| = 0, 1$$

in units of the lattice step. The following identity holds:

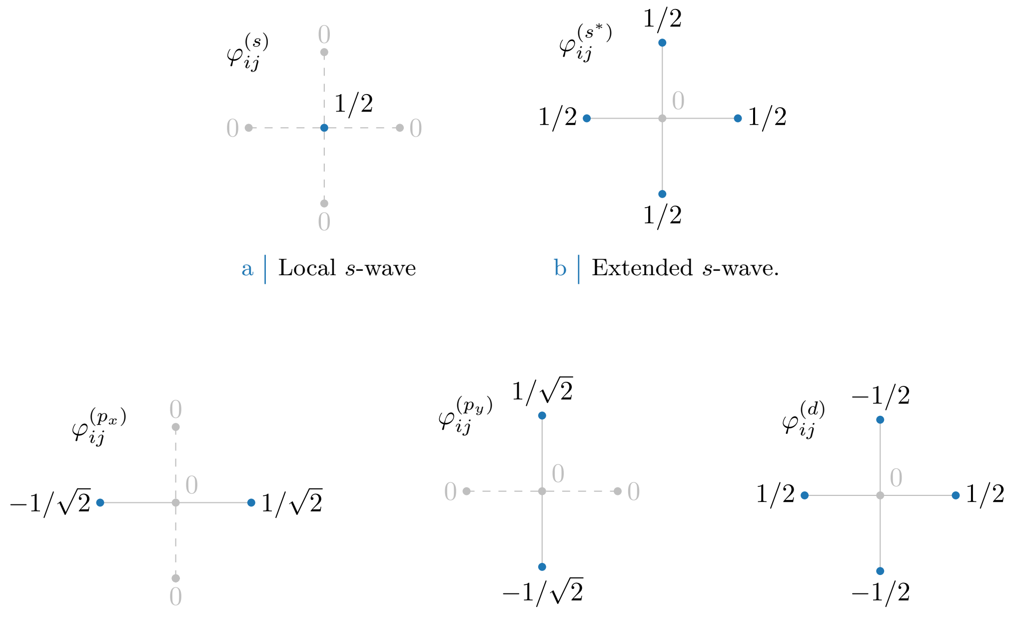
$$\begin{aligned} f(\mathbf{r}, \mathbf{r} + \underline{\mathbf{x}}) &= f(\mathbf{r}, \mathbf{r}') \delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} \\ &= f(\mathbf{r}, \mathbf{r}') \left(\varphi^{(s^*)}_{\mathbf{r}\mathbf{r}'} + \varphi^{(d)}_{\mathbf{r}\mathbf{r}'} + \sqrt{2}\varphi^{(p_x)}_{\mathbf{r}\mathbf{r}'}\right)\end{aligned}$$

where we defined the set of orthonormal functions $\varphi^{(\gamma)}_{\mathbf{r}\mathbf{r}'}$ listed explicitly in Tab. 1.4 and sketched in Fig. 1.4. The orthonormality condition reads:

$$\frac{1}{L_x L_y} \sum_{\mathbf{r}\mathbf{r}'} \varphi^{(\gamma)}_{\mathbf{r}\mathbf{r}'} \varphi^{(\gamma')*}_{\mathbf{r}\mathbf{r}'} = \delta_{\gamma\gamma'}$$

Similarly, also the following relations hold:

$$\begin{aligned} f(\mathbf{r}, \mathbf{r}) &= f(\mathbf{r}, \mathbf{r}') \varphi^{(s)}_{\mathbf{r}\mathbf{r}'} \\ f(\mathbf{r}, \mathbf{r} - \underline{\mathbf{x}}) &= f(\mathbf{r}, \mathbf{r}') \left(\varphi^{(s^*)}_{\mathbf{r}\mathbf{r}'} + \varphi^{(d)}_{\mathbf{r}\mathbf{r}'} - \sqrt{2}\varphi^{(p_x)}_{\mathbf{r}\mathbf{r}'}\right) \\ f(\mathbf{r}, \mathbf{r} + \underline{\mathbf{y}}) &= f(\mathbf{r}, \mathbf{r}') \left(\varphi^{(s^*)}_{\mathbf{r}\mathbf{r}'} - \varphi^{(d)}_{\mathbf{r}\mathbf{r}'} + \sqrt{2}\varphi^{(p_y)}_{\mathbf{r}\mathbf{r}'}\right) \\ f(\mathbf{r}, \mathbf{r} - \underline{\mathbf{y}}) &= f(\mathbf{r}, \mathbf{r}') \left(\varphi^{(s^*)}_{\mathbf{r}\mathbf{r}'} - \varphi^{(d)}_{\mathbf{r}\mathbf{r}'} - \sqrt{2}\varphi^{(p_y)}_{\mathbf{r}\mathbf{r}'}\right)\end{aligned}$$



a | Local s-wave b | Extended s-wave.

c | p_x-wave. d | p_y-wave. e | $d_{x^2-y^2}$-wave.

Figure 1.4 | Form factors at different topologies, as listed in Tab. 1.4. In figures five sites are represented: the hub site i and its four NN. Solid lines represent non-zero values for $\varphi_{\boldsymbol{\delta}}$, while dashed lines represent vanishing factors.

Structure	**Form factor**	**Graph**
s-wave	$\varphi^{(s)}_{\mathbf{rr'}} = \delta_{\mathbf{rr'}}$	Fig. 1.4a
Extended s-wave	$\varphi^{(s^*)}_{\mathbf{rr'}} = \frac{1}{2}\left[\delta_{\mathbf{r'}=\mathbf{r}+\underline{\mathbf{x}}} + \delta_{\mathbf{r'}=\mathbf{r}-\underline{\mathbf{x}}} + \delta_{\mathbf{r'}=\mathbf{r}+\underline{\mathbf{y}}} + \delta_{\mathbf{r'}=\mathbf{r}-\underline{\mathbf{y}}}\right]$	Fig. 1.4b
p_x-wave	$\varphi^{(p_x)}_{\mathbf{rr'}} = \frac{1}{\sqrt{2}}\left[\delta_{\mathbf{r'}=\mathbf{r}+\underline{\mathbf{x}}} - \delta_{\mathbf{r'}=\mathbf{r}-\underline{\mathbf{x}}}\right]$	Fig. 1.4c
p_y-wave	$\varphi^{(p_y)}_{\mathbf{rr'}} = \frac{1}{\sqrt{2}}\left[\delta_{\mathbf{r'}=\mathbf{r}+\underline{\mathbf{y}}} - \delta_{\mathbf{r'}=\mathbf{r}-\underline{\mathbf{y}}}\right]$	Fig. 1.4d
$d_{x^2-y^2}$-wave	$\varphi^{(d)}_{\mathbf{rr'}} = \frac{1}{2}\left[\delta_{\mathbf{r'}=\mathbf{r}+\underline{\mathbf{x}}} + \delta_{\mathbf{r'}=\mathbf{r}-\underline{\mathbf{x}}} - \delta_{\mathbf{r'}=\mathbf{r}+\underline{\mathbf{y}}} - \delta_{\mathbf{r'}=\mathbf{r}-\underline{\mathbf{y}}}\right]$	Fig. 1.4e

Table 1.4 | Form factors for a function defined on a square lattice, with a prefactor included for normalization. We use lattice vector notation here. The last column indicates the graph representation of each contribution given in Fig. 1.4. Subscript $x^2 - y^2$ is omitted for notational clarity.

Thus, defining:

$$
\begin{aligned}
f^{(s)} &\equiv f(\mathbf{r},\mathbf{r}) \\
f^{(s^*)} &\equiv f(\mathbf{r},\mathbf{r}+\underline{\mathbf{x}}) + f(\mathbf{r},\mathbf{r}-\underline{\mathbf{x}}) + f(\mathbf{r},\mathbf{r}+\underline{\mathbf{y}}) + f(\mathbf{r},\mathbf{r}-\underline{\mathbf{y}}) \\
f^{(p_x)} &\equiv f(\mathbf{r},\mathbf{r}+\underline{\mathbf{x}}) - f(\mathbf{r},\mathbf{r}-\underline{\mathbf{x}}) \\
f^{(p_y)} &\equiv f(\mathbf{r},\mathbf{r}+\underline{\mathbf{y}}) - f(\mathbf{r},\mathbf{r}-\underline{\mathbf{y}}) \\
f^{(d)} &\equiv f(\mathbf{r},\mathbf{r}+\underline{\mathbf{x}}) + f(\mathbf{r},\mathbf{r}-\underline{\mathbf{x}}) - f(\mathbf{r},\mathbf{r}+\underline{\mathbf{y}}) - f(\mathbf{r},\mathbf{r}-\underline{\mathbf{y}})
\end{aligned}
$$

we get the decomposition:

$$
f(\mathbf{r},\mathbf{r}') = \sum_{\gamma} f^{(\gamma)} \varphi^{(\gamma)}_{\mathbf{rr'}}
$$

Structure	Structure factor	Graph
s-wave	$\varphi_{\mathbf{k}}^{(s)} = 1$	Fig. 1.4a
Extended s-wave	$\varphi_{\mathbf{k}}^{(s^*)} = \cos k_x + \cos k_y$	Fig. 1.4b
p_x-wave	$\varphi_{\mathbf{k}}^{(p_x)} = i\sqrt{2}\sin k_x$	Fig. 1.4c
p_y-wave	$\varphi_{\mathbf{k}}^{(p_y)} = i\sqrt{2}\sin k_y$	Fig. 1.4d
$d_{x^2-y^2}$-wave	$\varphi_{\mathbf{k}}^{(d)} = \cos k_x - \cos k_y$	Fig. 1.4e

Table 1.5 Structure factors derived from the correlation structures of Tab. 1.4. The functions hereby defined are orthonormal, and are plotted in Fig. 1.5.

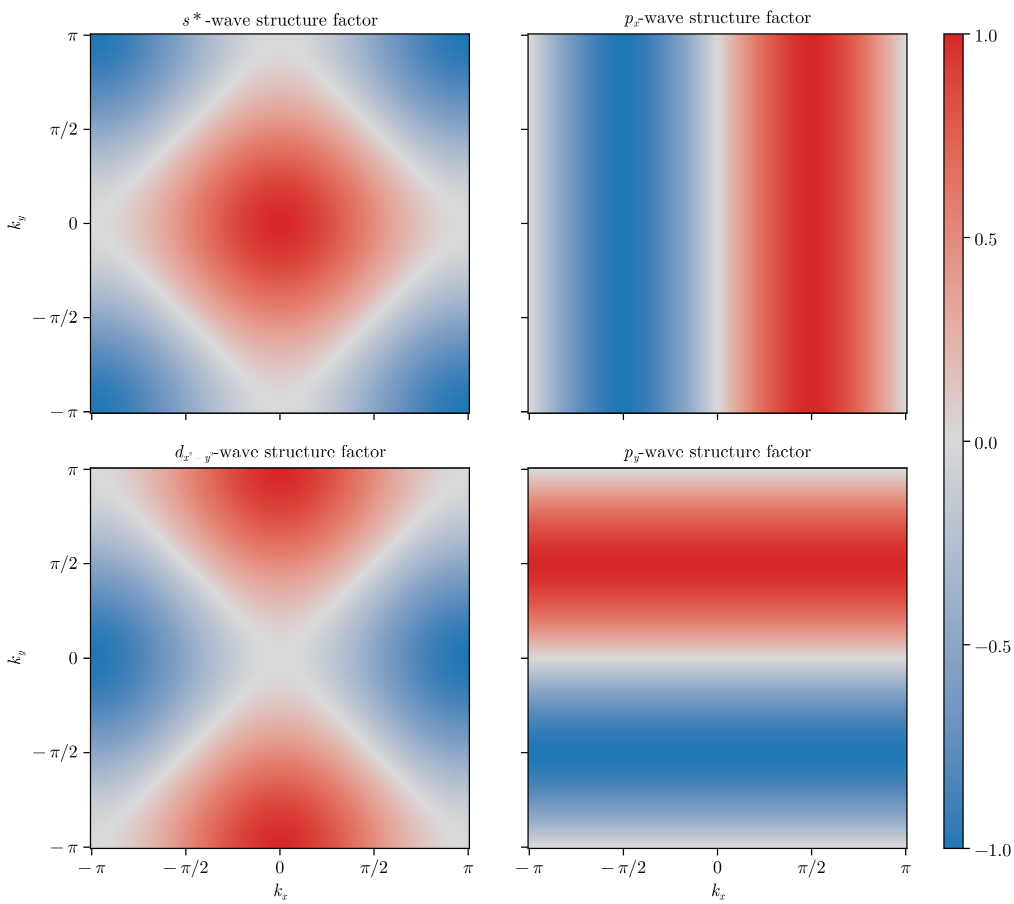


Figure 1.5 Representation of the various structure factors listed in Tab. 1.5 on the BZ. For the s^* and $d_{x^2-y^2}$-wave factors, the real part is plotted; for the p_ℓ factors, the imaginary part is plotted.

We decomposed the original function in its harmonics. Moving to reciprocal space, we get the $\mathcal{F}$-transformed functions of Tab. 1.5, also satisfying the orthonormality condition:

$$\frac{1}{L_x L_y} \sum_{\mathbf{k}} \varphi_{\mathbf{k}}^{(\gamma)} \varphi_{\mathbf{k}}^{(\gamma')*} = \delta_{\gamma\gamma'}$$

The functions of Tab. 1.5 are plotted in the BZ in Fig. 1.5.

We will refer to these harmonics as "symmetry channels", indicating the various spatial components of the order parameter associated to a given phase transition. As we will see, indeed, we consider order parameters that have specific transformation properties under planar rotations.

To distinguish the various spatial harmonics of the order parameter is useful in order to apply symmetry-based selection rules.

Finally, a very useful mathematical identity is given by the following argument. Consider Eq. (1.31), where is present the factor $\cos(\delta k_x) + \cos(\delta k_y)$ (with $\delta k_x = k_x - k'_x$ and $\delta k_y = k_y - k'_y$). Expanding,

$$\cos(\delta k_x) + \cos(\delta k_y) = c_x c'_x + c_y c'_y + s_x s'_x + s_y s'_y$$

In last passage, for the sake of readability, we used the notations

$$c_\ell \equiv \cos k_\ell \qquad s_\ell \equiv \sin k_\ell \qquad c'_\ell \equiv \cos k'_\ell \qquad s'_\ell \equiv \sin k'_\ell$$

Group the four terms,

$$\underbrace{(c_x c'_x + c_y c'_y)}_{\text{Symmetric}} + \underbrace{(s_x s'_x + s_y s'_y)}_{\text{Anti-symmetric}} \tag{1.34}$$

All four terms are invariant under simultaneous inversion of both momenta, $(\mathbf{k}, \mathbf{k}') \to (-\mathbf{k}, -\mathbf{k}')$. Instead, if we invert just one momentum of the pair, the first two are symmetric for both arguments $\mathbf{k}$, $\mathbf{k}'$; the second two are anti-symmetric. Decoupling the symmetric part, we obtain

$$c_x c'_x + c_y c'_y = \frac{1}{2}(c_x + c_y)(c'_x + c'_y) + \frac{1}{2}(c_x - c_y)(c'_x - c'_y)$$

which finally gives:

$$\begin{aligned}
\cos(\delta k_x) + \cos(\delta k_y) = {} & \frac{1}{2}(c_x + c_y)(c'_x + c'_y) && (s^*\text{-wave}) \\
& + s_x s'_x && (p_x\text{-wave}) \\
& + s_y s'_y && (p_y\text{-wave}) \\
& + \frac{1}{2}(c_x - c_y)(c'_x - c'_y) && (d_{x^2-y^2}\text{-wave})
\end{aligned}$$

or, in a more compact form

$$\cos(\delta k_x) + \cos(\delta k_y) = \frac{1}{2}\sum_\gamma \varphi_{\mathbf{k}}^{(\gamma)} \varphi_{\mathbf{k}'}^{(\gamma)*} \quad \text{for } \gamma \in \{s^*, p_x, p_y, d_{x^2-y^2}\} \tag{1.35}$$

$$= \frac{1}{2}\sum_\gamma \varphi_{\mathbf{k}}^{(\gamma)*} \varphi_{\mathbf{k}'}^{(\gamma)} \tag{1.36}$$

This decomposition will result particularly handy in later calculations.

Chapter 2

The normal phase

ToDo

- Clarify the Mean-Field decomposition of quartic operators OR use the self-consistent fields

In this chapter we apply the HF methods sketched in Sec. ?? to the "normal phase". We are studying the model in a state were no symmetry of those enumerated in Tab. 1.1 is broken. Even though this phase may appear not so interesting, it actually shows a significant physical difference with respect to the conventional perfectly degenerate Fermi gas.

HF methods applied in this context will lead us to a self-consistent hopping renormalisation effect. For certain values of the non-local attraction V and the doping level, a complete Mott localisation effect takes place. The resulting bands shrinking and flattening is the main net effect of the presence of $\hat{H}_V$ in the EHM.

This chapter is divided in three main parts. First, we discuss the symmetries of the normal phase, following the general discussion of Chap. 1, and apply HF methods to derive the self-consistent approximated hamiltonian. Second, we impose some theoretical constraints on the self-consistent solution. Lastly, we present numeric results.

2.1 The "normal" phase and its symmetries

In the normal phase no symmetry is broken. Thus, the HF wavefunction approximating the ground state must exhibit the full $\mathrm{U}(2) \otimes \mathrm{C}_4 \otimes \mathrm{T}_1^{(x)} \otimes \mathrm{T}_1^{(y)}$ symmetry discussed in Sec. 1.1. We take a slight detour from this, and instead allow for

$$\mathrm{C}_4 \overset{\mathrm{SSB}}{\to} \mathrm{C}_2$$

This is the only potential SSB we consider. It will turn out in Sec. 2.5.2 that even allowing for breaking of rotational symmetry, the ground-state actually does not break C_4. This can be seen as a convolute way to confirm that there is not a normal-nematic phase. While this may seem as an useless complication, the derivation will set grounds for actual nematic effects found in the superconducting phase.

Our HF method relies on the application of Wick's theorem on the quartic part of the interacting hamiltonian, expressed in Eq. (??). [It's not that immediate how Wick's theorem gives HF decomposition.]

Recalling Tabs. 1.2 and 1.3, for the normal phase we must consider the following contributes to the EVs of Eqns. (1.15), (1.20), (1.22):

$\hat{H}_U$	Hartree contraction
$\hat{H}_V$ (o.s. sector)	Hartree contraction
$\hat{H}_V$ (s.s. sector)	Hartree and Fock contractions

As we will see in detail and is summarised in Tab. 2.1, Hartree contractions give a chemical potential shift, while the unique Fock contraction effectively renormalises the bare bands.

Operator	**Sector**	**RWCs**	**Net effect**
$\hat{H}_U$		Hartree	μ shift by $-nU$
$\hat{H}_V$	o.s.	Hartree	μ shift by nzV
	s.s.	Hartree	μ shift by nzV
		Fock	Bands renormalization and resymmetrization

Table 2.1 | Relevant Wick's contractions and net effect in the normal phase.

In the normal phase, spin is a pure physical degeneracy, being the $\uparrow$ and $\downarrow$ sectors separated; moreover, $\hat{H}$ is $\mathbb{Z}_2$ invariant with respect to spin inversion, $\sigma \leftrightarrow \overline{\sigma}$. This implies that all physical quantities, as is for the normal phase, should not depend on the spin index. We will perform all computations leaving σ explicit and using this consideration at the end.

2.2 Hartree effects: chemical potential renormalization

Let us start by the analysis of the Hartree channel. The Hartree channel is defined as operator resulting from summing Hartree-like terms originating from Wick's decomposition of the quartic parts of $\hat{H}$. Each quartic term decomposes as a sum of fully contracted pairs, as in Eq. (**??**).

In the final hamiltonian, the net effect originating from Hartree terms will be a renormalisation of μ. To derive analytically this effect is only important to the ends of computing correctly the free energy density of the approximated ground state. As is discussed in Sec. **??**, in the context of numerical simulations at fixed density, the chemical potential is determined self-consistently at each step; thus any net effect that shifts μ is effectively ignored within (and limitedly to) the iterative algorithm.

2.2.1 Hartree channel in the normal phase: analytic derivation

The quartic part of $\hat{H}$ is given by the the local repulsion $\hat{H}_U$ and the non-local attraction $\hat{H}_V$. [To be continued...]

Local repulsion. In the normal phase, the local repulsion $\hat{H}_U$ decomposes in Hartree channel as

$$\hat{H}_U \simeq U \sum_i \left[\langle \hat{n}_{i\uparrow} \rangle \hat{n}_{i\downarrow} + \hat{n}_{i\uparrow} \langle \hat{n}_{i\downarrow} \rangle - \langle \hat{n}_{i\downarrow} \rangle \langle \hat{n}_{i\uparrow} \rangle \right]$$

For a translationally invariant phase such as the normal phase, it holds

$$\langle \hat{n}_{i\sigma} \rangle = n \tag{2.1}$$

which gives

$$\begin{aligned} \hat{H}_U &\simeq nU \sum_i \left[\hat{n}_{i\uparrow} + \hat{n}_{i\downarrow} \right] - U \sum_i n^2 \\ &= nU \times \hat{N} - E^{(\mathrm{N})}_{\mathrm{H}/U} \end{aligned}$$

being $\hat{N}$ the total number operator and $E^{(\mathrm{N})}_{\mathrm{H}/U}$ the "contraction energy" (also known as double counting term). In Sec. 2.2.2 we deal with it. The chemical potential is corrected by

$$\mu \to \mu - nU \tag{2.2}$$

The non-local attraction further corrects μ.

Non-local attraction. The non-local repulsion is divided in two sectors, as in Eq. (1.17). Its Hartree-like terms are:

$$\hat{H}_V \simeq \overbrace{-V\sum_{\langle ij\rangle}\sum_{\sigma}\left[\langle\hat{n}_{i\sigma}\rangle\hat{n}_{j\sigma}+\hat{n}_{i\sigma}\langle\hat{n}_{j\sigma}\rangle-\langle\hat{n}_{i\sigma}\rangle\langle\hat{n}_{j\sigma}\rangle\right]}^{\text{s.s.}} \underbrace{-V\sum_{\langle ij\rangle}\sum_{\sigma}\left[\langle\hat{n}_{i\sigma}\rangle\hat{n}_{j\bar{\sigma}}+\hat{n}_{i\sigma}\langle\hat{n}_{j\bar{\sigma}}\rangle-\langle\hat{n}_{i\sigma}\rangle\langle\hat{n}_{j\bar{\sigma}}\rangle\right]}_{\text{o.s.}} + (\text{all the rest}) \tag{2.3}$$

where "all the rest" collects all non-Hartree terms. Recalling the Ansatz of Eq. (2.1), we get

$$\hat{H}_V \simeq \underbrace{-nV\sum_{\langle ij\rangle}\sum_{\sigma}\left[\hat{n}_{j\sigma}+\hat{n}_{i\sigma}\right]}_{\text{s.s.}}\underbrace{-nV\sum_{\langle ij\rangle}\sum_{\sigma}\left[\hat{n}_{j\bar{\sigma}}+\hat{n}_{i\sigma}\right]}_{\text{o.s.}}-E^{(\mathrm{N})}_{\mathrm{H}/V}+(\text{all the rest})$$

with $E^{(\mathrm{N})}_{\mathrm{H}/V}$ the normal state shift to energy brought by $\hat{H}_V$. Both sums above give a number operator. Note that we sum over NNs: each site is connected to $z = 4$ sites. Including this combinatorial factor, we have

$$\hat{H}_V \simeq -2nzV \times \hat{N} - E^{(\mathrm{N})}_{\mathrm{H}/V} + (\text{all the rest})$$

This accounts for the final Hartree shift of μ,

$$\mu \to \mu + 2znV \tag{2.4}$$

Considering the net shift to μ due to both interactions by combining Eqns. (2.2), (2.4), we get the final RMP anticipated in Tab. 2.1,

$$\tilde{\mu} \equiv \mu + n(2zV - U) \tag{2.5}$$

This result will remain valid throughout all phases discussed in this text. For the AF phase, Sec. 3.2.1, the SDW character leaves this shift untouched. The SC phase, Sec. **??**, we will discuss is inherently translational invariant, thus the computation is the same as here.

2.2.2 Hartree contraction energy

The double counting terms accounting for energy shifts in the Hartree channel are, as anticipated:

$$\begin{aligned} E^{(\mathrm{N})}_{\mathrm{H}/U} &= \sum_i \langle\hat{n}_{i\downarrow}\rangle\langle\hat{n}_{i\uparrow}\rangle \\ &= L_x L_y \times n^2 U \end{aligned} \tag{2.6}$$

for the local repulsion and

$$\begin{aligned} E^{(\mathrm{N})}_{\mathrm{H}/V} &= -V\sum_{\langle ij\rangle}\sum_{\sigma}\left[\langle\hat{n}_{i\sigma}\rangle\langle\hat{n}_{j\sigma}\rangle+\langle\hat{n}_{i\sigma}\rangle\langle\hat{n}_{j\bar{\sigma}}\rangle\right] \\ &= -2V\sum_{\langle ij\rangle}\sum_{\sigma} n^2 \\ &= -L_x L_y \times 2zn^2 V \end{aligned} \tag{2.7}$$

for the non-local attraction. In the last passage, we used Eq. (1.14) and spin degeneracy. The contraction energy is an extensive quantity. When computing total free energy density, we will use the contraction energy per site.

2.3 Fock effects: bare bands renormalization

Analysis of Fock terms will lead us to the effect that makes the normal phase interesting, when compared to the free Fermi gas on the square lattice. From Wick's decomposition of $\hat{H}_V$, the unique allowed Fock term comes from the same-spin sector. This selection rules is implied by the SU(2) symmetry of the normal phase. Then:

$$\hat{H}_V \simeq (\text{Hartree channel}) + V \sum_{\langle ij\rangle} \sum_{\sigma} \left[\langle \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} \rangle \hat{c}^{\dagger}_{j\sigma} \hat{c}_{i\sigma} + \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} \langle \hat{c}^{\dagger}_{j\sigma} \hat{c}_{i\sigma} \rangle - \langle \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} \rangle \langle \hat{c}^{\dagger}_{j\sigma} \hat{c}_{i\sigma} \rangle \right] \tag{2.8}$$

Note the $+$ sign in front of the displayed term: it originates from the $-$ sign in Fock term of Eq. (**??**). A bond-wise hopping amplitude can be defined,

$$\tilde{t}_{ij\sigma} \equiv t - V \langle \hat{c}^{\dagger}_{j\sigma} \hat{c}_{i\sigma} \rangle$$

The effective kinetic hamiltonian is given by

$$\begin{aligned} \hat{H}_{\tilde{t}} &= \hat{H}_t + V \sum_{\langle ij\rangle} \sum_{\sigma} \left[\langle \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} \rangle \hat{c}^{\dagger}_{j\sigma} \hat{c}_{i\sigma} + \text{h.c.} \right] \\ &= - \sum_{\langle ij\rangle} \sum_{\sigma} \left[\tilde{t}_{ij\sigma} \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} + \text{h.c.} \right] \end{aligned}$$

To proceed, it is useful to move to reciprocal space.

2.3.1 Fock channel in the normal phase: analytic derivation

In reciprocal space, the effective hopping must be transformed as well. Consider the Fourier Transform $\mathcal{F}$ given in Eq. (1.33), applied to the displayed part of Eq. (2.8),

$$\begin{aligned} & V \sum_{\langle ij\rangle} \sum_{\sigma} \left[\langle \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} \rangle \hat{c}^{\dagger}_{j\sigma} \hat{c}_{i\sigma} + \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} \langle \hat{c}^{\dagger}_{j\sigma} \hat{c}_{i\sigma} \rangle - \langle \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} \rangle \langle \hat{c}^{\dagger}_{j\sigma} \hat{c}_{i\sigma} \rangle \right] \\ & \qquad \overset{\mathcal{F}}{=} \frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \sum_{\sigma} \left[\cos(\delta k_x) + \cos(\delta k_y) \right] \langle \hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}'\sigma} \rangle \hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} - E^{(\mathrm{N})}_{\mathrm{F}/V} \end{aligned} \tag{2.9}$$

Here, the 2 prefactor comes from recognizing that the first two operators in square brackets in the first line generate identical contributions to the full sum (as is easy to see by performing trivial variables changes); the double counting energy shift is given by

$$E^{(\mathrm{N})}_{\mathrm{F}/V} \equiv \frac{V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \sum_{\sigma} \left[\cos(\delta k_x) + \cos(\delta k_y) \right] \langle \hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}'\sigma} \rangle \langle \hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} \rangle \tag{2.10}$$

Up to this point, this derivation is general and will be re-used in next chapters. Here the specificity of the physical phase settles in: if we assume the normal state to be a free degenerate Fermi gas with renormalized bands $\tilde{\epsilon}_{\mathbf{k}}$, we get

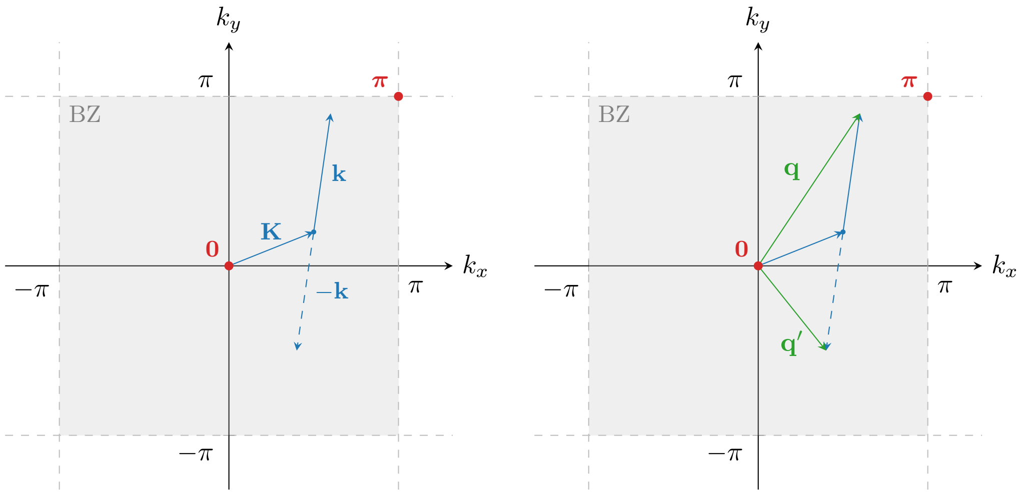
$$\langle \hat{c}^{\dagger}_{\mathbf{k}_1\sigma_1} \hat{c}_{\mathbf{k}_2\sigma_2} \rangle = \delta_{\mathbf{k}_1=\mathbf{k}_2} \delta_{\sigma_1=\sigma_2} f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \tag{2.11}$$

being f the Fermi-Dirac distribution,

$$f(\epsilon; \beta, \mu) \equiv \frac{1}{e^{\beta(\epsilon-\mu)} + 1}$$

It follows that Eq. (2.9) becomes

$$\begin{aligned} & V \sum_{\langle ij\rangle} \sum_{\sigma} \left[\langle \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} \rangle \hat{c}^{\dagger}_{j\sigma} \hat{c}_{i\sigma} + \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} \langle \hat{c}^{\dagger}_{j\sigma} \hat{c}_{i\sigma} \rangle - \langle \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} \rangle \langle \hat{c}^{\dagger}_{j\sigma} \hat{c}_{i\sigma} \rangle \right] \\ & \qquad \overset{\mathcal{F}}{=} \frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k}} \sum_{\sigma} \left[\cos(2k_x) + \cos(2k_y) \right] \langle \hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}\sigma} \rangle \hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}\sigma} - E^{(\mathrm{N})}_{\mathrm{F}/V} \end{aligned} \tag{2.12}$$



a | Generic vectors.   b | Change of variables.

Figure 2.1 | Representation of the variable change employed to pass from Eq. (2.12) to (2.13). In Fig. 2.1a generic vectors are considered, cycling over all values of $\mathbf{K}, \mathbf{k} \in$ BZ. In Fig. 2.1b is depicted the variables change to the new vectors $\mathbf{q}, \mathbf{q}'$.

since the only contribution comes from $\mathbf{k}' = -\mathbf{k}$.

Let now $\mathbf{q} \equiv \mathbf{K} + \mathbf{k}$, $\mathbf{q}' \equiv \mathbf{K} - \mathbf{k}$. A representation of this variables change is given in Fig. 2.1. Being for $\ell = x, y$

$$\begin{aligned} \delta q_\ell &\equiv q_\ell - q'_\ell \\ &= (K_\ell + k_\ell) - (K_\ell - k_\ell) \\ &= 2k_\ell \end{aligned}$$

we reduce Eq. (2.9) to

$$V\sum_{\langle ij\rangle}\sum_{\sigma}\left[\langle\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\rangle\hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma} + \hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\langle\hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}\rangle - \langle\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\rangle\langle\hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}\rangle\right]$$
$$\overset{\mathcal{F}}{=} \frac{2V}{L_x L_y}\sum_{\mathbf{q},\mathbf{q}'}\sum_{\sigma}\left[\cos(\delta q_x) + \cos(\delta q_y)\right]\langle\hat{c}^\dagger_{\mathbf{q}\sigma}\hat{c}_{\mathbf{q}\sigma}\rangle\hat{c}^\dagger_{\mathbf{q}'\sigma}\hat{c}_{\mathbf{q}'\sigma} - E^{(\mathrm{N})}_{\mathrm{F}/V} \tag{2.13}$$

Recall now the result of the symmetry channels decomposition of Eq. (1.35),

$$\cos(\delta q_x) + \cos(\delta q_y) = \frac{1}{2}\sum_{\gamma}\varphi^{(\gamma)}_{\mathbf{q}}\varphi^{(\gamma)*}_{\mathbf{q}'} \qquad \text{for } \gamma \in \{s^*, p_x, p_y, d_{x^2-y^2}\}$$

Thanks to this handy property, we reduce Eq. (2.13) to

$$V\sum_{\langle ij\rangle}\sum_{\sigma}\left[\langle\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\rangle\hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma} + \hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\langle\hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}\rangle - \langle\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\rangle\langle\hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}\rangle\right]$$
$$\overset{\mathcal{F}}{=} \sum_{\gamma,\sigma}\left[\frac{1}{L_x L_y}\sum_{\mathbf{q}\in\mathrm{BZ}}\varphi^{(\gamma)}_{\mathbf{q}}u_{\mathbf{q}\sigma}\right] \times V\sum_{\mathbf{q}'\in\mathrm{BZ}}\varphi^{(\gamma)*}_{\mathbf{q}'}\hat{c}^\dagger_{\mathbf{q}'\sigma}\hat{c}_{\mathbf{q}'\sigma} - E^{(\mathrm{N})}_{\mathrm{F}/V} \tag{2.14}$$

where we defined the EV:

$$u_{\mathbf{q}\sigma} \equiv \langle\hat{c}^\dagger_{\mathbf{q}\sigma}\hat{c}_{\mathbf{q}\sigma}\rangle \qquad u_{\mathbf{q}\sigma} = u^*_{\mathbf{q}\sigma} \tag{2.15}$$

The numeric value of $u_{\mathbf{q}\sigma}$ is given by the Fermi-Dirac distribution, but leaving it implicit we can re-use this computation later. Let $u^{(\gamma)}_{\sigma}$ be its γ-wave component,

$$u^{(\gamma)}_{\sigma} = \frac{1}{L_x L_y}\sum_{\mathbf{q}\in\mathrm{BZ}}\varphi^{(\gamma)}_{\mathbf{q}}\langle\hat{c}^\dagger_{\mathbf{q}\sigma}\hat{c}_{\mathbf{q}\sigma}\rangle$$

Symmetry	Self-consistency equation	Graph
s^*-wave	$u^{(s^*)} = \dfrac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}(\cos k_x + \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})$	Fig. 1.4b
$d_{x^2-y^2}$-wave	$u^{(d)} = \dfrac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}(\cos k_x - \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})$	Fig. 1.4e

Table 2.2 | Symmetry resolved self-consistency equations for the harmonics of $u_{\mathbf{k}}$.

The bare bands $\epsilon_{\mathbf{k}}$ are s^*-wave symmetric,

$$\epsilon_{\mathbf{k}} = -2t(\cos k_x + \cos k_y) = -4t\varphi_{\mathbf{k}}^{(s^*)}$$

We used one function of those listed in Tab. 1.5. The renormalised bands $\tilde{\epsilon}_{\mathbf{k}}$ can exhibit a more complex symmetry. In the normal phase inversion symmetry is unbroken, thus

$$u_{\mathbf{q}\sigma} = u_{-\mathbf{q}\sigma}$$

It follows that antisymmetric harmonics are zero,

$$u_\sigma^{(p_\ell)} = 0 \quad \text{for} \quad \ell \in \{x, y\}$$

The remaining s^* and d symmetries give legitimate HFPs with the respective self-consistency equations, reported in Tab. 2.2.Spin index is omitted due to spin degeneracy.

The two HFPs $u_\sigma^{(s^*)}$ and $u_\sigma^{(d)}$ are in general non-zero, since we are allowing for $\mathrm{C}_4 \overset{\mathrm{SSB}}{\to} \mathrm{C}_2$. Note that when imposing C_4 invariance the $u_\sigma^{(d)}$ HFP vanishes, while in general $u_\sigma^{(s^*)}$ is non-zero. From these considerations Eq. (2.14) becomes

$$\begin{aligned} V\sum_{\langle ij\rangle}\sum_\sigma \Big[\langle \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma}\rangle \hat{c}_{j\sigma}^\dagger \hat{c}_{i\sigma} + \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma}\langle \hat{c}_{j\sigma}^\dagger \hat{c}_{i\sigma}\rangle - \langle \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma}\rangle\langle \hat{c}_{j\sigma}^\dagger \hat{c}_{i\sigma}\rangle\Big] \\ \overset{\mathcal{F}}{=} V\sum_{\mathbf{q}\in\mathrm{BZ}}\sum_\sigma \Big[u_\sigma^{(s^*)}\varphi_{\mathbf{q}}^{(s^*)*} + u_\sigma^{(d)}\varphi_{\mathbf{q}}^{(d)*}\Big]\hat{c}_{\mathbf{q}\sigma}^\dagger \hat{c}_{\mathbf{q}\sigma} - E_{\mathrm{F}/V}^{(\mathrm{N})} \end{aligned} \tag{2.16}$$

Let

$$\tilde{\epsilon}_{\mathbf{k}\sigma} \equiv \epsilon_{\mathbf{k}\sigma} + u_\sigma^{(s^*)}V(\cos k_x + \cos k_y) + u_\sigma^{(d)}V(\cos k_x - \cos k_y) \tag{2.17}$$

This equation defines the bands renormalisation anticipated in Tab. 2.1, and is actually spin-independent. The fact that we have both s^* and a d-wave components introduces an anisotropy that is discussed in next section. However, we anticipate here that, as will result from Sec. 2.5.2, the d-wave harmonic will be suppressed. For now we keep this component ignoring these anticipations.

2.3.2 Fock contraction energy

Finally, let us derive the Fock contraction energy by expanding Eq. (2.10):

$$\begin{aligned} E_{\mathrm{F}/V}^{(\mathrm{N})} &= \frac{V}{L_xL_y}\sum_{\mathbf{q},\mathbf{q}'}\sum_\sigma [\cos(\delta q_x) + \cos(\delta q_y)]\langle \hat{c}_{\mathbf{q}\sigma}^\dagger \hat{c}_{\mathbf{q}'\sigma}\rangle\langle \hat{c}_{\mathbf{q}\sigma}^\dagger \hat{c}_{\mathbf{q}'\sigma}\rangle \\ &= L_xL_y \times \frac{V}{2}\sum_{\gamma,\sigma}\left[\frac{1}{L_xL_y}\sum_{\mathbf{q}\in\mathrm{BZ}}\varphi_{\mathbf{q}}^{(\gamma)}u_{\mathbf{q}\sigma}\right]\left[\frac{1}{L_xL_y}\sum_{\mathbf{q}'\in\mathrm{BZ}}\varphi_{\mathbf{q}'}^{(\gamma)*}u_{\mathbf{q}'\sigma}^*\right] \\ &= L_xL_y \times \frac{V}{2}\sum_\sigma\Big[\big|u_\sigma^{(s^*)}\big|^2 + \big|u_\sigma^{(d)}\big|^2\Big] \end{aligned} \tag{2.18}$$

This expression gives the Fock contraction energy coming from the $\hat{H}_V$ operator. The squared spectral amplitude of $u_{\mathbf{k}\sigma}$ here is summed, and contributes to contraction energy linearly in V. This suggests that a mixed-symmetry band could be energetically preferred.

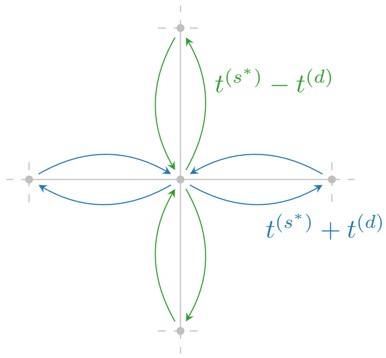


Figure 2.2 Graphic rendition of anisotropic hopping, with a s^*-wave component (symmetric by $\pi/$ rotations) and a d-wave component (antisymmetric).

2.3.3 Nematicity: extending hopping to d-wave bands

For the normal phase, there is no reason to expect nematic effects, nor to exclude them. They are observed in superconducting phase [22]. Thus, we allow the possibility.

From Tabs. 1.4 and 1.5 we can inverse-Fourier transform the form factors to get the full effective hopping,

$$\tilde{t}_{ij\sigma} = \sum_{\gamma\in\{s^*,d\}} t_\sigma^{(\gamma)} \varphi_{ij}^{(\gamma)}$$

with

$$t_\sigma^{(s^*)} = t - \frac{u_\sigma^{(s^*)} V}{2} \qquad t_\sigma^{(d)} = -\frac{u_\sigma^{(d)} V}{2}$$

This is what we intend for "nematic effects". The possibility that planar rotational symmetry is broken in such a way that the kinetic operators in x and y directions differ. A schematics of anisotropic hopping is given in Fig. 2.2: for each bond linking two sites, the hopping parameter can be a positive or negative combination of the two components, whether the link direction is vertical or horizontal. The solution needs to be invariant under exchange $x \leftrightarrow y$, thus the actual SSB is $\mathrm{C}_4 \overset{\mathrm{SSB}}{\to} \mathrm{C}_2/\mathbb{Z}_2$.

The fact that hopping is here anisotropic does not break full translational invariance $\mathrm{T}_1^{(x)} \otimes \mathrm{T}_1^{(y)}$. Imposing a free degenerate Fermi gas ground state in Eq. (2.11) leads to a uniform density. This is proven by Fourier-transforming the site number operator according to the convention of Eq. (1.24),

$$\hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}\sigma} \overset{\mathcal{F}}{=} \frac{1}{L_x L_y} \sum_{\mathbf{k},\mathbf{k}'} e^{i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{r}} \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}'\sigma}$$

Then the on-site density EV is

$$\langle \hat{n}_{\mathbf{r}\sigma} \rangle = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} f(\tilde{\epsilon}_\mathbf{k}; \beta, \tilde{\mu})$$

A perfectly degenerate Fermi gas is always translational invariant and thus cannot exhibit non-uniform symmetry. The fact that hopping is allowed to be anisotropic does not interfere with that.

Note that for null HFPs,

$$u_\sigma^{(s^*)} = 0 \qquad u_\sigma^{(d)} = 0$$

we have, recalling Tab. 1.4:

$$\tilde{t}_{\mathbf{r}\mathbf{r}'\sigma} = \frac{t}{2} \left[\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{y}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{y}}} \right]$$

The fact that $t/2$ appears here instead of t is a consequence of the equivalence:

$$-t \sum_{\langle ij \rangle} \sum_\sigma \left[\hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} + \mathrm{h.c.} \right] = -\frac{t}{2} \sum_{i,j\in\mathcal{S}} \sum_\sigma \left[\hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} + \mathrm{h.c.} \right]$$

where we including a prefactor that corrects double-counting. The RMP $\tilde{t}_{ij\sigma}$ must be interpreted like the right-hand side of the above equation, with the kinetic operator becoming:

$$\hat{H}_{\tilde{t}} \equiv -\frac{1}{2}\sum_{i,j\in\mathcal{S}}\sum_{\sigma}\left[\tilde{t}_{ij\sigma}\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma}+\text{h.c.}\right]$$

To limit the sum to NNs is a matter of the symmetry structure of the function $\tilde{t}_{ij\sigma}$.

We can define the direction-dependent hopping as

$$\tilde{t}^{(x)} \equiv t - \frac{u^{(s^*)}-u^{(d)}}{2}V \tag{2.19}$$

$$\tilde{t}^{(y)} \equiv t - \frac{u^{(s^*)}+u^{(d)}}{2}V \tag{2.20}$$

Both components depend linearly on V. The HFP $u^{(s^*)}$ is a measure of the direction-averaged hopping shift with respect to V, while $u^{(d)}$ measures the anisotropy,

$$u^{(s^*)} = \frac{\left(t-\tilde{t}^{(x)}\right)+\left(t-\tilde{t}^{(y)}\right)}{V} \qquad u^{(d)} = \frac{\tilde{t}^{(y)}-\tilde{t}^{(x)}}{V}$$

This discussion is general, and not specific to the normal phase. In next chapters we will interpret phase renormalisation as in this section.

2.4 Normal free energy density

In the end, we want to compare phases. Thus we need a way to extract the free energy density of the approximated ground-state for each phase. The procedure sketched here will be re-used in next chapters. Let

$$f^{(\mathrm{N})}_{\mathrm{MFT}} \equiv \frac{F^{(\mathrm{N})}_{\mathrm{MFT}}}{L_x L_y}$$

being F_{MFT} the total free energy. Consider the complete hamiltonian

$$\hat{H}-\mu\hat{N} \simeq \sum_{\mathbf{k},\sigma}\tilde{\epsilon}_{\mathbf{k}\sigma}\hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} + E^{(\mathrm{N})}_{\mathrm{c}} - \tilde{\mu}\hat{N}$$

where $E^{(\mathrm{N})}_{\mathrm{c}}$ is the (negative) total contraction energy for the normal phase,

$$E^{(\mathrm{N})}_{\mathrm{c}} \equiv -\left[E^{(\mathrm{N})}_{\mathrm{H}/U}+E^{(\mathrm{N})}_{\mathrm{H}/V}+E^{(\mathrm{N})}_{\mathrm{F}/V}\right]$$

having included the various contraction energies from Eqns. (2.6), (2.7) and (2.18).

Let us now take together all terms. From now on, we will omit spin index σ, making use of the always present $\mathbb{Z}_2$ spin inversion symmetry discussed in Sec. 2.1. For a free Fermi gas, the free energy density is made of four contributions:

1. The free particles ground state energy density,

$$e^{(\mathrm{N})}_{\mathrm{g}} = \frac{2}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}}(\tilde{\epsilon}_{\mathbf{k}}-\tilde{\mu})f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})$$

2. The contraction energy density,

$$\begin{aligned} e^{(\mathrm{N})}_{\mathrm{c}} &\equiv \frac{E^{(\mathrm{N})}_{\mathrm{c}}}{L_x L_y} \\ &= -n^2 U + V\left[2zn^2 - \left|u^{(s^*)}\right|^2 - \left|u^{(d)}\right|^2\right] \end{aligned}$$

3. The entropic contribution to free energy

$$e_s^{(\mathrm{N})} \equiv -Ts$$

with $T = 1/k_B\beta$ and s the free Fermi gas entropy density for a single spin sector [1],

$$s = -\frac{k_B}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \left[(1 - f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu}))\ln(1 - f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})) + f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\ln f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\right] \tag{2.21}$$

We use the identity:

$$\ln\left(1 - \frac{1}{e^x + 1}\right) = x + \ln\left(\frac{1}{e^x + 1}\right)$$

and, substituting $x = \beta(\tilde{\epsilon}_{\mathbf{k}} - \tilde{\mu})$, reduce the above expression to:

$$\begin{aligned} s = -\frac{k_B}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \Big[&\ln(1 - f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})) + f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\ln f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu}) \\ &- \beta(\tilde{\epsilon}_{\mathbf{k}} - \tilde{\mu}) f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu}) - f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\ln f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\Big] \end{aligned}$$

Second and fourth terms cancel out, and multiplying by $-2T$ (taking care of spin degeneracy) we have

$$e_s^{(\mathrm{N})} = \frac{2}{\beta L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \ln(1 - f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})) - \frac{2}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\tilde{\epsilon}_{\mathbf{k}} - \tilde{\mu}) f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})$$

4. The chemical potential correction

$$e_n^{(\mathrm{N})} = \mu n$$

Note here that we are using the "physical" value of μ, not its RMP counterpart $\tilde{\mu}$. The reason for this is discussed in Sec. ??.

Thus, composing everything

$$\begin{aligned} f_{\mathrm{MFT}}^{(\mathrm{N})} &\equiv e_{\mathrm{g}}^{(\mathrm{N})} + e_{\mathrm{c}}^{(\mathrm{N})} + e_s^{(\mathrm{N})} + e_n^{(\mathrm{N})} \\ &= \frac{2}{\beta L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \ln(1 - f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})) - V\left[\left|u^{(s^*)}\right|^2 + \left|u^{(d)}\right|^2\right] + \left[\mu + n(2zV - U)\right] n \end{aligned}$$

In Sec. ?? we argued that our iterative scheme computes $\tilde{\mu}$, not μ. Using Eq. (2.5), we see

$$\mu + n(2zV - U) = \tilde{\mu}$$

Then, the final expression is:

$$f_{\mathrm{MFT}}^{(\mathrm{N})} = \frac{2}{\beta L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \ln(1 - f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})) - V\left[\left|u^{(s^*)}\right|^2 + \left|u^{(d)}\right|^2\right] + \tilde{\mu} n \tag{2.22}$$

including all effects.

2.5 Theoretical constraints on the normal phase HFPs

Within this section we discuss some theoretical constraints on the two HFPs of the normal phase. The discussion is divided in two parts: first we consider the simpler case of unviolated C_4 symmetry, then we move to anisotropic hopping and $C_4 \overset{\mathrm{SSB}}{\to} C_2$.

Before starting, it is useful to spend a few words on the spatial structure of a mixed $s^* \otimes d$-wave band. Let in general for notational simplicity

$$\tilde{\epsilon}_{\mathbf{k}} = \epsilon^{(s^*)}(\cos k_x + \cos k_y) + \epsilon^{(d)}(\cos k_x - \cos k_y)$$

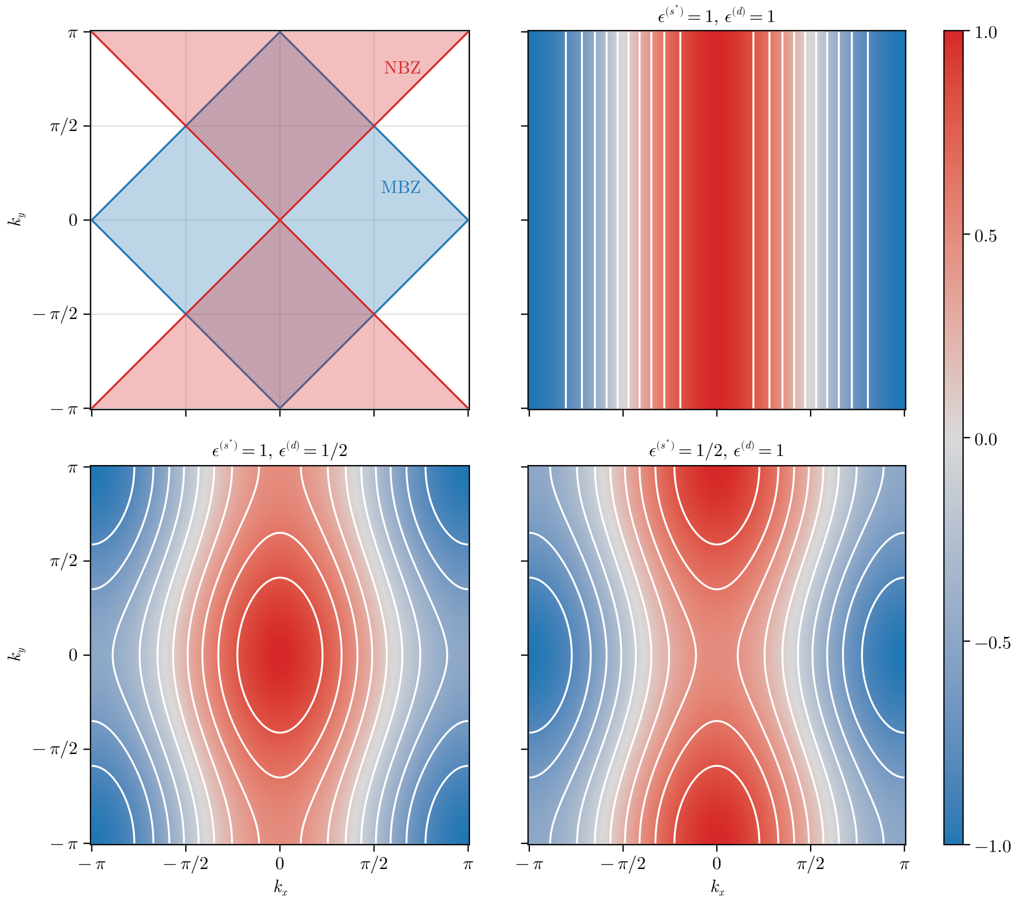


Figure 2.3 Theoretical representation of mixed-symmetry nematic bands for the normal phase. The white contour represent level lines. In top-left panel, the MBZ and NBZ are depicted. While these plots only account for positive-defined components, in text is described how to map each one on its counterpart with different sign.

It is the balance between the two components $\epsilon^{(s^*)}$ and $\epsilon^{(d)}$ that gives the main features of the renormalised band. Consider the s^*-wave and d-wave structure factors already presented in Fig. 1.5, and define the Magnetic Brillouin Zone (MBZ) and Nematic Brillouin Zone (NBZ) as follows:

$$\mathrm{MBZ} \equiv \left\{ \mathbf{k} \in \mathrm{BZ} \mid \varphi^{(s^*)}_{\mathbf{k}} \geq 0 \right\}$$
$$\mathrm{NBZ} \equiv \left\{ \mathbf{k} \in \mathrm{BZ} \mid \varphi^{(d)}_{\mathbf{k}} \geq 0 \right\}$$

represented in the top-left panel of Fig. 2.3. The reason for the name of the MBZ will be clear in Chap. 3. In the rest of the panels of Fig. 2.3 are reported examples of mixed symmetry bands, for positive components. The respective opposite sign plots are obtained by the following rules:

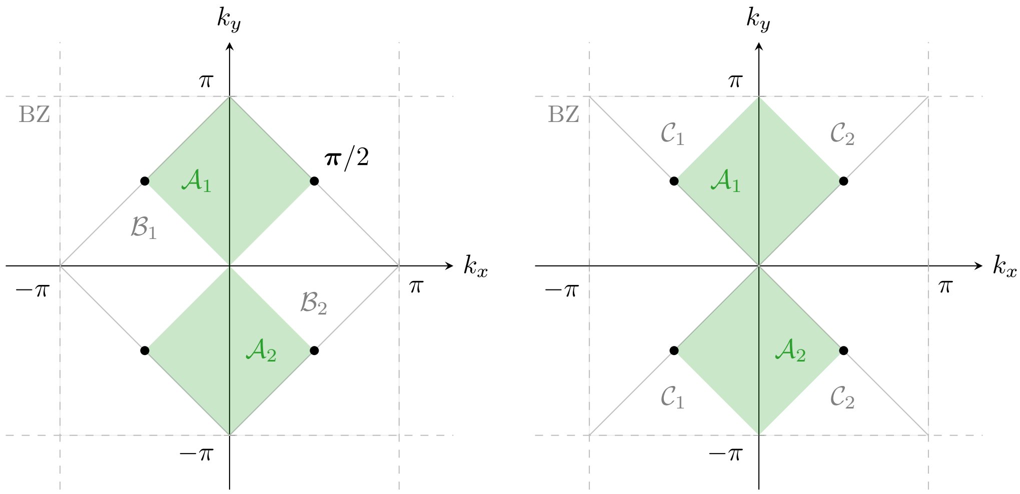
$$\begin{aligned}
&\epsilon^{(s^*)} < 0 \quad \text{and} \quad \epsilon^{(d)} < 0 \quad \Longrightarrow \quad \text{Invert colors in Fig. 2.3} \\
&\epsilon^{(s^*)} > 0 \quad \text{and} \quad \epsilon^{(d)} < 0 \quad \Longrightarrow \quad \text{Rotate plots by } \pi/2 \text{ in Fig. 2.3} \\
&\epsilon^{(s^*)} < 0 \quad \text{and} \quad \epsilon^{(d)} > 0 \quad \Longrightarrow \quad \text{Rotate plots by } \pi/2 \text{ and invert colors in Fig. 2.3}
\end{aligned}$$

Let us focus on the MBZ and the NBZ. Starting from the intersections of the two zones, both can be factored in two sectors of interest,

$$\mathrm{MBZ} = \mathcal{A} \cup \mathcal{B} \qquad \mathrm{NBZ} = \mathcal{A} \cup \mathcal{C}$$

being, accordingly to Fig. 2.4,

$$\mathcal{A} \equiv \mathcal{A}_1 \cup \mathcal{A}_2 \qquad \mathcal{B} \equiv \mathcal{B}_1 \cup \mathcal{B}_2 \qquad \mathcal{C} \equiv \mathcal{C}_1 \cup \mathcal{C}_2$$



a | MBZ sectors.

b | NBZ sectors.

Figure 2.4 | Depiction of the sectors composing the MBZ (Fig. 2.4a) and the NBZ (Fig. 2.4b), based on the divisions shown in the top-left panel of Fig. 2.3. The full green zone composes $\mathcal{A}$, the region where both the s^*-wave and d-wave structure factors are positive defined; its MBZ complementary white counterpart in the left panel gives $\mathcal{B}$, for which the s^*-wave structure factors is positive-definite while the d-wave is negative. Similarly the NBZ complementary white counterpart in the right panel gives $\mathcal{C}$, for which signs are exchanged. The black dots, at the contours intersection, are band nodes.

These domains satisfy:

$$\begin{aligned}\mathcal{A} &\equiv \left\{\mathbf{k} \in \mathrm{BZ} \mid \varphi_{\mathbf{k}}^{(s^*)} \geq 0 \wedge \varphi_{\mathbf{k}}^{(d)} \geq 0\right\} \\ &= \mathrm{MBZ} \cap \mathrm{NBZ}\end{aligned} \tag{2.23}$$

$$\begin{aligned}\mathcal{B} &\equiv \left\{\mathbf{k} \in \mathrm{BZ} \mid \varphi_{\mathbf{k}}^{(s^*)} \geq 0 \wedge \varphi_{\mathbf{k}}^{(d)} \leq 0\right\} \\ &= \mathrm{MBZ} \setminus \mathrm{NBZ}\end{aligned} \tag{2.24}$$

$$\begin{aligned}\mathcal{C} &\equiv \left\{\mathbf{k} \in \mathrm{BZ} \mid \varphi_{\mathbf{k}}^{(s^*)} \leq 0 \wedge \varphi_{\mathbf{k}}^{(d)} \geq 0\right\} \\ &= \mathrm{NBZ} \setminus \mathrm{MBZ}\end{aligned} \tag{2.25}$$

and will be used later on. Whatever the shape of the mixed band, the following equation always holds

$$|k_x| = \frac{\pi}{2} \quad \text{and} \quad |k_y| = \frac{\pi}{2} \quad \implies \quad \tilde{\epsilon}_{(k_x,k_y)} = 0$$

since only cosines are involved. This identifies four node points, represented as black dots in Fig. 2.4.

2.5.1 Constraints in presence of a robust C_4 symmetry and bands shrinking

Consider the simpler situation where C_4 symmetry is imposed. This implies that the renormalised bands are

$$\tilde{\epsilon}_{\mathbf{k}} \equiv (u^{(s^*)}V - 2t)(\cos k_x + \cos k_y)$$

with a single self-consistency equation

$$u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathrm{BZ}} (\cos k_x + \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$$

The following chain of deductions leads us to a constraint on the possible values of $u^{(s^*)}$:

1. For $t > 0$, there is no self-consistent solution admitting $u^{(s^*)} \leq 0$. Let us prove this: suppose instead $u^{(s^*)} \leq 0$. For any function $g_{\mathbf{k}}$ defined on the BZ:

$$\sum_{\mathbf{k}\in\mathrm{BZ}} g_{\mathbf{k}} = \sum_{\mathbf{k}\in\mathrm{MBZ}} [g_{\mathbf{k}} + g_{\mathbf{k}+\boldsymbol{\pi}}]$$

where we defined the vector $\boldsymbol{\pi} \equiv (\pi, \pi)$. The self-consistency equation thus reads

$$\begin{aligned} u^{(s^*)} &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\cos k_x + \cos k_y) \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) + f(\tilde{\epsilon}_{\mathbf{k}+\boldsymbol{\pi}}; \beta, \tilde{\mu}) \right] \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\cos k_x + \cos k_y) \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \right] \end{aligned} \tag{2.26}$$

having we used the antisymmetry of s^*-wave functions by $\boldsymbol{\pi}$ shifts. Now, if $u^{(s^*)} \leq 0$ the renormalised band $\tilde{\epsilon}_{\mathbf{k}}$ is negative inside the MBZ and positive outside, implying

$$\forall \mathbf{k} \in \mathrm{MBZ} \quad : \quad f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \geq f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$$

the equality only being satisfied on the MBZ contour. Then, since inside the MBZ it holds by definition $\cos k_x + \cos k_y \geq 0$, the above equations imply

$$u^{(s^*)} \leq 0 \quad \Longrightarrow \quad u^{(s^*)} \geq 0$$

To exclude the possibility of $u^{(s^*)} = 0$, it suffices to notice that from the argument above such a situation only arises when the MBZ contour degenerates to the entirety of the MBZ itself – which is, when the band is flat. But that cannot be the case for $u^{(s^*)} = 0$, as long as $t > 0$. Thus we have proven that

$$u^{(s^*)} > 0$$

2. For $t > 0$, there is no self-consistent solution admitting $u^{(s^*)} \geq 2t/V$. We proceed exactly as did for the previous point, up to the passage:

$$u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\cos k_x + \cos k_y) \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \right]$$

Now, if $u^{(s^*)} \geq 2t/V$ this means the prefactor in front of the renormalised bands is positive. Then $\tilde{\epsilon}_{\mathbf{k}}$ is positive inside the MBZ and negative outside, implying

$$\forall \mathbf{k} \in \mathrm{MBZ} \quad : \quad f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \leq f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$$

thus we have

$$u^{(s^*)} \geq 2t/V \quad \Longrightarrow \quad u^{(s^*)} \leq 0$$

which is absurd. Thus

$$u^{(s^*)} < 2t/V$$

Collecting the two points above, we have the constraint on $u^{(s^*)}$

$$\mathrm{C}_4 \quad \Longrightarrow \quad 0 < u^{(s^*)} < 2t/V \tag{2.27}$$

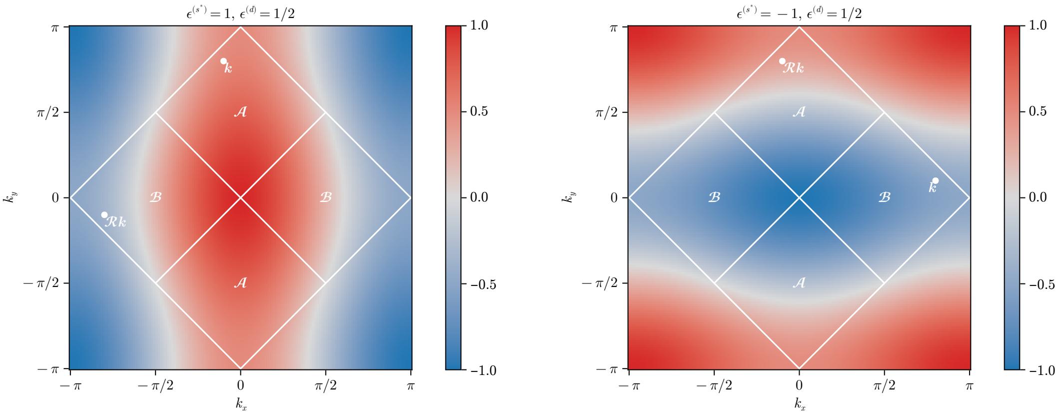
The effective hopping is smaller than the original hopping. We can denominate this net effect "bands shrinking".

2.5.2 Natural suppression of nematicity within $\mathrm{C}_2/\mathbb{Z}_2$ symmetry

Let us extend the arguments above to a weaker symmetry, allowing for

$$\mathrm{C}_4 \overset{\mathrm{SSB}}{\to} \mathrm{C}_2/\mathbb{Z}_2$$

We work with the full, anisotropic band of Eq. (2.17). From the following chain of deductions we conclude that C_4 is not broken:



a | Same-sign mixed bands.

b | Opposite-sign mixed bands.

Figure 2.5 | Theoretical example of bands with $u^{(s^*)} > 2t/V$, $u^{(d)} > 0$ (top-left) and $u^{(s^*)} < 0$, $u^{(d)} > 0$ (top right). Two example points are reported, connected by a $\pi/2$ rotation as given in Eq. (**??**).

1. Let for instance

$$u^{(s^*)} \geq 2t/V > 0 \qquad u^{(d)} > 0 \qquad \Longrightarrow \qquad \epsilon^{(s^*)} \geq 0 \qquad \epsilon^{(d)} > 0$$

In Fig. 2.5a an example band is shown for $\epsilon^{(s^*)} = 1$, $\epsilon^{(d)} = 1/2$. Note that if $u^{(s^*)} = 2t/V$, the renormalised band is purely d-wave since $\epsilon^{(s^*)} = 0$.

Recall Eq. (2.26), and separate in two sums over $\mathcal{A}$ and $\mathcal{B}$:

$$u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathcal{A}} (\cos k_x + \cos k_y) \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \right] + \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathcal{B}} (\cos k_x + \cos k_y) \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \right]$$

Consider the sum over $\mathcal{B}$. Looking at the bottom-left panel of Fig. 2.3, such a region contains nodal curves across which bands change sign. Let $\mathcal{R}$ be the clockwise $\pi/2$ rotation: then, any point in $\mathcal{B}$ can be seen as the rotation of a point in $\mathcal{A}$

$$\forall \mathbf{k}' \in \mathcal{B} \quad \exists \quad \mathbf{k} \in \mathcal{A} \,|\, \mathbf{k}' = \mathcal{R}\mathbf{k} \tag{2.28}$$

A graphical rendition of this idea is given in Fig. 2.5a. Then we can rewrite $u^{(s^*)}$ as a single sum over $\mathcal{A}$,

$$u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathcal{A}} (\cos k_x + \cos k_y) \times \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) + f(\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) \right] \tag{2.29}$$

From Eq. (2.23), $\forall \mathbf{k} \in \mathcal{A}$ the form factors $\cos k_x \pm \cos k_y$ are positive by construction. By hypothesis $\epsilon^{(s^*)} \geq 0$ and $\epsilon^{(d)} > 0$. Hence:

$$\forall \mathbf{k} \in \mathcal{A} \quad : \quad \begin{aligned} \tilde{\epsilon}_{\mathbf{k}} &= \overbrace{\epsilon^{(s^*)} (\cos k_x + \cos k_y)}^{\geq 0} + \overbrace{\epsilon^{(d)} (\cos k_x - \cos k_y)}^{\geq 0} \\ \tilde{\epsilon}_{\mathcal{R}\mathbf{k}} &= \underbrace{\epsilon^{(s^*)} (\cos k_x + \cos k_y)}_{\geq 0} - \underbrace{\epsilon^{(d)} (\cos k_x - \cos k_y)}_{\geq 0} \end{aligned}$$

Summing the two equations above we have:

$$\forall \mathbf{k} \in \mathcal{A} \quad : \quad \tilde{\epsilon}_{\mathbf{k}} + \tilde{\epsilon}_{\mathcal{R}\mathbf{k}} \geq 0$$

thus $\tilde{\epsilon}_{\mathbf{k}} \geq -\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}$. The Fermi-Dirac distribution is monotonically descendent. Then:

$$\forall \mathbf{k} \in \mathcal{A} \quad : \quad \begin{array}{r} f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \leq f(-\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) \\ f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \geq f(\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) \end{array}$$

This condition, inserted in Eq. (2.29), leads to an absurd:

$$u^{(s^*)} \geq 2t/V \qquad u^{(d)} > 0 \qquad \Longrightarrow \qquad u^{(s^*)} \leq 0$$

2. An identical argument, exchanging the roles of $\mathcal{A}$ and $\mathcal{B}$, can be carried out to show that

$$u^{(s^*)} \geq 2t/V > 0 \qquad u^{(d)} < 0 \qquad \Longrightarrow \qquad u^{(s^*)} \leq 0$$

3. The two points above imply that it is possible to have $u^{(s^*)} \geq 2t/V$ if and only if $u^{(d)} = 0$. But from Sec. 2.5.1, we conclude that the upper boundary $u^{(s^*)} < 2t/V$ always holds.

4. Let for instance

$$u^{(s^*)} \leq 0 \qquad u^{(d)} > 0 \qquad \Longrightarrow \qquad \epsilon^{(s^*)} < 0 \qquad \epsilon^{(d)} > 0$$

In Fig. 2.5b an example band is shown for $\epsilon^{(s^*)} = -1$, $\epsilon^{(d)} = 1/2$.

Proceed as for the first point. Similarly to Eq. (2.28), Any point in $\mathcal{A}$ can be obtained by rotating some point in $\mathcal{B}$,

$$\forall \mathbf{k}' \in \mathcal{A} \quad \exists \quad \mathbf{k} \in \mathcal{B} \mid \mathbf{k}' = \mathcal{R}\mathbf{k} \tag{2.30}$$

An equation similar to Eq. (2.29) holds, the sum being performed over $\mathcal{B}$

$$\begin{aligned} u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathcal{B}} &(\cos k_x + \cos k_y) \\ &\times [f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) + f(\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu})] \end{aligned} \tag{2.31}$$

From Eq. (2.24), $\forall \mathbf{k} \in \mathcal{B}$ the form factor $\cos k_x + \cos k_y$ is positive by construction; the form factor $\cos k_x - \cos k_y$ is negative by construction. By hypothesis $\epsilon^{(s^*)} < 0$ and $\epsilon^{(d)} > 0$. Hence:

$$\forall \mathbf{k} \in \mathcal{B} \quad : \quad \begin{aligned} \tilde{\epsilon}_{\mathbf{k}} &= \overbrace{\epsilon^{(s^*)} (\cos k_x + \cos k_y)}^{\leq 0} + \overbrace{\epsilon^{(d)} (\cos k_x - \cos k_y)}^{\leq 0} \\ \tilde{\epsilon}_{\mathcal{R}\mathbf{k}} &= \underbrace{\epsilon^{(s^*)} (\cos k_x + \cos k_y)}_{\leq 0} - \underbrace{\epsilon^{(d)} (\cos k_x - \cos k_y)}_{\leq 0} \end{aligned}$$

Summing the two equations above we have:

$$\forall \mathbf{k} \in \mathcal{B} \quad : \quad \tilde{\epsilon}_{\mathbf{k}} + \tilde{\epsilon}_{\mathcal{R}\mathbf{k}} \leq 0$$

thus $\tilde{\epsilon}_{\mathbf{k}} \leq -\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}$. The Fermi-Dirac distribution is monotonically descendent. Then:

$$\forall \mathbf{k} \in \mathcal{B} \quad : \quad \begin{array}{r} f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \geq f(-\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) \\ f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \leq f(\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) \end{array}$$

This condition, inserted in Eq. (2.29), leads to an absurd:

$$u^{(s^*)} \leq 0 \qquad u^{(d)} > 0 \qquad \Longrightarrow \qquad u^{(s^*)} > 0$$

5. can be carried out to show that

$$u^{(s^*)} \leq 0 \qquad u^{(d)} < 0 \qquad \Longrightarrow \qquad u^{(s^*)} > 0$$

6. The two points above imply that it is possible to have $u^{(s^*)} \leq 0$ if and only if $u^{(d)} = 0$. But from Sec. 2.5.1, we conclude that the lower boundary $u^{(s^*)} > 0$ always holds.

From the chain of derivations above, we can conclude self-consistently that also in the nematic structure the constraint of Eq. (2.27),

$$0 < u^{(s^*)} < 2t/V$$

holds. In the s^*-wave channel, partial breaking of rotational symmetry is not enough to modify the main effect identified under C_4 symmetry, "bands shrinking". What is missing is the behaviour of the d-wave channel when said effect takes place. Continuing the enumeration above,

7. Let $0 < u^{(s^*)} < 2t/V$ and $u^{(d)} > 0$. Then $\epsilon^{(d)} > 0$. The self-consistency equation for $u^{(d)}$ reads

$$\begin{aligned} u^{(d)} &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in \mathrm{BZ}} (\cos k_x - \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in \mathrm{NBZ}} (\cos k_x - \cos k_y) \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) \right] \end{aligned}$$

   where we recognized, as is shown in Fig. 2.4b, that BZ $\setminus$ NBZ $= \mathcal{R}$NBZ. It holds:

$$\forall \mathbf{k} \in \mathrm{NBZ} \quad : \quad \begin{aligned} \tilde{\epsilon}_{\mathbf{k}} &= \epsilon^{(s^*)} (\cos k_x + \cos k_y) + \overbrace{\epsilon^{(d)} (\cos k_x - \cos k_y)}^{\geq 0} \\ \tilde{\epsilon}_{\mathcal{R}\mathbf{k}} &= \epsilon^{(s^*)} (\cos k_x + \cos k_y) - \underbrace{\epsilon^{(d)} (\cos k_x - \cos k_y)}_{\geq 0} \end{aligned}$$

   thus $\tilde{\epsilon}_{\mathcal{R}\mathbf{k}} < \tilde{\epsilon}_{\mathbf{k}}$, and inserting this fact in the self-consistency equation above we get

$$0 < u^{(s^*)} < 2t/V \quad \text{and} \quad u^{(d)} > 0 \quad \Longrightarrow \quad u^{(d)} < 0$$

   which is absurd.

8. An identical argument, with inverted inequalities, can be carried out to show that

$$0 < u^{(s^*)} < 2t/V \quad \text{and} \quad u^{(d)} < 0 \quad \Longrightarrow \quad u^{(d)} > 0$$

9. The two points above imply that it is possible to satisfy the constraint $0 < u^{(s^*)} < 2t/V$ if and only if $u^{(d)} = 0$.

All the deductions above indicate that $u^{(d)} = 0$ everywhere. Nematic effects are always absent,

$$0 < u^{(s^*)} < 2t/V \quad \text{and} \quad u^{(d)} = 0 \tag{2.32}$$

This is physically expected but non trivial. The normal phase does not break C_4 and we can ignore $u^{(d)}$ at all. In next section we identify the expected self-consistent value for $u^{(s^*)}$.

2.5.3 Expected dependence on δ at low temperature

We understood that $u^{(d)} = 0$. Moreover, the unique HFP $u^{(s^*)}$ is independent of U, but in general it will depend in some way on the doping level. We work at low-temperature, $\beta \to \infty$. Consider the self-consistency equation of Tab. 2.2 as a function of temperature and chemical potential,

$$u^{(s^*)}(\beta, \tilde{\mu}) = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in \mathrm{BZ}} (\cos k_x + \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$$

in the zero-temperature limit, as long as the band is not flat, it holds

$$\lim_{\beta\to\infty} u^{(s^*)}(\beta, \tilde{\mu}) = \frac{1}{L_x L_y} \sum_{\epsilon_{\mathbf{k}} \leq \tilde{\mu}} (\cos k_x + \cos k_y)$$

Let $\mathrm{FS}(\delta)$ be the Fermi surface for a given doping level δ. We denote with $\mathrm{Int}(\mathrm{FS})$ be its internal part, the Fermi sea:

$$\mathrm{FS} = \{\mathbf{k} \in \mathrm{BZ} \mid \tilde{\epsilon}_{\mathbf{k}} = \tilde{\mu}\}$$
$$\mathrm{Int}(\mathrm{FS}) = \{\mathbf{k} \in \mathrm{BZ} \mid \tilde{\epsilon}_{\mathbf{k}} \leq \tilde{\mu}\}$$

At half filling:

$$\delta = 0 \quad \implies \quad \mathrm{FS} = \partial\mathrm{MBZ}$$

At $\delta > 0$ the FS expands in the region $\mathrm{BZ} \setminus \mathrm{MBZ}$ where the form factor $\cos k_x + \cos k_y$ becomes negative. This gives the self-consistency equation

$$\lim_{\beta\to\infty} u^{(s^*)}[\beta, \tilde{\mu}] \simeq \frac{1}{L_x L_y} \sum_{\epsilon_{\mathbf{k}} \leq 0} |\cos k_x + \cos k_y| - \frac{1}{L_x L_y} \sum_{0 < \epsilon_{\mathbf{k}} \leq \tilde{\mu}} |\cos k_x + \cos k_y|$$

which, being the sum of a positive and a negative term, is maximised whenever $\tilde{\mu} = 0$, at half filling. This holds as long as the band is not flat. Indeed, if the effective hopping is not zero, to rise the doping level means to

$$\delta_1 > \delta_2 \quad \implies \quad \mathrm{Int}(\mathrm{FS}_1) \subset \mathrm{Int}(\mathrm{FS}_2)$$

with obvious meaning of the notation. Using this property we conclude

$$\tilde{t} > 0 \quad \text{and} \quad \delta_1 > \delta_2 \quad \implies \quad \lim_{\beta\to\infty} u^{(s^*)}[\beta, \tilde{\mu}]\Big|_{\delta_1} < \lim_{\beta\to\infty} u^{(s^*)}[\beta, \tilde{\mu}]\Big|_{\delta_2}$$

This is the first expected behaviour of the self-consistent functional $u^{(s^*)}$: if there exists a region of parametric space where the shrinking effect is not strong enough to flatten the band, a rise in the doping level must lower $u^{(s^*)}$. Such a region must exist: at least all the segment $(V = 0, \delta)$ is included, since on said region of parametric space the model is just the common free electronic model.

Following this line of reasoning, we can extract the two low-temperature limits of $u^{(s^*)}$ at half-filling and unitary filling.

1. At half-filling, the self-consistency equation reduces to

$$u^{(s^*)}[\beta, 0]\Big|_{\beta \gg 1} \simeq \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathrm{MBZ}} (\cos k_x + \cos k_y)$$

In thermodynamic limit, performing integration on the MBZ shown in the top-left panel of Fig. 2.3:

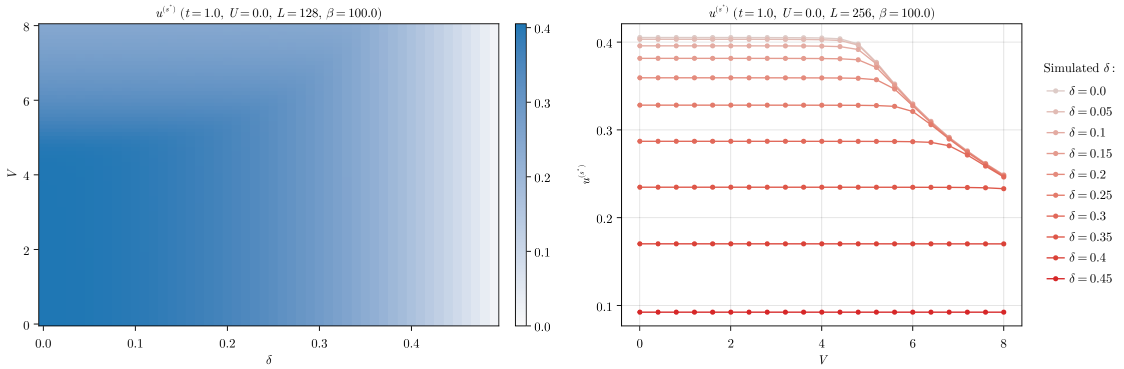
$$\begin{aligned}
\lim_{\beta\to\infty} u^{(s^*)}[\beta, 0] &\simeq \frac{1}{(2\pi)^2} \int_{-\pi}^{\pi} \mathrm{d}k_y \int_{-\pi+|k_y|}^{\pi-|k_y|} \mathrm{d}k_x \cos k_x + \frac{1}{(2\pi)^2} \int_{-\pi}^{\pi} \mathrm{d}k_x \int_{-\pi+|k_x|}^{\pi-|k_x|} \mathrm{d}k_y \cos k_y \\
&= \frac{1}{2\pi^2} \int_{-\pi}^{0} \mathrm{d}k_y \int_{-\pi-k_y}^{\pi+k_y} \mathrm{d}k_x \cos k_x + \frac{1}{2\pi^2} \int_{0}^{\pi} \mathrm{d}k_y \int_{-\pi+k_y}^{\pi-k_y} \mathrm{d}k_x \cos k_x \\
&= \frac{1}{\pi^2} \int_{-\pi}^{0} \mathrm{d}k_y \sin(\pi + k_y) + \frac{1}{\pi^2} \int_{0}^{\pi} \mathrm{d}k_y \sin(\pi - k_y) \\
&= \left(\frac{2}{\pi}\right)^2 \approx 0.405
\end{aligned} \tag{2.33}$$

2. At unitary filling we have

$$\lim_{\beta\to\infty} u^{(s^*)}\left[\beta, 2\tilde{t}^{(s^*)}\right] = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathrm{BZ}} (\cos k_x + \cos k_y) = 0$$

which vanishes due to the structure factor periodicity by $\boldsymbol{\pi}$.

To summarise: around $V = 0$, $u^{(s^*)}$ is expected to descend monotonically from $4/\pi^2$ to 0 at increasing doping.



a | Heatmap of $u^{(s^*)}$ in δV plane. b | V-dependency of $u^{(s^*)}$ varying δ.

Figure 2.6 | Plot of the self-consistent convergence values for the HFP $u^{(s^*)}$ in the normal phase. To extract the data on the right side coarse-graining precision of the left-hand side data has been exchanged with higher lattice dimensions. As is evident, based on doping level $u^{(s^*)}$ exhibits non-trivial regions of V-independence.

2.6 Discussion of numeric results

In this section we present and discuss the numeric results, the physical relevance of the found phenomena and the boundaries of the algorithm. Let us start by sketching the algorithm used to determine self-consistently the mean-field structure of the normal phase in parameter space. For the normal phase, the HFPs "vector" defined in Sec. **??** has just one component:

$$\mathbf{v} = \left[u^{(s^*)}\right]$$

As before, we treat spin as a pure degeneracy. The only HFP is characterised by the first self-consistency equation of Tab. 2.2. The resulting model has two RMPs:

$$\tilde{t}_{ij} \equiv \left(t - \frac{u^{(s^*)}V}{2}\right)\varphi_{ij}^{(s^*)}$$
$$\tilde{\mu} \equiv \mu + n(2zV - U)$$

the second one being only dependent on the model parameters and not on HFPs. Finally, the phase free energy is given by Eq. (2.22). The normal phase is completely independent of U. Because of this we present the results varying the other parameters, keeping everywhere $U = 0$. We ran simulations in the (V, δ) plane, with the setup:

$$U = 0 \qquad \Delta n = 10^{-2} \qquad \Delta\mathbf{v} = \left[u^{(s^*)} : 5 \times 10^{-4}\right]$$

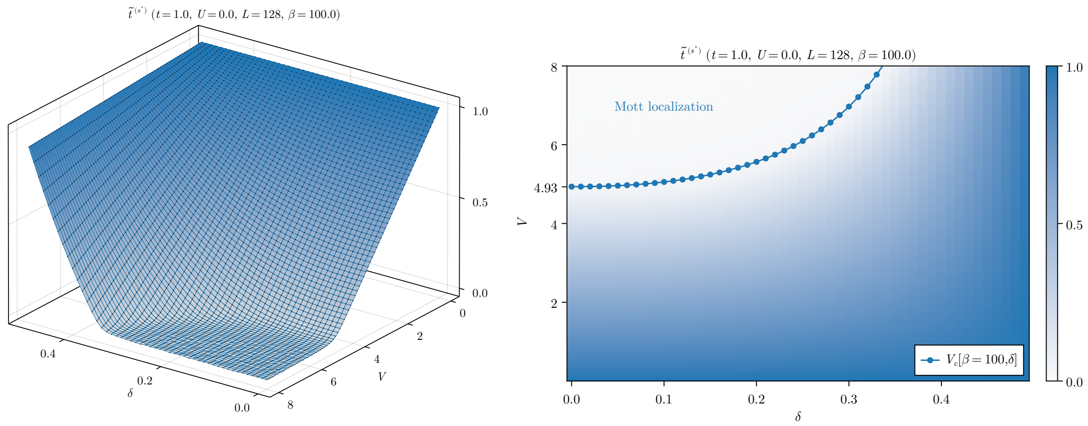
with $\Delta\mathbf{v}$ the tolerance vector defined as in Sec. **??**.

In Fig. 2.6a are reported the results obtained for the HFP $u^{(s^*)}$ on a 128×128 lattice and "low" temperature $\beta = 100$, varying the non-local attraction and the doping level

$$V = 0,\ 0.1,\ 0.2,\ \ldots,\ 4.0 \qquad \delta = 0,\ 0.01,\ 0.02,\ \ldots,\ 0.49$$

The expectations of Sec. 2.5.3 are satisfied: at low V (indicatively $V \lesssim 5$) the trend is monotonically descendent at increasing doping. As V gets bigger, however, $u^{(s^*)}$ starts decreasing in the top-left corner of Fig. 2.6a. This effect is more visible in Fig. 2.6b, where we doubled L and ran simulations with a smaller δ resolution: the convergence value of $u^{(s^*)}$ is approximately independent of V up to some point, where this behaviour is broken. Note that, as was expected from the discussion of Sec. 2.5.3, the half-filled convergence value of $u^{(s^*)}$ in the V-independent region is approximately given by Eq. (2.33).

This effect can only be interpreted in one way, as we discuss in next section: $u^{(s^*)}$ maintains a constant value at fixed doping, as is described in Sec. 2.5.3, as long as $\tilde{t}^{(s^*)} > 0$. When V gets large enough, $u^{(s^*)}$ starts decreasing because of its constraints. The net effect is the nullification of $\tilde{t}^{(s^*)}$, which is, a total localisation effect.



a | Simulated values of $\tilde{t}^{(s^*)}$ b | Critical Mott boundary.

Figure 2.7 | Plot of the self-consistent convergence values for the HFP $u^{(s^*)}$ in the normal phase and the Mott-boundary V_c, computed as is described in text. In the right panel, the tick $V = 4.93$ is specified explicitly as a graphic confirm of the computation of Eq. (2.34). For graphic reasons the points at $V_c > 8$ have been omitted.

2.6.1 Mott localisation effect

The most interesting result we obtained for the normal phase is the possibility of localising electrons by tuning the V/δ ratio. In Fig. 2.6b, at low V the convergence value of $u^{(s^*)}$ is approximately V-independent at sufficiently low temperatures. Let $V_c[\beta,\delta]$ be the critical value such that

$$0 \leq V < V_c[\beta,\delta] \quad : \quad u^{(s^*)} \text{ is } V \text{ independent.}$$

Then within the entire segment $[0, V_c]$ the convergence value is (approximately) the same obtained for $V = 0$. We can define in first instance the critical V as the value such that the hopping correction is maximized,

$$V_c[\beta,\delta] \equiv \frac{2t}{u^{(s^*)}[\beta,\tilde{\mu}(\delta)]\big|_{V=0}}$$

When V becomes larger than this critical value, the resulting RMP $\tilde{t}^{(s^*)}$ drops to 0 and due to the constraint of Eq. (2.27) the convergence value of $u^{(s^*)}$ is forced to decrease. Note that the constraint of Eq. (2.27) does not prevent $u^{(s^*)}$ from saturating to its upper boundary from below. In Fig. 2.7a the converged values of $\tilde{t}^{(s^*)}$, extracted from the same data of Fig. 2.6a, are plotted in parametric space.

While to extract the value of $V_c[\beta,\delta]$ is a non-trivial analytic problem due to the insurgence of elliptic integrals and the smearing effects of temperature, for the high-β half-filled case it is immediate from Eq. (2.33) to see that

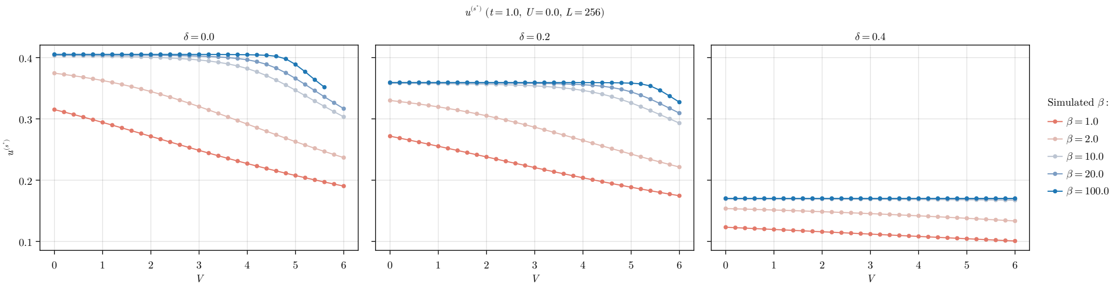
$$V_c[\beta \to \infty, 0] = \frac{2t}{(2/\pi)^2} \approx 4.93t \tag{2.34}$$

At half-filling, the effective bands amplitude reduction is so strong that small values of V (compared to t) are sufficient to nullify kinetics. In Fig. 2.7b we report, superimposed, the convergence values of the RMP $\tilde{t}^{(s^*)}$ and $V_c[\beta = 100,\delta]$ computed with the simple procedure hereby described. Our prediction of the critical value for V approximates reasonably the localisation region.

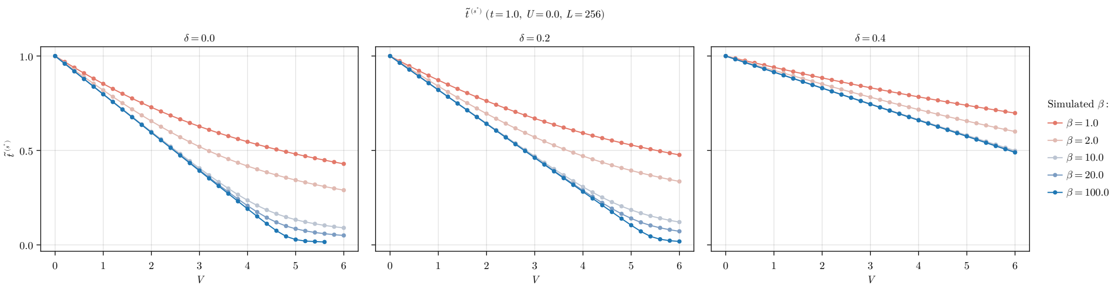
Increasing temperature Mott localisation is suppressed. Also, out precedent argument were valid only at low temperature. In Fig. 2.8a we report results for the HFP $u^{(s^*)}$ computed at decreasingly lower temperature and three example doping levels,

$$\beta = 1.0, 2.0, 10.0, 20.0, 100.0 \qquad \delta = 0.0, 0.2, 0.4$$

At high temperature the V-independence region that was visible in Fig. 2.6b completely disappears, making $u^{(s^*)}$ a monotonically descendent function. In Fig. 2.8a the same dataset is plotted with



a | Temperature dependence of $u^{(s^*)}$ plots at various dopings.



b | Temperature dependence of $\tilde{t}^{(s^*)}$ plots at various dopings.

Figure 2.8 | Comparative plots of the HFP $u^{(s^*)}$ (top panel) and RMP $\tilde{t}^{(s^*)}$ (bottom panel) in null-doped (left plots), "medium-doped" ($\delta = 0.2$, central plots) and "heavily-doped" ($\delta = 0.4$, right plots) contexts. Increasing temperature progressively destroys Mott localization, reducing the bands shrinking effect.

focus on the RMP $\tilde{t}^{(s^*)}$. Temperature increase disrupts localisation effects, at least at small values of V. However, the bands shrinking effect is still relevant at $\beta = 1.0$, the hopping parameter approximately halving at $V \approx 5 \div 6$.

2.6.2 Solution free energy and entropy densities

In order to compare phases, we need an estimation of free energy of each. The plots of Fig. 2.9 report our findings on the normal phase free energy, computed based on Eq. (2.22). For a non-flat band, doping increases energy because the band fills up and electrons populate higher energy levels.

Looking the data of Fig. 2.9a, at fixed δ free energy density remains pretty much constant up to the Mott boundary. This is simply because as long as a band with non-zero amplitude is present, the HFP correction of Eq. (2.22) compensates the free energy increase due to bands shrinking,

$$
\begin{aligned}
-V\left|u^{(s^*)}\right|^2 &= -\frac{V u^{(s^*)*}}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x + \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \left(\epsilon_{\mathbf{k}} - \tilde{\epsilon}_{\mathbf{k}}\right) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})
\end{aligned}
$$

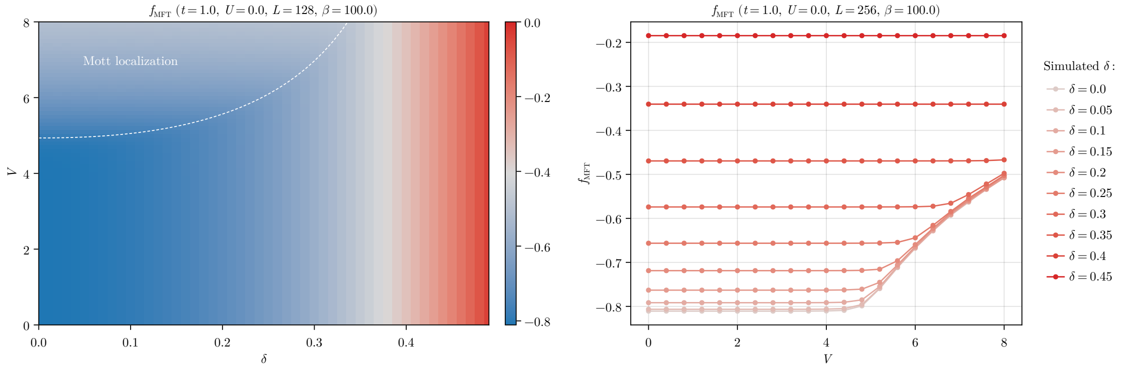
having we used in the second passage

$$
\tilde{\epsilon}_{\mathbf{k}} = \tilde{\epsilon}_{\mathbf{k}} + u^{(s^*)*} V (\cos k_x + \cos k_y)
$$

Recalling the first point of the derivation of Sec. 2.4, to add the HFP correction to free energy is the same as removing the *tilde* sign from single-state energies, thus using the free model ($V = 0$) free energy density expression.

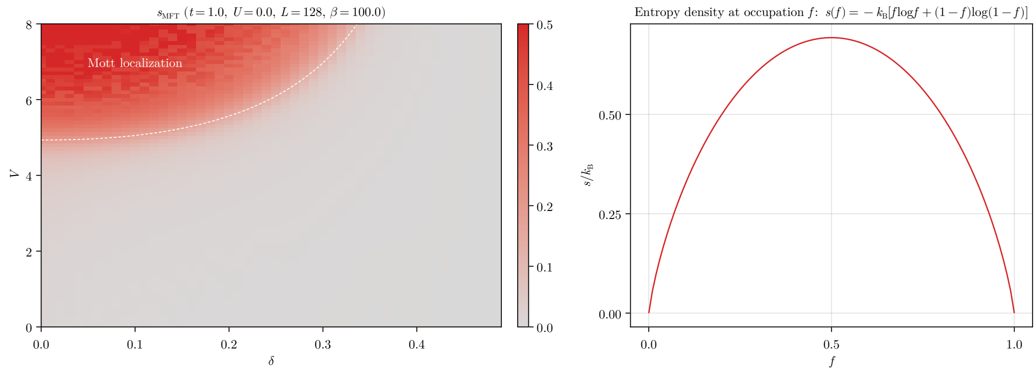
Mott localisation is energetically expensive. This is visible in both panels of Fig. 2.9. The V-dependent energy increase is an entropic effect: indeed, for a flat band:

$$
f(0; \beta, \tilde{\mu}(\delta)) = \frac{1}{2}
$$



a | Heatmap of f_{MFT} in δV plane. b | V-dependency of f_{MFT} varying δ.

Figure 2.9 | Free energy density of the normal phase. On the left-side panel, the dashed white line has the same meaning as the dotted line of Fig. 2.7b, indicating the Mott transition.



a | Heatmap of s_{MFT} in δV plane. b | V-dependency of f_{MFT} varying δ.

Figure 2.10 | Entropy density of the normal phase (left panel) and theoretical contribution to entropy for a state with occupation f. On the left-side panel, the dashed white line has the same meaning as the dotted line of Fig. 2.7b, indicating the Mott transition.

at any temperature. The expression for entropy density of a free Fermi gas [1] is given by

$$s(f) = -k_{\mathrm{B}}\left[f \log f + (1-f)\log(1-f)\right]$$

represented in Fig. 2.10b. The maximum is located at $f = 1/2$, then a flat band realises the maximally entropic condition. The Mott insulating region can be seen as a degeneration of the FS interface to the entire BZ, a process that maximises entropy.

All of this holds, as all the rest, at low temperature: by lowering β, finite-temperature effects become more and more relevant and the statements above need to be corrected.

2.6.3 Convergence problems and mixing fine-tuning

The normal phase is simple enough to describe some of the typical convergence problems arising in HF iterative schemes anticipated in Sec. **??**. Bands flattening is a source of algorithmic instability: numerics can lead to values of $u^{(s^*)}$, during iterations, slightly out of bounds with respect to Eq. (2.27). Even if the numeric effect is small, the consequence can be vast. To heal this effect and extract the actual convergence values, it is necessary to optimise the mixing parameter $g \in [0, 1]$.

We defined g in Sec. **??**, and operatively in Eq. (**??**). This parameter controls deterministically how, from an initialiser (the HFPs vector we plug in the algorithm) and an output (the HFPs vector we get back from the algorithm) the new initialiser for the following step is generated. We show in Fig. 2.11 the evolutions at different mixing parameters of a simulation at

$$V = 6.0 \qquad \delta = 0.1 \qquad \beta = 100.0$$

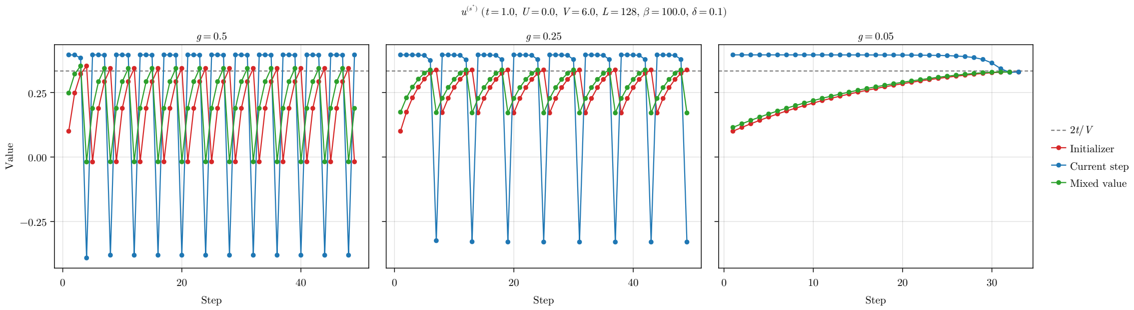


Figure 2.11 | Comparison of three example tracks for the single-valued HFPs vector of the normal phase, simulated at a "Mott-deep" point.

This point is inside the Mott region, as can be checked in Fig. 2.6a. The HFP $u^{(s^*)}$ needs to converge at $2t/V = t/3$ in order to give complete localisation. The initialiser

$$\text{Step 1} \quad : \quad u^{(s^*)} = 0.1$$

was arbitrarily chosen. In each panel of Fig. 2.11, the red track represents the value of $u^{(s^*)}$ we plug in the self-consistency equations, the blue the result of the self-consistent equation, the green one the mixed value of the two based on different gs. The gray dashed line is the theoretical convergence target.

In the first two panels, there is an evident infinite loop occurring whenever the green line crosses the threshold $2t/V$. At those steps, the green point is out of bounds with respect to Eq. (2.27). When this happens, the following red point gives a band with inverted sign, a feature that "breaks" the self-consistency equation and generates a blue point with no physical significance.

To overcome this problem it is necessary to move "slowly". As is shown in the right panel, the choice of a small enough mixing parameter ensures convergence. In the track, each blue dot is above the target and would produce a band with inverted sign. Mixing blue and red dots with a 95% unbalance on the red ones, we ensure that the green track approaches the target from below. Note, however, that lowering g cannot be reiterated too many times. Indeed, for $g \to 0$, any point of our map becomes a fixed point.

This kind of problems can turn out to be pretty common for iHF schemes, and to understand how to knob the algorithmic parameters without sacrificing accurateness and precision, as well as avoiding computationally costly and frustrating infinite loops, was an essential part of this algorithm design.

Chapter 3

Antiferromagnetic instability

ToDo

- Clarify the Mean-Field decomposition of quartic operators OR use the self-consistent fields

In this chapter we apply HF method to the AF phase. In the context of conventional HM, antiferromagnetism is widely studied. We focus on one particular realisation of antiferromagnetism, referred to as "commensurate". It is inherently metallic and only relevant at half-filling.

In this phase, HF methods give two main self-consistent effects. First, the presence of the non-local attraction redefines the bands gap, an effect we denote "gap complexification" (Sec. 3.4) and later show is actually suppressed (Sec. 3.7). Second, bands shrinking (Sec. 3.3) similarly to what found for the normal phase.

As for the normal phase, this chapter is equally divided in the same three main parts. First: discussion of AF phase symmetries, based on Chap. 1, and implementation of HF methods. Second: theoretical constraints on the self-consistent solution. Third: numeric results.

3.1 The AF phase and its symmetries

The AF phase is not uniquely defined. In the context of single-band Hubbard models, antiferromagnetism indicates a spin-density wave (SDW) structure. The two spin populations $\sigma =\uparrow$, $\sigma =\downarrow$ are identical, with a given periodicity and displaced by a finite amount.

Fig. 3.1 presents three (of many) types of antiferromagnets. Left panel shows the one we deal with: SDW structures are 2-sites periodic, making the AF phase "commensurate". The central one presents another simple commensurate AF with higher periodicity. The right panel shows a possible "striped" antiferromagnet, where AF is effectively established along one particular direction; the example we report is 2 sites periodic in y direction, but any other periodicity could work. Striped antiferromagnetism is studied in HM physics [14] as a possible long-range ordering energetically competing with anisotropic superconductivity. Nematic antiferromagnets, establishing long-range magnetic ordering along one particular direction, are interesting due to competition and interplay against superconducting ordering at finite doping [23]. Geometrically non-trivial magnetic ordering are at the core of SDW formation in copper oxides at finite doping [24, 25].

The AF phase is described by simultaneous global spin rotation and translational symmetry SSBs:

$$\mathrm{SU}(2) \stackrel{\mathrm{SSB}}{\to} \mathrm{U}^z(1)$$

$$\mathrm{T}_1^{(x)} \otimes \mathrm{T}_1^{(y)} \stackrel{\mathrm{SSB}}{\to} \mathrm{T}_2^{(x)} \otimes \mathrm{T}_2^{(y)}$$

We describe a SDW whose period is of two sites, as in the left panel of Fig. 3.1. As done for the normal phase, we allow

$$\mathrm{C}_4 \stackrel{\mathrm{SSB}}{\to} \mathrm{C}_2$$

in order to capture the same theoretical nematic effects seen in Sec. 2.3.1. The theoretical reason to do so for AF phase relies in our previous discussion. As said, planar Hubbard lattices exhibits tendency towards nematic magnetic ordering, thus (even if our Ansatz is the simplest and non evidently nematic) nematic effects are not to be excluded *a priori*.

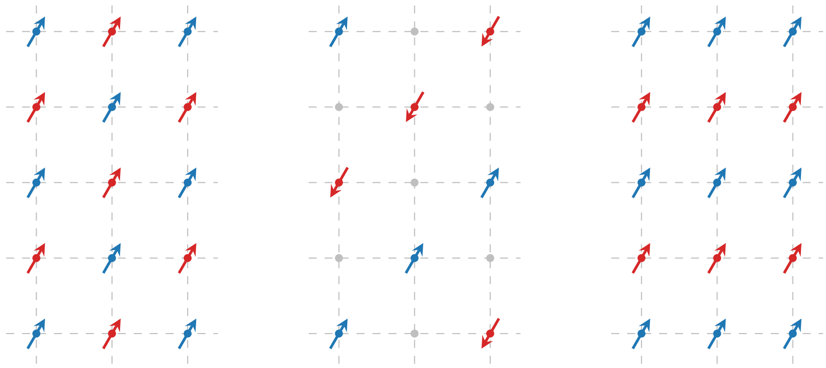


Figure 3.1 Three possible antiferromagnetic structures. Dots and arrows indicate spin abundance on a given site, while empty sites are intended as spin-balanced. Within this text we deal with the lowest-periodicity commensurate phase, the antiferromagnet on left panel.

Operator	**Sector**	**RWCs**	**Net effect**
$\hat{H}_U$		Hartree	μ shift by $-nU$
	o.s.	Hartree	μ shift by nzV
$\hat{H}_V$	s.s.	Hartree	μ shift by nzV
		Fock	Bands renormalization and resymmetrization, gap complexification

Table 3.1 Relevant Wick's contractions and net effect in the AF phase. Note the similarity with Tab. 2.1.

Finally, recall Tabs. 1.2 and 1.3: the relevant Wick's contractions are, for each sector,

$\hat{H}_U$	Hartree contraction
$\hat{H}_V$ (o.s. sector)	Hartree contraction
$\hat{H}_V$ (s.s. sector)	Hartree and Fock contractions

We anticipate the role of all these in Tab. 3.1.

Commensurate antiferromagnetism is specified by the Ansatz:

$$\langle \hat{n}_{\mathbf{r}\sigma} \rangle = n - (-1)^{x+y+\delta_{\sigma=\uparrow}} m \qquad n, m \in [0, 1] \tag{3.1}$$

This Ansatz captures the idea of the staggered distribution of spin populations shown in the left panel of Fig. 3.1. At site $\mathbf{r} = (x, y)$ the two spin populations average to n and differ by $2m$. A detailed justification for this Ansatz in the HM is given in Sec. A.1.2. Note that $\langle \hat{n}_{\mathbf{r}\sigma} \rangle \leq 1$ due to Pauli exclusion principle. This argument is already sufficient to claim that doping disrupts commensurate antiferromagnetism.

The Ansats of Eq. (3.1) breaks translational invariance in each spin sector, but preserves $\mathrm{U}^z(1)$ and $\mathrm{U}^c(1)$ symmetries. Let us prove this. The Ansatz is defined by the following EV being non-zero,

$$\langle \hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow} \rangle = 2m \tag{3.2}$$

SSB happens establishes when $m \neq 0$. Then:

- Recall the definitions of $\hat{\Psi}_{\mathbf{r}}$ of Eq. (1.6)

$$\hat{\Psi}_{\mathbf{r}} = \begin{bmatrix} \hat{c}_{\mathbf{r}\uparrow} \\ \hat{c}_{\mathbf{r}\downarrow} \end{bmatrix}$$

and the local spin operators of Eq. (1.7),

$$\hat{S}^{+}_{\mathbf{r}} = \frac{\hat{c}^{\dagger}_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\downarrow}}{2} \qquad \hat{S}^{-}_{\mathbf{r}} = \frac{\hat{c}^{\dagger}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow}}{2} \qquad \hat{S}^{z}_{\mathbf{r}} = \frac{\hat{c}^{\dagger}_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\uparrow} + \hat{c}^{\dagger}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\downarrow}}{2}$$

Consider Eq. (1.16) and its conjugate,

$$\left[\hat{c}^{\dagger}_{\mathbf{r}\sigma}, \hat{n}_{\mathbf{r}\sigma}\right] = -\hat{c}^{\dagger}_{\mathbf{r}\sigma} \qquad \left[\hat{c}_{\mathbf{r}\sigma}, \hat{n}_{\mathbf{r}\sigma}\right] = \hat{c}_{\mathbf{r}\sigma}$$

The operators $\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}$ and $\hat{S}^{\pm}_{\mathbf{r}}$ do not commutate. Indeed,

$$\begin{aligned}\left[\hat{S}^{+}_{\mathbf{r}}, \hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}\right] &= \frac{1}{2}\left[\hat{c}^{\dagger}_{\mathbf{r}\uparrow}, \hat{n}_{\mathbf{r}\uparrow}\right]\hat{c}_{\mathbf{r}\downarrow} - \frac{1}{2}\hat{c}^{\dagger}_{\mathbf{r}\uparrow}[\hat{c}_{\mathbf{r}\downarrow}, \hat{n}_{\mathbf{r}\downarrow}] \\ &= -\hat{c}^{\dagger}_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\downarrow} \\ \left[\hat{S}^{-}_{\mathbf{r}}, \hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}\right] &= \frac{1}{2}\hat{c}^{\dagger}_{\mathbf{r}\downarrow}[\hat{c}_{\mathbf{r}\uparrow}, \hat{n}_{\mathbf{r}\uparrow}] - \frac{1}{2}\left[\hat{c}^{\dagger}_{\mathbf{r}\downarrow}, \hat{n}_{\mathbf{r}\downarrow}\right]\hat{c}_{\mathbf{r}\uparrow} \\ &= \hat{c}^{\dagger}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\end{aligned}$$

Instead, $\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}$ and $\hat{S}^{z}_{\mathbf{r}}$ commutate, because all on-site density operators always do. Then, in order to have $m \neq 0$ in Eq. (3.2), SU(2) needs to be broken while $\mathrm{U}^z(1)$ needs not.

- From Eq. (1.5) we know that on-site density operators are gauge-invariant separately in both spin sectors. Then no selection rule applies to $\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}$: its EV can be non-zero without breaking $\mathrm{U}^z(1)$.

The Ansatz of Eq. (3.1) is used in literature to describe antiferromagnetism in Hubbard lattice. Before moving to the EHM, we sketch in next section the theoretical description of commensurate antiferromagnetism in the HM.

3.1.1 Antiferromagnetism in the HM

Commensurate antiferromagnetism in conventional HM is widely studied [2]. FROM HERE

The detailed derivation is given in Sec. A.1, and here we just sketch the most important steps. Let us define:

$$\hat{n}_{\mathbf{k}\sigma} \equiv \hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} \tag{3.3}$$

$$\hat{\psi}_{\mathbf{k}\sigma} \equiv \hat{c}^{\dagger}_{\mathbf{k}+\boldsymbol{\pi}\sigma}\hat{c}_{\mathbf{k}\sigma} \tag{3.4}$$

Recall also the γ-wave form factors of Tab. 1.5. Then:

$$\begin{aligned}\hat{\psi}^{\dagger}_{\mathbf{k}+\boldsymbol{\pi}\sigma} &= \hat{\psi}_{\mathbf{k}\sigma} \\ \varphi^{(\gamma)}_{\mathbf{k}+\boldsymbol{\pi}} &= -\varphi^{(\gamma)}_{\mathbf{k}}\end{aligned}$$

which imply the property

$$\varphi^{(\gamma)}_{\mathbf{k}+\boldsymbol{\pi}}\psi^{\dagger}_{\mathbf{k}+\boldsymbol{\pi}\sigma} = -\varphi^{(\gamma)}_{\mathbf{k}}\psi_{\mathbf{k}\sigma} \tag{3.5}$$

As reported in Tab. **??**, the only RWC comes in Hartree channel, since the local two-body attraction couples operators at opposite spin index. The hamiltonian $\hat{H}$ is reduced thereby to the form of Eq. (A.5)

$$\begin{aligned}\hat{H}_t + \hat{H}_U \overset{\mathrm{MFT}}{\simeq} -t\sum_{\langle\mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}\left[\hat{c}^{\dagger}_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}'\sigma} + \mathrm{h.c.}\right] + nU\sum_{\mathbf{r}}\left[\hat{n}_{\mathbf{r}\uparrow} + \hat{n}_{\mathbf{r}\downarrow}\right] \\ - mU\sum_{\mathbf{r}}(-1)^{x+y}\left[\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}\right] - E^{(\mathrm{AF})}_{\mathrm{H}/U}\end{aligned}$$

with $E^{(\mathrm{AF})}_{\mathrm{H}/U}$ the contraction energy resulting from double counting terms, given explicitly in Eq. (A.6),

$$E^{(\mathrm{AF})}_{\mathrm{H}/U} = L_x L_y \times U\left(n^2 - m^2\right)$$

EV	Meaning	Equation
$\langle \hat{s}^x_{\mathbf{k}\sigma} \rangle$	$\left\langle \hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}^\dagger_{\mathbf{k}\sigma} \right\rangle$	(A.20)
$\langle \hat{s}^y_{\mathbf{k}\sigma} \rangle$	$i\left\langle \hat{\psi}_{\mathbf{k}\sigma} - \hat{\psi}^\dagger_{\mathbf{k}\sigma} \right\rangle$	(A.21)
$\langle \hat{s}^z_{\mathbf{k}\sigma} \rangle$	$\langle \hat{n}_{\mathbf{k}\sigma} + \hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \rangle$	(A.22)

Table 3.2 List of the various expectation values (EVs) of each pseudo-spin component, their meaning in terms of fermionic operators and the reference equations from the pure HM.

In reciprocal space, the hamiltonian decomposes as in Eq. (A.11),

$$\hat{H}_t + \hat{H}_U \overset{\text{MFT}}{\simeq} \sum_{\mathbf{k}\in\text{MBZ}} \sum_{\sigma} \hat{\Psi}^\dagger_{\mathbf{k}\sigma} h_{\mathbf{k}\sigma} \hat{\Psi}_{\mathbf{k}\sigma} + nU \times \hat{N} - E^{(\text{AF})}_{\text{H}/U} \quad \text{being} \quad h_{\mathbf{k}\sigma} \equiv \begin{bmatrix} \epsilon_{\mathbf{k}\sigma} & -\Delta^*_{\mathbf{k}\sigma} \\ -\Delta_{\mathbf{k}\sigma} & -\epsilon_{\mathbf{k}\sigma} \end{bmatrix} \tag{3.6}$$

and $\Delta_{\mathbf{k}\sigma} = (-1)^{\delta_{\sigma=\downarrow}} mU$. Nambu spinorial formulation is used,

$$\hat{\Psi}_{\mathbf{k}\sigma} \equiv \begin{bmatrix} \hat{c}_{\mathbf{k}\sigma} \\ \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \end{bmatrix}$$

and free electrons energy is simply the tight binding energy of Eq. (1.29) written with an explicit spin index,

$$\epsilon_{\mathbf{k}\sigma} = -2t\left(\cos k_x + \cos k_y\right)$$

which is spin-invariant. The MFT description of the model reduces to a gas of free "γ-fermions", described by the Nambu spinor of Eq. (A.24),

$$\hat{\Gamma}_{\mathbf{k}\sigma} = W_{\mathbf{k}\sigma} \hat{\Psi}_{\mathbf{k}\sigma} = \begin{bmatrix} \hat{\gamma}^{(+)}_{\mathbf{k}\sigma} \\ \hat{\gamma}^{(-)}_{\mathbf{k}\sigma} \end{bmatrix}$$

where

$$W_{\mathbf{k}\sigma} = \begin{bmatrix} \cos\theta_{\mathbf{k}\sigma} & -\sin\theta_{\mathbf{k}\sigma} \\ \sin\theta_{\mathbf{k}\sigma} & \cos\theta_{\mathbf{k}\sigma} \end{bmatrix} \quad \text{and} \quad \sin 2\theta_{\mathbf{k}\sigma} \equiv \frac{\Delta_{\mathbf{k}\sigma}}{E_{\mathbf{k}\sigma}}$$

These fermions populate the two bands $\pm E_{\mathbf{k}\sigma} = \sqrt{\epsilon^2_{\mathbf{k}\sigma} + |\Delta_{\mathbf{k}\sigma}|^2}$. The entire system is mapped onto an ensemble of pseudo-spins, each subject to a pseudo-field, as in Fig. 3.2. To diagonalize the system essentially means to anti-align each pseudo-spin with the z axis. Each pseudo-spin component expectation value (EV) is described by a simple equation and has a precise meaning, listed in Tab. 3.2.

3.1.2 Expansion to EHM

Consider now the non-local interaction $\hat{H}_V$. Being RWCs the same as for the normal phase, the present discussion is essentially the same as of Chap. 2, with the only relevant difference being the Ansatz we use in Eq. (2.1), substituted by the commensurate oscillatory Ansatz of Eq. (3.1). By itself, even if it may seem the contrary, said Ansatz does not imply C_4 symmetry preservation. It is possible to reduce point group symmetry having still a rotation-symmetric magnetization, as well as it was possible to do in the normal phase with keeping a uniform density. As for the normal phase, by design the AF MFT solution to the model is described via a redefinition of various physical quantities and mathematical objects,

$$\epsilon_{\mathbf{k}\sigma} \to \tilde{\epsilon}_{\mathbf{k}\sigma} \qquad E_{\mathbf{k}\sigma} \to \tilde{E}_{\mathbf{k}\sigma} \qquad \Delta_{\mathbf{k}\sigma} \to \tilde{\Delta}_{\mathbf{k}\sigma} \qquad \mu \to \tilde{\mu}$$

The band energies renormalization is simply

$$\tilde{E}_{\mathbf{k}\sigma} \equiv \sqrt{\tilde{\epsilon}^2_{\mathbf{k}\sigma} + |\tilde{\Delta}_{\mathbf{k}\sigma}|^2}$$

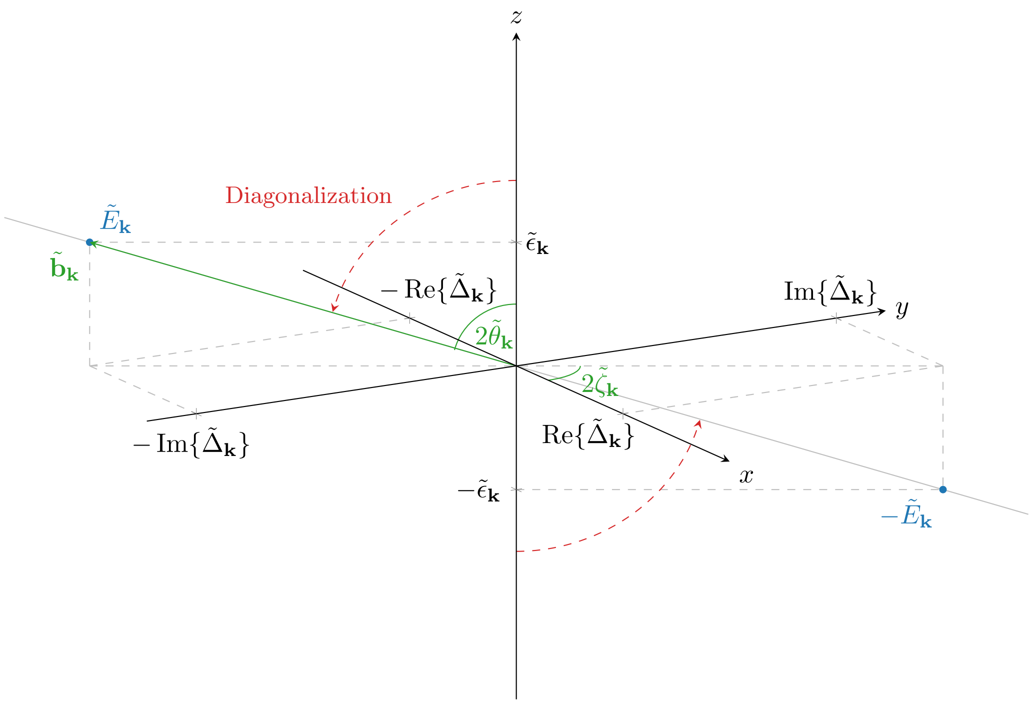


Figure 3.2 Sketch of the diagonalization of the pseudo-spin problem. The dashed line represents the diagonalization given by the z-axis alignment with the field direction. In the top-right side of the picture are listed the expectation values of the original Nambu spinor. For simplicity everywhere the index $\sigma = \uparrow$ has been omitted.

and the following relations hold (*tilde* sign indicates the renormalized quantity):

$$\langle \hat{\Psi}^{\dagger}_{\mathbf{k}\sigma} \tau^x \hat{\Psi}_{\mathbf{k}\sigma} \rangle = -\sin 2\tilde{\theta}_{\mathbf{k}\sigma} \cos 2\tilde{\zeta}_{\mathbf{k}\sigma} \langle \hat{\Gamma}^{\dagger}_{\mathbf{k}\sigma} \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle \tag{3.7}$$

$$\langle \hat{\Psi}^{\dagger}_{\mathbf{k}\sigma} \tau^y \hat{\Psi}_{\mathbf{k}\sigma} \rangle = -\sin 2\tilde{\theta}_{\mathbf{k}\sigma} \sin 2\tilde{\zeta}_{\mathbf{k}\sigma} \langle \hat{\Gamma}^{\dagger}_{\mathbf{k}\sigma} \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle \tag{3.8}$$

$$\langle \hat{\Psi}^{\dagger}_{\mathbf{k}\sigma} \tau^z \hat{\Psi}_{\mathbf{k}\sigma} \rangle = \cos 2\tilde{\theta}_{\mathbf{k}\sigma} \langle \hat{\Gamma}^{\dagger}_{\mathbf{k}\sigma} \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle \tag{3.9}$$

equivalent to the ones for the pure HM, Eqns. (A.20), (A.21) and (A.22). Moving to the tilted frame, the z component of the pseudo-spin is just a thermal superposition of "up" and "down" states

$$\langle \hat{\Gamma}^{\dagger}_{\mathbf{k}\sigma} \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle = f\left(\tilde{E}_{\mathbf{k}\sigma}; \beta, \tilde{\mu}\right) - f\left(-\tilde{E}_{\mathbf{k}\sigma}; \beta, \tilde{\mu}\right) \tag{3.10}$$

Let us define the (renormalized) thermal factors as in Eq. (A.19),

$$\tilde{\mathrm{Th}}^{(\pm)}_{\mathbf{k}\sigma}[\beta, \tilde{\mu}] \equiv f\left(-\tilde{E}_{\mathbf{k}\sigma}; \beta, \tilde{\mu}\right) \pm f\left(\tilde{E}_{\mathbf{k}\sigma}; \beta, \tilde{\mu}\right) \tag{3.11}$$

Angles satisfy:

$$\sin 2\tilde{\xi}_{\mathbf{k}\sigma} = \frac{\mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}}{|\tilde{\Delta}_{\mathbf{k}\sigma}|} \quad \cos 2\tilde{\xi}_{\mathbf{k}\sigma} = \frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}}{|\tilde{\Delta}_{\mathbf{k}\sigma}|} \quad \sin 2\tilde{\theta}_{\mathbf{k}\sigma} = \frac{|\tilde{\Delta}_{\mathbf{k}\sigma}|}{\tilde{E}_{\mathbf{k}\sigma}} \quad \cos 2\tilde{\theta}_{\mathbf{k}\sigma} = \frac{\tilde{\epsilon}_{\mathbf{k}\sigma}}{\tilde{E}_{\mathbf{k}\sigma}} \tag{3.12}$$

Physical behaviour is the same as for the pure HM. In terms of MFT discussion, the AF phase is similar enough to the normal phase of Chap. 2: RWCs are the same of Sec. 2.1, listed in Tabs. **??** and **??**: all three Hartree contractions, responsible in normal phase for chemical potential shift, and the s.s. Fock contraction, that originated the hopping renormalization. Let us start with Hartree terms: due to the "staggered" nature of commensurate AF phase, their role is here more complex. Then we will move to the Fock term, discussing MFT approximations effects on bands. Next sections are basically an extension of the calculations of Chap. 2.

3.2 Non-local Hartree effects: chemical potential renormalization

As in Eq. (2.3), the same-spin and opposite-spin non-local Hartree terms are

$$\hat{H}_V \simeq \overbrace{-V\sum_{\langle ij\rangle}\sum_{\sigma}\left[\langle\hat{n}_{i\sigma}\rangle\hat{n}_{j\sigma}+\hat{n}_{i\sigma}\langle\hat{n}_{j\sigma}\rangle-\langle\hat{n}_{i\sigma}\rangle\langle\hat{n}_{j\sigma}\rangle\right]}^{\text{s.s.}}$$
$$\underbrace{-V\sum_{\langle ij\rangle}\sum_{\sigma}\left[\langle\hat{n}_{i\sigma}\rangle\hat{n}_{j\bar{\sigma}}+\hat{n}_{i\sigma}\langle\hat{n}_{j\bar{\sigma}}\rangle-\langle\hat{n}_{i\sigma}\rangle\langle\hat{n}_{j\bar{\sigma}}\rangle\right]}_{\text{o.s.}}+(\text{all the rest})$$

Let $i\to\mathbf{r}=(x,y)$ and $j\to\mathbf{r}'=(x',y')$. We use Eq. (3.1), describing oscillatory density for each spin sector. This Ansatz is composed of two parts: the offset n, and the oscillatory part m. The algebraic derivation relative to n part is, of course, the same as for the normal phase. We now treat separately the "operatorial" part of the braced terms of Eq. (2.3),

$$-V\sum_{\langle\mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}\left[\langle\hat{n}_{\mathbf{r}\sigma}\rangle\hat{n}_{\mathbf{r}'\sigma}+\hat{n}_{\mathbf{r}\sigma}\langle\hat{n}_{\mathbf{r}'\sigma}\rangle\right]-V\sum_{\langle\mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}\left[\langle\hat{n}_{\mathbf{r}\sigma}\rangle\hat{n}_{\mathbf{r}'\bar{\sigma}}+\hat{n}_{\mathbf{r}\sigma}\langle\hat{n}_{\mathbf{r}'\bar{\sigma}}\rangle\right] \tag{3.13}$$

as well as the its scalar counterpart

$$V\sum_{\langle\mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}\langle\hat{n}_{\mathbf{r}\sigma}\rangle\langle\hat{n}_{\mathbf{r}'\sigma}\rangle+V\sum_{\langle\mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}\langle\hat{n}_{\mathbf{r}\sigma}\rangle\langle\hat{n}_{\mathbf{r}'\bar{\sigma}}\rangle \tag{3.14}$$

accounting for energy shift originated as "leftovers" of Wick's contractions.

3.2.1 Hartree channel in the AF phase: analytic derivation

Let us start from the operator of Expr. (3.13). By inserting the Ansatz of Eq. (A.4), we get

$$\overbrace{-nV\sum_{\langle\mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}\left[\hat{n}_{\mathbf{r}'\sigma}+\hat{n}_{\mathbf{r}\sigma}\right]+mV\sum_{\langle\mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}(-1)^{\delta_{\sigma=\uparrow}}\left[(-1)^{x+y}\hat{n}_{\mathbf{r}'\sigma}+(-1)^{x'+y'}\hat{n}_{\mathbf{r}\sigma}\right]}^{\text{s.s.}}$$
$$\underbrace{-nV\sum_{\langle\mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}\left[\hat{n}_{\mathbf{r}'\bar{\sigma}}+\hat{n}_{\mathbf{r}\sigma}\right]+mV\sum_{\langle\mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}\left[(-1)^{x+y+\delta_{\sigma=\uparrow}}\hat{n}_{\mathbf{r}'\bar{\sigma}}+(-1)^{x'+y'+\delta_{\bar{\sigma}=\uparrow}}\hat{n}_{\mathbf{r}\sigma}\right]}_{\text{o.s.}}$$

For a square lattice, if $\mathbf{r}=(x,y)$ and $\mathbf{r}'=(x',y')$ are NNs, evidently

$$(-1)^{x'+y'}=(-1)^{x+y+1}$$
$$(-1)^{\delta_{\bar{\sigma}=\uparrow}}=(-1)^{\delta_{\sigma=\uparrow}+1}$$

We obtain

$$\overbrace{-nV\sum_{\langle\mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}\left[\hat{n}_{\mathbf{r}\sigma}+\hat{n}_{\mathbf{r}'\sigma}\right]}^{\text{s.s. (density)}}\overbrace{-mV\sum_{\langle\mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}(-1)^{x+y+\delta_{\sigma=\uparrow}}\left[\hat{n}_{\mathbf{r}\sigma}-\hat{n}_{\mathbf{r}'\sigma}\right]}^{\text{s.s. (magnetization)}}$$
$$\underbrace{-nV\sum_{\langle\mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}\left[\hat{n}_{\mathbf{r}\sigma}+\hat{n}_{\mathbf{r}'\bar{\sigma}}\right]}_{\text{o.s. (density)}}\underbrace{+mV\sum_{\langle\mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}(-1)^{x+y+\delta_{\sigma=\uparrow}}\left[\hat{n}_{\mathbf{r}\sigma}+\hat{n}_{\mathbf{r}'\bar{\sigma}}\right]}_{\text{o.s. (magnetization)}} \tag{3.15}$$

In the above expressions we separated in "density" contributions and "magnetization" ones. Let us deal with them separately.

Density terms. Consider s.s. and o.s. (density) terms of Expr. (3.15),

$$-nV\sum_{\langle\mathbf{rr'}\rangle}\sum_{\sigma}\left[\hat{n}_{\mathbf{r}\sigma}+\hat{n}_{\mathbf{r'}\sigma}\right]-nV\sum_{\langle\mathbf{rr'}\rangle}\sum_{\sigma}\left[\hat{n}_{\mathbf{r}\sigma}+\hat{n}_{\mathbf{r'}\overline{\sigma}}\right] \tag{3.16}$$

The discussion is identical to Sec. 2.2.1 for the normal phase, with renormalization of μ being the same of Eq. (2.5).

$$\tilde{\mu}\equiv\mu+n(2zV-U) \tag{3.17}$$

Within this equation we are also already implicitly considering the μ shift effect due to $\hat{H}_U$., discussed both in Sec. 2.2 (and Sec. A.1.1, specifically for antiferromagnetism in conventional HM).

Magnetization terms. When it comes to magnetization, s.s. and o.s. terms of Expr. (3.15) perfectly cancel out:

$$mV\sum_{\langle\mathbf{rr'}\rangle}\sum_{\sigma}(-1)^{x+y+\delta_{\sigma=\uparrow}}\left[(\hat{n}_{\mathbf{r}\sigma}-\hat{n}_{\mathbf{r'}\sigma})-(\hat{n}_{\mathbf{r}\sigma}+\hat{n}_{\mathbf{r'}\overline{\sigma}})\right]=0$$

thus Hartree RWCs only affects μ. This is somewhat to be expected: in the HM context, gap function emerges in its purely s-wave structure from this precise computation on Hartree terms. However, $\hat{H}_V$ non-local interaction cannot provide a pure s-wave potential component. Effects on the gap function are to be observed somewhere else.

3.2.2 Hartree contraction energy

We now collect double counting terms originated by Wick's contraction in Hartree sector. Consider the scalar Expr. (3.14): it's the sum of two parts, the first coming from s.s. sector, the second from o.s. sector. In computing the contraction energy from Hartree channel, being the operatorial result identical to the one obtained for the normal phase, we expect the same to happen for double counting terms.

s.s. sector double counting energy. For s.s. terms we get

$$\begin{aligned}V\sum_{\langle\mathbf{rr'}\rangle}\sum_{\sigma}\langle\hat{n}_{\mathbf{r}\sigma}\rangle\langle\hat{n}_{\mathbf{r'}\sigma}\rangle=&\overbrace{V\sum_{\langle\mathbf{rr'}\rangle}\left[(n+(-1)^{x+y}m)(n+(-1)^{x'+y'}m)\right]}^{\sigma=\uparrow}\\&+\underbrace{V\sum_{\langle\mathbf{rr'}\rangle}\left[(n-(-1)^{x+y}m)(n-(-1)^{x'+y'}m)\right]}_{\sigma=\downarrow}\end{aligned}$$

Evidently the mixed terms (those of order $\mathcal{O}(n)\times\mathcal{O}(m)$) cancel out, and since sign factors necessarily alternate on neighboring sites, the remainder is just

$$V\sum_{\langle\mathbf{rr'}\rangle}\sum_{\sigma}\langle\hat{n}_{i\sigma}\rangle\langle\hat{n}_{j\sigma}\rangle=V\sum_{\langle\mathbf{rr'}\rangle}2(n^2-m^2)$$

o.s. sector double counting energy. Proceeding identically on the o.s. sector, we get

$$\begin{aligned}V\sum_{\langle\mathbf{rr'}\rangle}\sum_{\sigma}\langle\hat{n}_{\mathbf{r}\sigma}\rangle\langle\hat{n}_{\mathbf{r'}\overline{\sigma}}\rangle=&\overbrace{V\sum_{\langle\mathbf{rr'}\rangle}\left[(n+(-1)^{x+y}m)(n-(-1)^{x'+y'}m)\right]}^{\sigma=\uparrow}\\&+\underbrace{V\sum_{\langle\mathbf{rr'}\rangle}\left[(n-(-1)^{x+y}m)(n+(-1)^{x'+y'}m)\right]}_{\sigma=\downarrow}\end{aligned}$$

which gives immediately

$$V\sum_{\langle ij\rangle}\sum_{\sigma}\langle\hat{n}_{i\sigma}\rangle\langle\hat{n}_{j\bar{\sigma}}\rangle = V\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}2(n^2+m^2)$$

Hence, the total "contraction energy", given by sum of both contributions, is identical to the normal one of Eq. (2.7) indeed:

$$E^{(\mathrm{AF})}_{\mathrm{H}/V} = -2V\sum_{\langle ij\rangle}\sum_{\sigma}n^2 = 4n^2V\times\frac{z}{2}L_xL_y \tag{3.18}$$

the minus sign coming from the fact that we are subtracting this energy shift.

3.3 Fock effects: gap and bands renormalization

We explored thoroughly the Hartree contractions effects on the MFT hamiltonian. Let us now move to the effects originated by Fock contractions. We can use the very beginning of Sec. 2.3 derivation, up to Eq. (2.9),

$$\begin{aligned}&V\sum_{\langle ij\rangle}\sum_{\sigma}\left[\langle\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\rangle\hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}+\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\langle\hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}\rangle-\langle\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\rangle\langle\hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}\rangle\right]\\ &\quad\stackrel{\mathcal{F}}{=}\frac{2V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\sum_{\sigma}\left[\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)\right]\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma}\rangle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}-E^{(\mathrm{N})}_{\mathrm{F}/V}\end{aligned} \tag{3.19}$$

where $E^{(\mathrm{AF})}_{\mathrm{F}/V}$ is defined as in Eq. (2.10), with the expectation value being computed on the AF (thermal) ground-state. As before, we discuss the operatorial part and the scalar part separately.

3.3.1 Fock channel in the AF phase: analytic derivation

The commensurate AF phase is essentially described by two EVs being in principle non-zero,

$$\left|\langle\hat{n}_{\mathbf{k}\sigma}\rangle\right|\geq 0 \qquad \left|\langle\hat{\psi}_{\mathbf{k}\sigma}\rangle\right|\geq 0$$

Because of this, for conventional HM, AF ground state is a degenerate Fermi gas of $\gamma^{(\pm)}$ quasiparticles, and for null temperature acquires the simple form of Eq. (A.30). Assuming the new ground state to be similar and going back to Eq. (3.19), this implies only two contributions are non-zero:

$$\mathbf{k}=-\mathbf{k}' \quad \text{or} \quad \mathbf{k}+\boldsymbol{\pi}=-\mathbf{k}'$$

The first case gives us normal fermionic EVs, $\langle\hat{n}_{\mathbf{k}\sigma}\rangle$; the second, staggered AF order parameter: $\langle\hat{\psi}_{\mathbf{k}\sigma}\rangle$. Then Eq. (3.19) is reduced to:

$$\begin{aligned}(\cdots)&\stackrel{\mathcal{F}}{=}\frac{2V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k}}\sum_{\sigma}\left[\cos\left(2k_x\right)+\cos\left(2k_y\right)\right]\\ &\qquad\left[\underbrace{\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}\sigma}\rangle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}\sigma}}_{\text{Diagonal terms}}-\underbrace{\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma}\rangle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma}}_{\text{Off-diagonal terms}}\right]\end{aligned} \tag{3.20}$$

Now, the above equation presents *diagonal* and *off-diagonal* terms.

Diagonal terms. Diagonal terms of Eq. (3.20) are simple density interactions with mean density field. Consider Fig. 2.1a: density at vector $\mathbf{q}\equiv\mathbf{K}+\mathbf{k}$ interacts with mean density at $\mathbf{q}'\equiv\mathbf{K}-\mathbf{k}$. These variables are depicted in Fig. 2.1b. Apply in the diagonal part of Eq. (3.20) this variables

Symmetry	Self-consistency equation	Graph
s^*-wave	$u^{(s^*)} = -\frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}(\cos k_x + \cos k_y)\frac{\tilde{\epsilon}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}}\tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}}[\beta,\tilde{\mu}]$	Fig. 1.4b
$d_{x^2-y^2}$-wave	$u^{(d)} = -\frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}(\cos k_x - \cos k_y)\frac{\tilde{\epsilon}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}}\tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}}[\beta,\tilde{\mu}]$	Fig. 1.4e

Table 3.3 | Symmetry and spin resolved self-consistency equations for the harmonics of $u_{\mathbf{k}}$.

change (which is the same of Eq. (2.13), represented in Fig. 2.1)

$$
\begin{aligned}
\frac{2V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k}}\sum_{\sigma}&[\cos(2k_x)+\cos(2k_y)]\langle\hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}\sigma}\rangle\hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}\sigma} \\
&= \frac{2V}{L_xL_y}\sum_{\mathbf{q},\mathbf{q}'}\sum_{\sigma}[\cos(\delta q_x)+\cos(\delta q_y)]\langle\hat{c}^{\dagger}_{\mathbf{q}\sigma}\hat{c}_{\mathbf{q}\sigma}\rangle\hat{c}^{\dagger}_{\mathbf{q}'\sigma}\hat{c}_{\mathbf{q}'\sigma} \\
&= \sum_{\gamma,\sigma}\left[\frac{1}{L_xL_y}\sum_{\mathbf{q}}\varphi^{(\gamma)}_{\mathbf{q}}\langle\hat{n}_{\mathbf{q}\sigma}\rangle\right]\times V\sum_{\mathbf{q}'}\varphi^{(\gamma)*}_{\mathbf{q}'}\hat{n}_{\mathbf{q}'\sigma}
\end{aligned}
\tag{3.21}
$$

where in the last passage we substituted the decomposition of Eq. (1.35). Define as in Eq. (2.15) the EV

$$u_{\mathbf{q}\sigma} \equiv \langle\hat{n}_{\mathbf{q}\sigma}\rangle \tag{3.22}$$

The expression in square brackets of Eq. (3.21) is just its γ component, and is expanded in

$$
\begin{aligned}
u^{(\gamma)}_{\sigma} &= \frac{1}{L_xL_y}\sum_{\mathbf{q}\in\mathrm{BZ}}\varphi^{(\gamma)}_{\mathbf{q}}\langle\hat{n}_{\mathbf{q}\sigma}\rangle \\
&= \frac{1}{L_xL_y}\sum_{\mathbf{q}\in\mathrm{MBZ}}\left[\varphi^{(\gamma)}_{\mathbf{q}}\langle\hat{n}_{\mathbf{q}\sigma}\rangle + \varphi^{(\gamma)}_{\mathbf{q}+\boldsymbol{\pi}}\langle\hat{n}_{\mathbf{q}+\boldsymbol{\pi}\sigma}\rangle\right] \\
&= \frac{1}{L_xL_y}\sum_{\mathbf{q}\in\mathrm{MBZ}}\varphi^{(\gamma)}_{\mathbf{q}}\left[\langle\hat{n}_{\mathbf{q}\sigma}\rangle - \langle\hat{n}_{\mathbf{q}+\boldsymbol{\pi}\sigma}\rangle\right] \\
&= \frac{1}{L_xL_y}\sum_{\mathbf{q}\in\mathrm{MBZ}}\varphi^{(\gamma)}_{\mathbf{q}}\langle\hat{\Psi}^{\dagger}_{\mathbf{q}\sigma}\tau^z\hat{\Psi}_{\mathbf{q}\sigma}\rangle \\
&= -\frac{1}{L_xL_y}\sum_{\mathbf{q}\in\mathrm{MBZ}}\varphi^{(\gamma)}_{\mathbf{q}}\frac{\tilde{\epsilon}_{\mathbf{q}\sigma}}{\tilde{E}_{\mathbf{q}\sigma}}\tilde{\mathrm{Th}}^{(-)}_{\mathbf{q}\sigma}[\beta,\tilde{\mu}]
\end{aligned}
\tag{3.23}
$$

where, in the last passage, Eqns. (3.9), (3.10) and (3.11) have been used. Now, the AF phase we realize does not break inversion symmetry (and as a consequence, neither C_2): this implies that, whatever the shape of the renormalized bands, it must be

$$\tilde{\epsilon}_{\mathbf{q}\sigma} = \tilde{\epsilon}_{-\mathbf{q}\sigma} \quad \Longrightarrow \quad u^{(p_\ell)}_{\sigma} = 0 \quad \text{with} \quad \ell\in\{x,y\}$$

Renormalized bands need to be symmetric functions of the moment. Hence for $\gamma\in\{p_x,p_y\}$ the above sum vanishes. The remaining s^* and d symmetries are legitimate HFPs with relative self-consistency equations, reported in Tab. 3.3 using spin degeneracy.

Recognising harmonics, Eq. (3.21) becomes

$$
\begin{aligned}
\frac{2V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k}}\sum_{\sigma}&[\cos(2k_x)+\cos(2k_y)]\langle\hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}\sigma}\rangle\hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}\sigma} \\
&= V\sum_{\mathbf{q},\sigma}\left[u^{(s^*)}_{\sigma}(\cos q_x+\cos q_y) + u^{(d)}_{\sigma}(\cos q_x-\cos q_y)\right]\hat{c}^{\dagger}_{\mathbf{q}\sigma}\hat{c}_{\mathbf{q}\sigma}
\end{aligned}
$$

These two factors, together, define the full bands renormalization

$$\tilde{\epsilon}_{\mathbf{k}\sigma} \equiv \epsilon_{\mathbf{k}\sigma} + u_\sigma^{(s^*)} V(\cos k_x + \cos k_y) + u_\sigma^{(d)} V(\cos k_x - \cos k_y) \tag{3.24}$$

as was for the normal phase in Eq. (2.17). In Sec. 2.3.3 we discussed appearance of anisotropic hopping with d-wave extension of bare bands: here theory predicts something identical. These RMPs are, as before, controlled linearly by V and in principle can lead to relevant changes in the kinetic sector of the hamiltonian.

Off-diagonal terms. Consider off-diagonal terms of Eq. (3.20). These contribute instead to full gap, being out of diagonal in the 2×2 hamiltonian matrix of Eq. (3.6). In pseudo-spin picture, they account for a transverse field. Define $\mathbf{q}$, $\mathbf{q}'$ as in Fig. 2.1b,

$$\mathbf{q} \equiv \mathbf{K} + \mathbf{k} \qquad \mathbf{q}' \equiv \mathbf{K} - \mathbf{k}$$

and rewrite as for Eq. (3.21)

$$\begin{aligned}
&-\frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k}} \sum_{\sigma} [\cos(2k_x) + \cos(2k_y)] \langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma} \rangle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma} \\
&\qquad\qquad = -\sum_{\gamma,\sigma} \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{BZ}} \varphi_{\mathbf{q}}^{(\gamma)} \langle \hat{\psi}^\dagger_{\mathbf{q}\sigma} \rangle \right] \times V \sum_{\mathbf{q}'\in\mathrm{BZ}} \varphi_{\mathbf{q}'}^{(\gamma)*} \hat{\psi}^\dagger_{\mathbf{q}'\sigma}
\end{aligned}$$

Now shift the first sum by an AF vector $\boldsymbol{\pi}$ defining $\mathbf{k}+\boldsymbol{\pi} \equiv \mathbf{q}$, and use Eq. (3.5) as well as reciprocal space periodicity by a BZ,

$$\begin{aligned}
\frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{BZ}} \varphi_{\mathbf{q}}^{(\gamma)} \langle \hat{\psi}^\dagger_{\mathbf{q}\sigma} \rangle &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \varphi_{\mathbf{k}+\boldsymbol{\pi}}^{(\gamma)} \langle \hat{\psi}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma} \rangle \\
&= -\frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \varphi_{\mathbf{k}}^{(\gamma)} \langle \hat{\psi}_{\mathbf{k}\sigma} \rangle
\end{aligned}$$

which finally gives

$$\begin{aligned}
&-\frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k}} \sum_{\sigma} [\cos(2k_x) + \cos(2k_y)] \langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma} \rangle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma} \\
&\qquad\qquad = \sum_{\gamma,\sigma} \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{BZ}} \varphi_{\mathbf{q}}^{(\gamma)} \langle \hat{\psi}_{\mathbf{q}\sigma} \rangle \right] \times V \sum_{\mathbf{q}'\in\mathrm{BZ}} \varphi_{\mathbf{q}'}^{(\gamma)*} \hat{\psi}^\dagger_{\mathbf{q}'\sigma}
\end{aligned} \tag{3.25}$$

Proceed as done previously: define the expectation value

$$v_{\mathbf{q}\sigma} \equiv i \langle \hat{\psi}_{\mathbf{q}\sigma} \rangle \tag{3.26}$$

(which is just the AF order parameter with a complex factor) and its γ-wave component, with $\gamma \in \{s^*, d, p_x, p_y\}$:

$$\begin{aligned}
v_\sigma^{(\gamma)} &= \frac{i}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{BZ}} \varphi_{\mathbf{q}}^{(\gamma)} \langle \hat{\psi}_{\mathbf{q}\sigma} \rangle \\
&= \frac{i}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{MBZ}} \left[\varphi_{\mathbf{q}}^{(\gamma)} \langle \hat{\psi}_{\mathbf{q}\sigma} \rangle + \varphi_{\mathbf{q}+\boldsymbol{\pi}}^{(\gamma)} \langle \hat{\psi}_{\mathbf{q}+\boldsymbol{\pi}\sigma} \rangle \right] \\
&= \frac{i}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{MBZ}} \varphi_{\mathbf{q}}^{(\gamma)} \left[\langle \hat{\psi}_{\mathbf{q}\sigma} \rangle - \langle \hat{\psi}^\dagger_{\mathbf{q}\sigma} \rangle \right] \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{MBZ}} \varphi_{\mathbf{q}}^{(\gamma)} \langle \hat{\Psi}^\dagger_{\mathbf{q}\sigma} \tau^y \hat{\Psi}_{\mathbf{q}\sigma} \rangle \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{MBZ}} \varphi_{\mathbf{q}}^{(\gamma)} \frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{q}\sigma}\}}{\tilde{E}_{\mathbf{q}\sigma}} \tilde{\mathrm{Th}}^{(-)}_{\mathbf{q}\sigma}[\beta, \tilde{\mu}]
\end{aligned} \tag{3.27}$$

Symmetry	Self-consistency equation	Graph
s^*-wave	$v^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\cos k_x + \cos k_y) \frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tilde{\mathrm{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]$	Fig. 1.4b
p_x-wave	$v^{(p_x)} = \frac{i\sqrt{2}}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \sin k_x \frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tilde{\mathrm{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]$	Fig. 1.4c
p_y-wave	$v^{(p_y)} = \frac{i\sqrt{2}}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \sin k_y \frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tilde{\mathrm{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]$	Fig. 1.4d
$d_{x^2-y^2}$-wave	$v^{(d)} = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\cos k_x - \cos k_y) \frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tilde{\mathrm{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]$	Fig. 1.4e

Table 3.4 | Symmetry and spin resolved self-consistency equations for the harmonics of $v_{\mathbf{k}}$.

where we made use of Eq. (3.5), Tab. 3.2 and Eq. (3.8). There is no particular reason to exclude one component or another. In Tab. 3.4 the four derived self-consistency equations are reported (already ignoring spin index).

From structure factors definitions of Tab. 1.5, it's immediate to see that:

$$v_\sigma^{(s^*)}, v_\sigma^{(d)} \in \mathbb{R} \qquad v_\sigma^{(p_x)}, v_\sigma^{(p_y)} \in i\mathbb{R}$$

Getting back to Eq. (3.25) and inserting these components, we get

$$-\frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k}} \sum_\sigma [\cos(2k_x) + \cos(2k_y)] \langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma} \rangle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma} = -iV \sum_{\gamma,\sigma} v_\sigma^{(\gamma)} \sum_{\mathbf{q}\in\mathrm{BZ}} \varphi_{\mathbf{q}}^{(\gamma)*} \hat{\psi}^\dagger_{\mathbf{q}\sigma} \tag{3.28}$$

We can restrict to MBZ as

$$-\frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k}} \sum_\sigma [\cos(2k_x) + \cos(2k_y)] \langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma} \rangle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma} = -iV \sum_{\gamma,\sigma} \sum_{\mathbf{q}\in\mathrm{MBZ}} \left[v_\sigma^{(\gamma)} \varphi_{\mathbf{q}}^{(\gamma)*} \hat{\psi}^\dagger_{\mathbf{q}\sigma} + v_\sigma^{(\gamma)} \varphi_{\mathbf{q}+\boldsymbol{\pi}}^{(\gamma)*} \hat{\psi}^\dagger_{\mathbf{q}+\boldsymbol{\pi}\sigma} \right]$$

and use Eq. (3.5), exchanging terms positions

$$-\frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k}} \sum_\sigma [\cos(2k_x) + \cos(2k_y)] \langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma} \rangle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma} = iV \sum_{\gamma,\sigma} \sum_{\mathbf{q}\in\mathrm{MBZ}} \left[v_\sigma^{(\gamma)} \varphi_{\mathbf{q}}^{(\gamma)*} \hat{\psi}_{\mathbf{q}\sigma} - v_\sigma^{(\gamma)} \varphi_{\mathbf{q}}^{(\gamma)*} \hat{\psi}^\dagger_{\mathbf{q}\sigma} \right] \tag{3.29}$$

Now: recalling Tab. 1.5, for the four symmetries we are considering, form factors are real-valued for symmetries s^* and d, and purely imaginary for p_x and p_y. As mentioned before, the same holds for the $v_\sigma^{(\gamma)}$ harmonics. Then

$$i v_\sigma^{(\gamma)} \varphi_{\mathbf{q}}^{(\gamma)*} \in i\mathbb{R} \tag{3.30}$$

and to conjugate a purely imaginary number it suffices to put a minus sign in front of it,

$$-i v_\sigma^{(\gamma)} \varphi_{\mathbf{q}}^{(\gamma)*} = \left(i v_\sigma^{(\gamma)} \varphi_{\mathbf{q}}^{(\gamma)*} \right)^*$$

Therefore, we conclude

$$-\frac{2V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k}}\sum_{\sigma}[\cos(2k_x)+\cos(2k_y)]\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}\sigma}\rangle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma}$$
$$= V\sum_{\gamma,\sigma}\sum_{\mathbf{q}\in\mathrm{MBZ}}\left[iv^{(\gamma)}_\sigma\varphi^{(\gamma)*}_{\mathbf{q}}\hat{\psi}_{\mathbf{q}\sigma}+\mathrm{h.c.}\right] \tag{3.31}$$

which is obvious, being the hamiltonian operator hermitian from the beginning.

3.3.2 Fock contraction energy

Let us now get back to $E^{(\mathrm{AF})}_{\mathrm{F}/V}$, the double counting energy of Eq. (3.19). As said before, its definition is the same as for the normal phase, Eq. (2.10), with averaging operation being performed on the AF ground state:

$$E^{(\mathrm{AF})}_{\mathrm{F}/V}\equiv\frac{V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\sum_{\sigma}[\cos(\delta k_x)+\cos(\delta k_y)]\,\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma}\rangle\langle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}\rangle \tag{3.32}$$

Proceeding exactly as in Eq. (3.20) and making changing variables as in Fig. 2.1, we get

$$E^{(\mathrm{AF})}_{\mathrm{F}/V}=\frac{V}{L_xL_y}\sum_{\mathbf{q},\mathbf{q}'}\sum_{\sigma}[\cos(\delta q_x)+\cos(\delta q_y)]\left[\underbrace{\langle\hat{n}_{\mathbf{q}\sigma}\rangle\langle\hat{n}_{\mathbf{q}'\sigma}\rangle}_{\text{Diagonal terms}}-\underbrace{\langle\hat{\psi}_{\mathbf{q}\sigma}\rangle\langle\hat{\psi}_{\mathbf{q}'\sigma}\rangle}_{\text{Off-diagonal terms}}\right]$$

Apply the decomposition of Eq. (1.35) and use Eqns. (3.22), (3.26),

$$E^{(\mathrm{AF})}_{\mathrm{F}/V}=\frac{V}{2L_xL_y}\sum_{\gamma,\sigma}\sum_{\mathbf{q},\mathbf{q}'}\varphi^{(\gamma)}_{\mathbf{q}}\varphi^{(\gamma)*}_{\mathbf{q}'}\left[u_{\mathbf{q}\sigma}u_{\mathbf{q}'\sigma}+v_{\mathbf{q}\sigma}v_{\mathbf{q}'\sigma}\right] \tag{3.33}$$

Now: it holds

$$u_{\mathbf{q}\sigma}=\langle\hat{n}_{\mathbf{q}\sigma}\rangle\quad\Longrightarrow\quad u_{\mathbf{q}\sigma}=u^*_{\mathbf{q}\sigma}$$

and then

$$\frac{1}{(L_xL_y)^2}\sum_{\mathbf{q},\mathbf{q}'}\varphi^{(\gamma)}_{\mathbf{q}}\varphi^{(\gamma)*}_{\mathbf{q}'}u_{\mathbf{q}\sigma}u_{\mathbf{q}'\sigma}=\left[\frac{1}{L_xL_y}\sum_{\mathbf{q}\in\mathrm{BZ}}\varphi^{(\gamma)}_{\mathbf{q}}u_{\mathbf{q}\sigma}\right]\left[\frac{1}{L_xL_y}\sum_{\mathbf{q}'\in\mathrm{BZ}}\varphi^{(\gamma)*}_{\mathbf{q}'}u^*_{\mathbf{q}'\sigma}\right]$$
$$=\left|u^{(\gamma)}_\sigma\right|^2$$

At the same time, being $\hat{\psi}_{\mathbf{q}+\boldsymbol{\pi}\sigma}=\hat{\psi}^\dagger_{\mathbf{q}\sigma}$,

$$v_{\mathbf{q}\sigma}=i\langle\hat{\psi}_{\mathbf{q}\sigma}\rangle\quad\Longrightarrow\quad v_{\mathbf{q}+\boldsymbol{\pi}\sigma}=-v^*_{\mathbf{q}\sigma}$$

Since the form factors satisfy $\varphi^{(\gamma)}_{\mathbf{q}+\boldsymbol{\pi}}=-\varphi^{(\gamma)}_{\mathbf{q}}$, we get:

$$\frac{1}{(L_xL_y)^2}\sum_{\mathbf{q},\mathbf{q}'}\varphi^{(\gamma)}_{\mathbf{q}}\varphi^{(\gamma)*}_{\mathbf{q}'}v_{\mathbf{q}\sigma}v_{\mathbf{q}'\sigma}=\left[\frac{1}{L_xL_y}\sum_{\mathbf{q}\in\mathrm{BZ}}\varphi^{(\gamma)}_{\mathbf{q}}v_{\mathbf{q}\sigma}\right]\left[\frac{1}{L_xL_y}\sum_{\mathbf{q}'\in\mathrm{BZ}}\varphi^{(\gamma)*}_{\mathbf{q}'}v_{\mathbf{q}'\sigma}\right]$$
$$=v^{(\gamma)}_\sigma\left[\frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\varphi^{(\gamma)*}_{\mathbf{k}+\boldsymbol{\pi}}v_{\mathbf{k}+\boldsymbol{\pi}\sigma}\right]$$
$$=v^{(\gamma)}_\sigma\left[\frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\varphi^{(\gamma)*}_{\mathbf{k}}v^*_{\mathbf{k}\sigma}\right]$$
$$=\left|v^{(\gamma)}_\sigma\right|^2$$

where in the second passage we employed the change of variables $\mathbf{q}' = \mathbf{k}+\boldsymbol{\pi}$ and used BZ periodicity to map the original sum onto a new sum for $\mathbf{k} \in \mathrm{BZ}$. Inserting above results in Eq. (3.33), the latter becomes:

$$E^{(\mathrm{AF})}_{\mathrm{F}/V} = L_x L_y \times \frac{V}{2} \sum_{\gamma,\sigma} \left[\left| u^{(\gamma)}_\sigma \right|^2 + \left| v^{(\gamma)}_\sigma \right|^2 \right] \tag{3.34}$$

Interestingly enough, the new HFPs amplitudes contribute positively to contraction energy. This fact is non-trivial and has consequences we will discuss in Sec. 3.6.1.

3.4 Full MFT hamiltonian and gap complexification

Finally, it is time to collect everything in a single hamiltonian operator. The MFT model is approximated by:

$$\begin{aligned} \hat{H} - \mu\hat{N} \simeq \sum_\sigma \sum_{\mathbf{k}\in\mathrm{MBZ}} \tilde{\epsilon}_{\mathbf{k}\sigma} \left[\hat{n}_{\mathbf{k}\sigma} - \hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \right] - \sum_\sigma \sum_{\mathbf{k}\in\mathrm{MBZ}} \left[\tilde{\Delta}_{\mathbf{k}\sigma} \hat{\psi}_{\mathbf{k}\sigma} + \tilde{\Delta}^*_{\mathbf{k}\sigma} \hat{\psi}^\dagger_{\mathbf{k}\sigma} \right] \\ - \tilde{\mu}\hat{N} - \left[E^{(\mathrm{AF})}_{\mathrm{H}/U} + E^{(\mathrm{AF})}_{\mathrm{H}/V} + E^{(\mathrm{AF})}_{\mathrm{F}/V} \right] \end{aligned}$$

where renormalized bands are given by:

$$\tilde{\epsilon}_{\mathbf{k}\sigma} = \epsilon_{\mathbf{k}\sigma} + V \sum_{\gamma\in\{s^*,d\}} u^{(\gamma)}_\sigma \varphi^{(\gamma)*}_{\mathbf{k}}$$

and full gap function by

$$\tilde{\Delta}_{\mathbf{k}\sigma} = \Delta_{\mathbf{k}\sigma} - iV \sum_\gamma v^{(\gamma)}_\sigma \varphi^{(\gamma)*}_{\mathbf{k}} \tag{3.35}$$

The gap correction is purely imaginary, compared to HM gap $\Delta_{\mathbf{k}\sigma} = (-1)^{\delta_{\sigma=\downarrow}} mU$. This has two obvious consequences: being no other contribution to the x pseudo-spin projections other than the pure HM ones, spin antisymmetry is preserved within such direction

$$\langle \hat{s}^x_{\mathbf{k}\sigma} \rangle = -\langle \hat{s}^x_{\mathbf{k}\bar{\sigma}} \rangle$$

which is relevant during computation of the physical magnetization, which (as is proven in Eq. (A.10)) is defined as

$$\begin{aligned} m &= \frac{1}{2L_x L_y} \sum_{\mathbf{r}} (-1)^{x+y} \langle \hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow} \rangle \\ &= \frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \langle \hat{s}^x_{\mathbf{k}\uparrow} - \hat{s}^x_{\mathbf{k}\downarrow} \rangle \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \langle \hat{s}^x_{\mathbf{k}\uparrow} \rangle \end{aligned}$$

A proof of the above derivation is given in Sec. A.1.7. Moreover, the net effect $\hat{H}_V$ is a "gap complexification". In other words,

$$\left| \mathrm{Im}\{\Delta_{\mathbf{k}\sigma}\} \right| \geq 0 \quad \Longrightarrow \quad \left| \langle s^y_{\mathbf{k}\sigma} \rangle \right| \geq 0$$

while for HM the y pseudo-spin component was identically null.

3.4.1 The relative phase in the zero temperature half-filled ground state

In Sec. A.1.4, we derive analytically the AF ground state on conventional HM at zero temperature, as a simple Bloch's transform of a polarized (tilted) pseudo-spins collective pure state. Eq. (A.29) reports the final form of said ground-state. Said quantum state is in general pretty similar to conventional BCS state.

At half-filling and zero temperature, it is rather easy to extend that state to the new complexified gap. By performing an identical calculation as in Sec. A.1.4 we here omit, it's easy to show that the new ground state for the EHM in the AF phase is identical, with RMPs juxtaposed in the various amplitudes and a relative phase taking care of gap function phase,

$$|\Psi\rangle = \bigotimes_{\sigma} \bigotimes_{\mathbf{k}\in\mathrm{MBZ}} \left[\sin\tilde{\theta}_{\mathbf{k}\sigma}\hat{c}^{\dagger}_{\mathbf{k}\sigma} + e^{2i\tilde{\zeta}_{\mathbf{k}\sigma}}\cos\tilde{\theta}_{\mathbf{k}\sigma}\hat{c}^{\dagger}_{\mathbf{k}+\boldsymbol{\pi}\sigma}\right]|\Omega_{\mathbf{k}\sigma}\rangle \tag{3.36}$$

which is identical to Eq. (A.29), apart from renormalizations. Let $|\mathbf{k}\sigma\rangle \equiv \hat{c}^{\dagger}_{\mathbf{k}\sigma}|\Omega_{\mathbf{k}\sigma}\rangle$. Then we can write

$$|\Psi\rangle = \bigotimes_{\sigma} \bigotimes_{\mathbf{k}\in\mathrm{MBZ}} \left[\sin\tilde{\theta}_{\mathbf{k}\sigma} + e^{2i\tilde{\zeta}_{\mathbf{k}\sigma}}\cos\tilde{\theta}_{\mathbf{k}\sigma}\hat{\psi}_{\mathbf{k}\sigma}\right]|\mathbf{k}\sigma\rangle$$

having we used $|\Omega_{\mathbf{k}\sigma}\rangle = \{\hat{c}_{\mathbf{k}\sigma}, \hat{c}^{\dagger}_{\mathbf{k}\sigma}\}|\Omega_{\mathbf{k}\sigma}\rangle$.

Angles appearing in above equations are the same of Eq. (3.12), which acquire a physical meaning: the relative phase $2\tilde{\zeta}_{\mathbf{k}\sigma}$ is the gap function angle in complex plane, since

$$\begin{aligned}
\langle\Psi|\hat{s}^{x}_{\mathbf{k}\sigma}|\Psi\rangle &= \langle\Psi|\hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}^{\dagger}_{\mathbf{k}\sigma}|\Psi\rangle \\
&= \sin\tilde{\theta}_{\mathbf{k}\sigma}\cos\tilde{\theta}_{\mathbf{k}\sigma}\left(e^{-2i\tilde{\zeta}_{\mathbf{k}\sigma}} + e^{2i\tilde{\zeta}_{\mathbf{k}\sigma}}\right) \\
&= \sin 2\tilde{\theta}_{\mathbf{k}\sigma}\cos 2\tilde{\zeta}_{\mathbf{k}\sigma}
\end{aligned}$$

$$\begin{aligned}
\langle\Psi|\hat{s}^{y}_{\mathbf{k}\sigma}|\Psi\rangle &= i\,\langle\Psi|\hat{\psi}_{\mathbf{k}\sigma} - \hat{\psi}^{\dagger}_{\mathbf{k}\sigma}|\Psi\rangle \\
&= i\sin\tilde{\theta}_{\mathbf{k}\sigma}\cos\tilde{\theta}_{\mathbf{k}\sigma}\left(e^{-2i\tilde{\zeta}_{\mathbf{k}\sigma}} - e^{2i\tilde{\zeta}_{\mathbf{k}\sigma}}\right) \\
&= \sin 2\tilde{\theta}_{\mathbf{k}\sigma}\sin 2\tilde{\zeta}_{\mathbf{k}\sigma}
\end{aligned}$$

These equations are coherent with Eqns. (3.7), (3.8), thus the involved angles (apart from irrelevant 2π shifts) coincide.

3.4.2 AF phase flavours and spin degeneracy

The AF phase we are studying is not allowed to break inversion symmetry, a core property of the system which is respected from all our approximations. This translates in the simple requirement that:

$$\tilde{E}_{\mathbf{k}\sigma} \overset{!}{=} \tilde{E}_{-\mathbf{k}\sigma}$$

We know that:

$$\tilde{E}_{\mathbf{k}\sigma} = \sqrt{\tilde{\epsilon}^{2}_{\mathbf{k}\sigma} + \mathrm{Re}^{2}\{\tilde{\Delta}_{\mathbf{k}\sigma}\} + \mathrm{Im}^{2}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}}$$

Renormalized bare bands as well as the real part of the gap satisfy said inversion symmetry requirement,

$$\tilde{\epsilon}_{\mathbf{k}\sigma} = \tilde{\epsilon}_{-\mathbf{k}\sigma} \qquad \mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}\sigma}\} = \mathrm{Re}\{\tilde{\Delta}_{-\mathbf{k}\sigma}\}$$

hence inversion symmetry is respected only if

$$\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\} = \pm\,\mathrm{Im}\{\tilde{\Delta}_{-\mathbf{k}\sigma}\}$$

which is: either the imaginary part of the gap is inversion symmetric, or is antisymmetric. Based on this trivial consideration, we are able two discard two of the four self-consistency equations for the v-related HFPs listed in Tab. 3.4: s^{*} and d-wave symmetries HFPs are nonzero if $\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}$ is momentum-even; p_x and p_y-wave symmetries HFPs are nonzero if $\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}$ is momentum-odd. The two options cannot coexist, and separate simulations can be implemented in the two symmetry sectors.

We name the two possible symmetries "sAF" (symmetric antiferromagnet) and "aAF" (antisymmetric antiferromagnet). The set of HFPs then restricts to these AF structures, and is reported for both cases in Tab. 3.5. The presence itself of two "AF flavours" is not-so-physical, given that up to now we have no clear physical interpretation for $\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}}\}$. As we will discuss in Sec. 3.7, it is possible to prove that almost all these new HFPs actually vanish – a fact that is not inferrable from MFT analysis now, but can be seen only delving into the detailed discussion of the iterative

Phase	HFPs
sAF	$\{m, u^{(s^*)}, u^{(d)}, v^{(s^*)}, v^{(d)}\}$
aAF	$\{m, u^{(s^*)}, u^{(d)}, -iv^{(p_x)}, -iv^{(p_y)}\}$

Table 3.5 Set of HFP for the two possible realization of the AF phase, respectively a symmetric and an antisymmetric imaginary part of the gap.

algorithm seen as a recursive parametric map. For now, let us proceed integrally with all the HFPs we found and derive the results.

Moreover, from now on we will omit σ index, working just in the $\uparrow$ sector. As said in Sec. 3.1, the AF phase does not break $\mathbb{Z}_2$ spin inversion symmetry, being the SDW ordering generated by a staggered but globally symmetric spin density distribution among sites. The MFT hamiltonian is invariant by a $\sigma \leftrightarrow \overline{\sigma}$ exchange, thus all measurable physical quantities must be. What is unmeasurable and has no effect on Physics is free to be σ-dependent, as for the sign of $\mathrm{Re}\{\Delta_{\mathbf{k}\sigma}\}$. Since all it's relevant of both the complex components of the gap function is their magnitude, we can safely treat σ as a pure degeneracy. This gives a simple equation for the last HFP,

$$\begin{aligned} m &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \langle \hat{s}^x_{\mathbf{k}\uparrow} \rangle \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \frac{\mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}}[\beta, \tilde{\mu}] \end{aligned} \tag{3.37}$$

which is the actual physical magnetization. From the first line we see that it can be interpreted as the average transverse pseudo-spin value.

3.5 (In)stability of commensurate AF phase

[Re-write and move up.]

The discussion above is formally correct and self-consistent, thus a HF algorithm can be sketched to extract the HFPs. However we cannot proceed without considering the instability effects of Sec. ??. As is known (and we would have liked to notice earlier) the AF-SSB at wavevector $\boldsymbol{\pi}$ we are studying here, responsible of halving the BZ down to the MBZ, can produce a stable phase on pure Hubbard lattices only at half filling [13]. For the EHM the situation is even worse than that: repeating the simple Linear Reponse Theory (LRT) computations of Sec. ?? and mimicking Eq. (A.34), we get a proper response function

$$\tilde{\chi}_{\hat{\mathbf{S}}^-\hat{\mathbf{S}}^+}(\mathbf{k}, \omega) = \frac{\chi_{\hat{\mathbf{S}}^-\hat{\mathbf{S}}^+}(\mathbf{k}, \omega)}{1 - [U - 2V(\cos k_x + \cos k_y)]\, \chi_{\hat{\mathbf{S}}^-\hat{\mathbf{S}}^+}(\mathbf{k}, \omega)} \tag{3.38}$$

[Double check]

As is well documented in literature [14], AF can establish in a wide range of spatial structures with particular shapes, for example stripes. In order to extract the precise AF phase, we should

1. Choose an initializer wavevector, say $\mathbf{k} = \boldsymbol{\pi}$;
2. Solve self-consistently for the HFPs at $\mathbf{k}$;
3. Compute the free energy F as a functional of the wavevector around $\mathbf{k}$;
4. Look for gradient descent of the free energy, $\boldsymbol{\nabla} F < 0$;
5. If the gradient descent is found, update $\mathbf{k} \to \mathbf{k} + \delta\mathbf{k}$ and repeat from step 2, otherwise halt.

Of course the sketched procedure can lead to a lot of complications, first the fact that at incommensurate wavevector $\mathbf{k} \neq \boldsymbol{\pi}$ the unit cell is not as simple as a pair of sites, but can become much larger requiring more refined computational solutions.

In this text we will show the self consistent result for AF-SSB at wavevector $\boldsymbol{\pi}$ even at finite doping as a first step in the above sketched procedure. The presented results are self-consistent, but unfortunately deeply unstable and unphysical at finite doping, a situation where an incommensurate and insulating AF phase is preferred to this commensurate metallic counterpart. Nevertheless we present them anyway.

3.6 AF free energy density

In order to derive the free energy density expression for AF phase, we proceed precisely as in Sec. 2.4. The only notable difference is in entropy: Eq. (2.21) is written for two opposite bands on MBZ

$$s = s^{(+)} + s^{(-)}$$

with

$$s^{(\pm)} \equiv -\frac{k_B}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \Big[\Big(1 - f(\pm\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\Big) \ln \Big(1 - f(\pm\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\Big) + f(\pm\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu}) \ln f(\pm\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\Big] \quad (3.39)$$

Consider also that we are computing free energy on the gran-canonical hamiltonian

$$\hat{H} - \mu\hat{N} \simeq \sum_{\sigma} \sum_{\mathbf{k}\in\mathrm{MBZ}} \Big[(\tilde{E}_{\mathbf{k}} - \tilde{\mu}) \hat{\gamma}^{(+)\dagger}_{\mathbf{k}\sigma} \hat{\gamma}^{(+)}_{\mathbf{k}\sigma} - (\tilde{E}_{\mathbf{k}} + \tilde{\mu}) \hat{\gamma}^{(-)\dagger}_{\mathbf{k}\sigma} \hat{\gamma}^{(-)}_{\mathbf{k}\sigma} \Big] + E_{\mathrm{c}}^{(\mathrm{AF})}$$

being $E_{\mathrm{c}}^{(\mathrm{AF})}$ the total AF contraction energy. Note that on the left-hand side we have μ and on the right-hand side its RMP counterpart $\tilde{\mu}$.

With identical meaning of the notation with respect to Sec. 2.4, the free energy contributions are:

1. Quasiparticles energy, taking care of spin degeneracy:
$$e_{\mathrm{g}}^{(\mathrm{AF})} = \frac{2}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\tilde{E}_{\mathbf{k}} - \tilde{\mu}) f(\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu}) - \frac{2}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\tilde{E}_{\mathbf{k}} + \tilde{\mu}) f(-\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})$$

2. Contraction energy:
$$e_{\mathrm{c}}^{(\mathrm{AF})} = U(m^2 - n^2) + V \left[2zn^2 - \sum_{\gamma} \left(\left|u^{(\gamma)}\right|^2 + \left|v^{(\gamma)}\right|^2 \right) \right]$$

3. Entropic energy
$$e_{s}^{(\mathrm{AF})} = \frac{2}{\beta L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \ln \Big(1 - f(\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\Big) - \frac{2}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\tilde{E}_{\mathbf{k}} - \tilde{\mu}) f(\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu}) + \frac{2}{\beta L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \ln \Big(1 - f(-\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\Big) + \frac{2}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\tilde{E}_{\mathbf{k}} + \tilde{\mu}) f(-\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})$$

4. Chemical potential correction
$$e_{n}^{(\mathrm{AF})} = \mu n$$

Compose everything

$$f_{\mathrm{MFT}}^{(\mathrm{AF})} \equiv e_{\mathrm{g}}^{(\mathrm{AF})} + e_{\mathrm{c}}^{(\mathrm{AF})} + e_{s}^{(\mathrm{AF})} + e_{n}^{(\mathrm{AF})}$$

and we get

$$f_{\mathrm{MFT}}^{(\mathrm{AF})} = \frac{2}{\beta L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \Big[\ln \Big(1 - f(\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\Big) + \ln \Big(1 - f(-\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\Big) \Big] + Um^2 - V \sum_{\gamma} \left[\left|u^{(\gamma)}\right|^2 + \left|v^{(\gamma)}\right|^2 \right] + n\left[\mu + n(2zV - U)\right]$$

As was for the normal phase, contraction energy and chemical potential energy corrections build up the RMP $\tilde{\mu}$ of Eq. (3.17),

$$f_{\mathrm{MFT}}^{(\mathrm{AF})} = \frac{2}{\beta L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \left[\ln\left(1 - f(\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\right) + \ln\left(1 - f(-\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\right)\right]$$
$$+ Um^2 - V\sum_{\gamma}\left[\left|u^{(\gamma)}\right|^2 + \left|v^{(\gamma)}\right|^2\right] + \tilde{\mu}n \quad (3.40)$$

3.6.1 The gap energy

It is useful to see here a property of the above expression. Collecting the terms directly related to the gap function, the following identity holds:

$$Um^2 - V\sum_{\gamma}\left|v^{(\gamma)}\right|^2 = \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{MBZ}} \frac{\left|\tilde{\Delta}_{\mathbf{k}}\right|^2}{\tilde{E}_{\mathbf{k}}}\tilde{\mathrm{Th}}^{(-)}[\beta,\tilde{\mu}] \quad (3.41)$$

The proof is simple and immediate: expanding right-hand side,

$$\frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{MBZ}} \frac{\left|\tilde{\Delta}_{\mathbf{k}}\right|^2}{\tilde{E}_{\mathbf{k}}}\tilde{\mathrm{Th}}^{(-)}[\beta,\tilde{\mu}] = \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{MBZ}} \frac{\mathrm{Re}^2\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}}\tilde{\mathrm{Th}}^{(-)}[\beta,\tilde{\mu}]$$
$$+ \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{MBZ}} \frac{\mathrm{Im}^2\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}}\tilde{\mathrm{Th}}^{(-)}[\beta,\tilde{\mu}]$$

Now:

$$\mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}}\} = Um$$

which implies

$$\begin{aligned}\frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{MBZ}} \frac{\mathrm{Re}^2\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}}\tilde{\mathrm{Th}}^{(-)}[\beta,\tilde{\mu}] &= Um \times \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{MBZ}} \frac{\mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}}\tilde{\mathrm{Th}}^{(-)}[\beta,\tilde{\mu}] \\ &= Um^2\end{aligned}$$

having we recognized the magnetization from Eq. (3.37). Equivalently,

$$\begin{aligned}\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}}\} &= -V\sum_{\gamma} v^{(\gamma)}\varphi_{\mathbf{k}}^{(\gamma)*} \\ &= -V\sum_{\gamma} v^{(\gamma)*}\varphi_{\mathbf{k}}^{(\gamma)}\end{aligned}$$

where we used Eq. (3.30) to exchange the conjugation operation in the last passage. This implies:

$$\begin{aligned}\frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{MBZ}} \frac{\mathrm{Im}^2\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}}\tilde{\mathrm{Th}}^{(-)}[\beta,\tilde{\mu}] &= -V\sum_{\gamma} v^{(\gamma)*}\sum_{\mathbf{k}\in\mathrm{MBZ}}\varphi_{\mathbf{k}}^{(\gamma)}\frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}}\tilde{\mathrm{Th}}^{(-)}[\beta,\tilde{\mu}] \\ &= -V\sum_{\gamma} v^{(\gamma)*}\times v^{(\gamma)} \\ &= -V\sum_{\gamma}\left|v^{(\gamma)*}\right|^2\end{aligned}$$

where in second passage we recognized $v^{(\gamma)}$ given in Eq. (3.27). Then Eq. (3.41) is verified. The latter is particularly interesting because it shows a non-intuitive result: due to gap complexification, gap amplitude is only increased by $\hat{H}_V$, thus by analogy we would expect the relative free energy contribution to have the same sign as the original gap energy from the real part. This is not the case: to open a gap in the s-wave real sector contributes positively to free energy, while to open a gap in the γ-wave imaginary sector contributes negatively.

An immediate consequence of Eq. (3.41) is that, being the right-hand side evidently positive, for $V > 0$ the total amplitude of the "complexifiers" vector is maximised,

$$\sum_{\gamma} \left|v^{(\gamma)}\right|^2 \leq M \quad \text{being} \quad M^{(\mathrm{AF})} \equiv m^2 \frac{U}{V} \tag{3.42}$$

The above constraint predicts that:

- In the strong coupling limit $U \ll V$, gap complexification is suppressed;
- Correctly, when the physical magnetization closes, so does the imaginary part of the gap

$$\lim_{m \to 0} v^{(\gamma)} = 0$$

which is coherent with the entire derivation and takes us back to the normal phase.

Eq. (3.41) is also solved by a null magnetization and thus null complexifiers. This can be interpreted by the fact that when the constraint is impossible to solve self-consistently, the gap must close.

Interestingly, Eq. (3.42) provides a neat graphical interpretation for the constraint on HFPs in both sAF and aAF phases. At any competition ratio U/V, system magnetization m defines a sphere whose radius is $m\sqrt{U/V}$, and inside which the vector defined by remaining $v^{(\gamma)}$ HFPs must lie. For our AF flavours, the sphere is actually a circle, being in both cases three the total HFPs number, counting also m. All of this is nice, but notice that the trivial solution

$$\forall \gamma \quad : \quad v^{(\gamma)} = 0$$

solves the constraint of Eq. (3.42). These considerations do not imply gap complexification is an actual, realized effect.

3.6.2 Free energy at half-filling

Half-filling condition is of particular interest, with $\tilde{\mu} = 0$, where PHS discussed in Sec. 1.1.3 is preserved in the AF phase. Since, as it can be shown easily,

$$\begin{aligned} \ln\left(1 - \frac{1}{e^x + 1}\right) + \ln\left(1 - \frac{1}{e^{-x} + 1}\right) &= \ln\left[\frac{e^x}{(e^x + 1)^2}\right] \\ &= x - 2\ln(1 + e^x) \end{aligned}$$

Let now $x = \beta \tilde{E}_{\mathbf{k}}$. At low temperature for a band with no nodes, we can approximate $x \to +\infty$, which gives

$$x - 2\ln(1 + e^x) \simeq -x$$

Hence we get the half-filling free energy density substituting ,

$$f_{\mathrm{MFT}}^{(\mathrm{AF})} \simeq -\frac{2}{L_x L_y} \sum_{\mathbf{k} \in \mathrm{MBZ}} \tilde{E}_{\mathbf{k}} + Um^2 - V \sum_{\gamma} \left[\left|u^{(\gamma)}\right|^2 + \left|v^{(\gamma)}\right|^2\right]$$

Starting from the full expression of the free energy density, we recovered the intuitive expression of the energy of a fully (double) occupied band at $-\tilde{E}_{\mathbf{k}}$, with HFPs correction.

This approximation makes sense when $\tilde{E}_{\mathbf{k}} \neq 0$ everywhere. Fortunately, this is always realized in AF phase: from the constraint of Eq. (3.42) we know magnetization is necessary for all gap HFPs; for a magnetized system, whatever the symmetries,

$$\tilde{E}_{\mathbf{k}} = \sqrt{\tilde{\epsilon}_{\mathbf{k}}^2 + (Um)^2 + \left|\operatorname{Im}\{\Delta_{\mathbf{k}}\}\right|^2} \geq Um$$

At half filling and low temperature, the system is well approximated by a completely filled electronic band $-\tilde{E}_{\mathbf{k}}$, and this correctly gives a simple $f_{\mathrm{MFT}}^{(\mathrm{AF})}$ approximation.

3.7 Theoretical constraints on the AF phase HFPs

Up to now, all we derived was the most general MFT-compatible mathematical description of a commensurate AF established on EHM, allowing for nematic spatial symmetry breaking as well as non-trivial renormalization effects. This section, basically a clone of Sec. 2.5, is dedicated to understanding, given the algorithmic problem, how its solutions should behave. As we will see, based on this theoretical analysis we are able to reduce a lot the algorithmic complexity and limit the solution search to a tailored subspace of HFPs space.

We start from considering a robust C_4 antiferromagnet, thus setting $u^{(d)} = 0$ by hand, and show how the constraints on $u^{(s^*)}$ mimic those posed for the normal phase in Eq. (2.27). Then we move to a rather convoluted but necessary analysis of the five components HFPs vector described for both AF flavours in Tab. 3.5. For the sake of generality, even if – as discussed in Sec. 3.5 – at finite doping the system is naturally unstable, we take on all calculations at generic $\tilde{\mu}$.

3.7.1 Robust C_4 bands shrinking extension to AF phases

For the normal phase, we saw that when C_4 symmetry is preserved (Sec. 2.5.1) we can conclude immediately that bands renormalization is actually a shrinking. An identical result holds here: if C_4 symmetry is imposed onto the renormalized bands,

$$\tilde{\epsilon}_{\mathbf{k}\sigma} \equiv (u^{(s^*)}V - 2t)(\cos k_x + \cos k_y)$$

of $u^{(\gamma)}$, just the $s*$-wave channel contributes with a single self-consistency equation

$$u^{(s^*)} = -\frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\cos k_x + \cos k_y) \frac{\tilde{\epsilon}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tilde{\mathrm{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]$$

In this situation, it's immediate to derive constraints for $u^{(s^*)}$. As we did for the normal phase:

1. If $t > 0$ and $u^{(s^*)} \leq 0$
$$\forall \mathbf{k} \in \mathrm{MBZ} \quad : \quad \tilde{\epsilon}_{\mathbf{k}} \leq 0$$
the equality only holding on the MBZ contour. Looking at the self-consistency equation, all other objects inside the sum are positive-defined. Which gives
$$u^{(s^*)} \leq 0 \quad \Longrightarrow \quad u^{(s^*)} \geq 0$$
But $u^{(s^*)} \neq 0$ as long as $t > 0$, as can be seen by the self-consistency equation. We conclude necessarily
$$u^{(s^*)} > 0$$

2. If $t > 0$ and $u^{(s^*)} \geq 2t/V$
$$\forall \mathbf{k} \in \mathrm{MBZ} \quad : \quad \tilde{\epsilon}_{\mathbf{k}} \geq 0$$
the equality only holding on the MBZ contour. Inverting inequalities from point above, we conclude necessarily
$$u^{(s^*)} < 2t/V$$

The final result,

$$0 < u^{(s^*)} < 2t/V \tag{3.43}$$

is identical to the constraint obtained for the normal phase. Also in AF phase the bare bands are effectively shrunk by $\hat{H}_V$.

3.7.2 HFPs vanishing in sAF phase

We now present a somewhat convoluted argument to show that, even if raw MFT presents an apparently rich scenario with a lot of HFPs, the problem can be proven to be actually much simpler. Let us start from the sAF phase, assuming

$$\mathrm{C}_4 \overset{\mathrm{SSB}}{\to} \mathrm{C}_2$$

The system is described by the five HFPs on the first row of Tab. 3.5 and their self-consistency equations of Tabs. 3.3 and 3.4.

Let us define the functional

$$F[f(\mathbf{k})] \equiv \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \frac{\tilde{\mathrm{Th}}_{\mathbf{k}}^{(-)}[\beta,\tilde{\mu}]}{\tilde{E}_{\mathbf{k}}} f(\mathbf{k}) \tag{3.44}$$

whose linearity properties are trivial,

$$F[af(\mathbf{k}) + bg(\mathbf{k})] = aF[f(\mathbf{k})] + bF[g(\mathbf{k})] \tag{3.45}$$

being $a, b \in \mathbb{C}$ and f, g two integrable functions in reciprocal space. Another handy property is the sign-preserving relation for sign-defined functions. Let for instance

$$\forall \mathbf{k} \quad : \quad f(\mathbf{k}) \geq 0 \quad \Longrightarrow \quad F[f(\mathbf{k})] \geq 0 \tag{3.46}$$

The relation above is implied by the fact that both $\tilde{\mathrm{Th}}_{\mathbf{k}}^{(-)}$ and $\tilde{E}_{\mathbf{k}}$ are positive-defined,

$$\tilde{\mathrm{Th}}_{\mathbf{k}}^{(-)} = \frac{1}{e^{\beta(\tilde{E}_{\mathbf{k}}-\mu)}+1} - \frac{1}{e^{\beta(-\tilde{E}_{\mathbf{k}}-\mu)}+1} \geq 0 \qquad \tilde{E}_{\mathbf{k}} = \sqrt{\tilde{\epsilon}_{\mathbf{k}}^2 + |\Delta_{\mathbf{k}\sigma}|^2} \geq 0$$

so, being the functional F essentially a weighted integral over MBZ, any positive-defined function has a positive image. An identical statement with inverted inequalities holds for negative-defined function.

Consider now the self-consistency equation for $u^{(s^*)}$: explicitly,

$$\begin{aligned} u^{(s^*)} &= -\frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (c_x + c_y) \frac{\tilde{\epsilon}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tilde{\mathrm{Th}}_{\mathbf{k}}^{(-)}[\beta,\tilde{\mu}] \\ &= \frac{\left[2t - Vu^{(s^*)}\right]}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \frac{\tilde{\mathrm{Th}}_{\mathbf{k}}^{(-)}[\beta,\tilde{\mu}]}{\tilde{E}_{\mathbf{k}}} (c_x + c_y)^2 - \frac{Vu^{(d)}}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \frac{\tilde{\mathrm{Th}}_{\mathbf{k}}^{(-)}[\beta,\tilde{\mu}]}{\tilde{E}_{\mathbf{k}}} (c_x^2 - c_y^2) \end{aligned}$$

having we used the notation $c_\ell \equiv \cos k_\ell$ and substituted the full bare bands

$$\tilde{\epsilon}_{\mathbf{k}} \equiv -\left[2t - u_\sigma^{(s^*)} V\right](c_x + c_y) + u_\sigma^{(d)} V (c_x - c_y)$$

Evidently, in terms of the functional F, we can rewrite:

$$\begin{aligned} u^{(s^*)} &= \left[2t - Vu^{(s^*)}\right] \times F\left[(c_x + c_y)^2\right] - Vu^{(d)} \times F\left[c_x^2 - c_y^2\right] \\ &= \left[2t - Vu^{(s^*)}\right] \times \left\{F\left[c_x^2\right] + 2F\left[c_x c_y\right] + F\left[c_y^2\right]\right\} - Vu^{(d)} \times \left\{F\left[c_x^2\right] - F\left[c_y^2\right]\right\} \end{aligned}$$

having we used the linearity property of Eq. (3.45) in last passage.

Let now:

$$A \equiv F\left[c_x^2\right] \qquad B \equiv 2F\left[c_x c_y\right] \qquad C \equiv F\left[c_y^2\right] \tag{3.47}$$

which allows us to rewrite the self-consistency equation as:

$$u^{(s^*)} = \left[2t - Vu^{(s^*)}\right](A + B + C) - Vu^{(d)}(A - C)$$

Repeating an identical procedure on the three self-consistency equations for $u^{(d)}$, $v^{(s^*)}$ and $v^{(d)}$, we get the full system:

$$u^{(s^*)} = \left[2t - Vu^{(s^*)}\right](A + B + C) - Vu^{(d)}(A - C) \tag{3.48}$$
$$u^{(d)} = \left[2t - Vu^{(s^*)}\right](A - C) - Vu^{(d)}(A - B + C) \tag{3.49}$$
$$v^{(s^*)} = -Vv^{(s^*)}(A + B + C) - Vv^{(d)}(A - C) \tag{3.50}$$
$$v^{(d)} = -Vv^{(s^*)}(A - C) - Vv^{(d)}(A - B + C) \tag{3.51}$$

These equations define a four-fold nonlinear map, whose parameters A, B, C depend on the HFPs themselves, being the functional F dependent on the HFPs. Because of this last point, the four equations are coupled to each other and the map does not decouple in two independent sub-maps.

Aim of this section is to prove that no self-consistent solution exists with non-zero $u^{(d)}$, $v^{(s^*)}$ and $v^{(d)}$. Being the derivation convoluted and abstract enough, let us proceed by enumerating the relevant facts.

1. Let us start by applying a linear transform on the map: define the new "sum" and "difference" variables

$$S_u \equiv u^{(s^*)} + u^{(d)} \qquad D_u \equiv u^{(s^*)} - u^{(d)} \qquad S_v \equiv v^{(s^*)} + v^{(d)} \qquad D_v \equiv v^{(s^*)} - v^{(d)}$$

whose matrix representation is

$$\begin{bmatrix} S_u \\ D_u \\ S_v \\ D_v \end{bmatrix} \equiv \begin{bmatrix} 1 & 1 & & \\ 1 & -1 & & \\ & & 1 & 1 \\ & & 1 & -1 \end{bmatrix} \begin{bmatrix} u^{(s^*)} \\ u^{(d)} \\ v^{(s^*)} \\ v^{(d)} \end{bmatrix}$$

The map transform is obtained by summing and subtracting two couples of equations: Eq. (3.48) and (3.49) (the first two) as well as Eq. (3.50) and (3.51) (the second two). By doing this we get the simplified system:

$$(1 + 2AV)S_u + BVD_u = 2t(2A + B) \tag{3.52}$$
$$(1 + 2CV)D_u + BVS_u = 2t(B + 2C) \tag{3.53}$$
$$(1 + 2AV)S_v + BVD_v = 0 \tag{3.54}$$
$$(1 + 2CV)D_v + BVS_v = 0 \tag{3.55}$$

2. Let $S_v = 0$. Then $D_v = 0$. From Eq. (3.55), using $C = F\left[c_y^2\right] \geq 0$ implied by Eq. (3.46):

$$\underbrace{(1 + 2CV)}_{>0} D_v = 0 \qquad \Longrightarrow \qquad D_v = 0$$

Identically, let $D_v = 0$. Then applying the same consideration on Eq. (3.54) with $C \leftrightarrow A$, we conclude $S_v = 0$. Either both are non-zero or both are zero.

3. Let $S_v \neq 0$ and $D_v \neq 0$. Then $B \neq 0$. To prove this, consider for example Eq. (3.54): we are working at $V > 0$, thus

$$B = -\frac{1 + 2AV}{V}\frac{S_v}{D_v}$$

which is non-zero because $A = F\left[c_x^2\right] \geq 0$, the latter following from Eq. (3.46). We could have used identically Eq. (3.55).

4. Let $S_v \neq 0$ and $D_v \neq 0$. Hence the following relation holds:

$$(BV)^2 = (1 + 2CV)(1 + 2AV) \tag{3.56}$$

To prove the latter, it is sufficient to multiply Eq. (3.54) by $BV \neq 0$ (see point above),

$$BV(1+2AV)S_v + (BV)^2 D_v = 0$$

and substitute from Eq. (3.55) $BVS_v = -(1+2CV)D_v$,

$$\left[-(1+2CV)(1+2AV) + (BV)^2\right] D_v = 0$$

Since $D_v \neq 0$ by hypothesis, Eq. (3.56) is implied.

5. Let $S_v \neq 0$ and $D_v \neq 0$. Hence the following relation holds:

$$B + 2C = V(B^2 - 4AC) \tag{3.57}$$

To prove this: multiply Eq. (3.52) by BV on both sides,

$$BV(1+2AV)S_u + (BV)^2 D_u = 2tBV(2A+B)$$

and substitute from Eq. (3.53) $BVS_v = -(1+2CV)D_v + 2t(B+2C)$,

$$\begin{aligned} -(1+2CV)(1+2AV)D_v + 2t(B+2C)(1+2AV) + (BV)^2 D_u &= 2tBV(2A+B) \\ 2t(B+2C)(1+2AV) &= 2tBV(2A+B) \\ B + 2ABV + 2C + 4ACV &= 2ABV + B^2 V \end{aligned}$$

Passing from first to second line we used Eq. (3.56), and in the passage below $t > 0$. Last line implies Eq. (3.57).

6. Let $S_v \neq 0$ and $D_v \neq 0$. Then $B^2 \neq 4AC$. Indeed, let $B^2 = 4AC$. Then from Eq. (3.56),

$$(BV)^2 = 1 + 2V(A+C) + 4ACV^2 \qquad \Longrightarrow \qquad 1 + 2V(A+C) = 0$$

which is obviously absurd, being $A \geq 0$, $C \geq 0$ from Eq. (3.46). A consequence of this implication on Eq. (3.57) is that, at a given V, the map parameters should satisfy

$$V = \frac{B+2C}{B^2 - 4AC} \tag{3.58}$$

7. Let $S_v \neq 0$ and $D_v \neq 0$. Then $A = C$. To prove this, substitute Eq. (3.58) in Eq. (3.56),

$$\begin{aligned} (B^2 - 4AC)\left(\frac{B+2C}{B^2-4AC}\right)^2 - 2\frac{B+2C}{B^2-4AC}(A+C) - 1 &= 0 \\ (B+2C)^2 - 2(B+2C)(A+C) - (B^2 - 4AC) &= 0 \\ B^2 + 4C^2 + 4BC - 2AB - 2BC - 4AC - 4C^2 - B^2 + 4AC &= 0 \end{aligned}$$

Simplifying all recurring terms, we get

$$2B(C - A) = 0$$

From previous points we know $B \neq 0$. Then $A = C$.

8. It's impossible to have $S_v \neq 0$ and $D_v \neq 0$. To prove this, assume the contrary and use the point above to reduce Eq. (3.58) to

$$A = C \qquad \Longrightarrow \qquad V = \frac{B+2C}{B^2 - 4C^2} = \frac{1}{B - 2C}$$

Inserting the latter in Eq. (3.55),

$$\begin{aligned} \left(1 + \frac{2C}{B-2C}\right) D_v + \frac{B}{B-2C} S_v &= 0 \\ \frac{B}{B-2C} D_v + \frac{B}{B-2C} S_v &= 0 \end{aligned}$$

which implies

$$D_v + S_v = 0 \quad \Longrightarrow \quad v^{(s^*)} = 0$$

Recalling from Tab. 3.4 the explicit self-consistency equation for $v^{(d)}$ and setting $v^{(s^*)} = 0$,

$$v^{(d)} = -Vv^{(d)} F\left[(c_x - c_y)^2\right]$$

and using the positivity rule of Eq. (3.46) we get

$$V > 0 \quad \text{and} \quad F\left[(c_x - c_y)^2\right] \geq 0 \quad \Longrightarrow \quad v^{(d)} = 0$$

because no finite positive number can equal any finite negative number, and *viceversa*. Thus

$$S_v \neq 0 \quad \text{and} \quad D_v \neq 0 \quad \Longrightarrow \quad v^{(s^*)} = v^{(d)} = 0 \quad \Longrightarrow \quad S_v = D_v = 0$$

Absurd. We explored any possibility for S_v, D_v: no stable solution with finite values of the HFPs $v^{(s^*)}$, $v^{(d)}$ exists, thus both must be zero and can be eliminated from the algorithm.

To gain full knowledge on the map fixed point's shape, we should analyze similarly the behaviour of the self-consistency equations for $u^{(s^*)}$, $u^{(d)}$. This is done later in Sec. 3.7.4. For now, it suffices to know that limitedly to the sAF phase no gap complexification aforementioned in Sec. 3.4 actually occurs. This effect is not a consequence of MFT derivation, which remains self-coherent and correct; it is instead of the particular form of the non-linear map defined by the set of model equations.

3.7.3 HFPs vanishing in aAF phase

We now deal with a much easier proof for an analogous thesis on the sAF phase described by the five HFPs on the second row of Tab. 3.5, as well as the self-consistency equations of Tabs. 3.3 and 3.4. We continue using the functional F as defined in Eq. (3.44). We also define A, B, C as in Eq. (3.47). Let $s_\ell \equiv \sin k_\ell$, and

$$P \equiv F\left[s_x^2\right] \qquad Q \equiv 2F\left[s_x s_y\right] \qquad R \equiv F\left[s_y^2\right] \tag{3.59}$$

Using these definitions, four of five self-consistency equations are given by

$$\begin{aligned}
u^{(s^*)} &= \left[2t - Vu^{(s^*)}\right](A + B + C) - Vu^{(d)}(A - C) &(3.60)\\
u^{(d)} &= \left[2t - Vu^{(s^*)}\right](A - C) - Vu^{(d)}(A - B + C) &(3.61)\\
v^{(p_x)} &= -Vv^{(p_x)}P - Vv^{(p_y)}Q &(3.62)\\
v^{(p_y)} &= -Vv^{(p_x)}Q - Vv^{(p_y)}R &(3.63)
\end{aligned}$$

To prove that both $v^{(p_\ell)} = 0$ is immediate.

1. For the aAF phase, $Q = 0$. Let us write Q explicitly

$$\begin{aligned}
Q &= F\left[s_x s_y\right]\\
&= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\text{MBZ}} \frac{\tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]}{\tilde{E}_{\mathbf{k}}} \sin k_x \sin k_y\\
&= \frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\text{BZ}} \frac{\tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]}{\tilde{E}_{\mathbf{k}}} \sin k_x \sin k_y\\
&= \frac{1}{2L_x L_y} \sum_{k_x=-\pi}^{+\pi} \sin k_x \sum_{k_y=-\pi}^{+\pi} \frac{\tilde{\text{Th}}_{(k_x,k_y)}^{(-)}[\beta, \tilde{\mu}]}{\tilde{E}_{(k_x,k_y)}} \sin k_y
\end{aligned}$$

From second to third line, we used MBZ periodicity. The k_y sum at last line is null: even if using antiperiodic harmonics for $\text{Im}\{\tilde{\Delta}_{\mathbf{k}}\}$, we did not break inversion symmetry neither in x nor y direction. Bogoliubov bands $\tilde{E}_{\mathbf{k}}$ and thermal factors must show said symmetry. The product of a symmetric function $(\tilde{\text{Th}}/\tilde{E})$ with $\sin k_y$ integrates to zero. This ends the point.

2. $v^{(p_x)} = 0$. To see this, just plug $Q = 0$ in Eq. (3.62),

$$v^{(p_x)} = -Vv^{(p_x)}F\left[s_x^2\right]$$

Using the positivity rule of Eq. (3.46) we get

$$V > 0 \quad \text{and} \quad F\left[s_x^2\right] \geq 0 \quad \Longrightarrow \quad v^{(p_x)} = 0$$

3. $v^{(p_x)} = 0$. Same proof as point above.

This ends this short section and shows that also for the aAF phase all gap complexifiers actually vanish. This is trivially coherent with the constraint of Eq. (3.42). As a consequence, there is no need to distinguish two AF phases,

$$\text{sAF} = \text{aAF}$$

and gap complexification is a completely "ghost" effect, even if theoretically allowed. All is left is to check for the behaviour of $u^{(d)}$.

3.7.4 Natural suppression of nematicity within $\mathrm{C}_2/\mathbb{Z}_2$ symmetry

A consequence of the two sections above is that the gap function is purely real, non-renormalized and constant across all reciprocal space,

$$\tilde{\Delta}_{\mathbf{k}} = mU \quad \Longrightarrow \quad \tilde{E}_{\mathbf{k}} = \sqrt{\tilde{\epsilon}_{\mathbf{k}}^2 + (mU)^2}$$

Bogoliubov bands maintain renormalization through the $u^{(\gamma)}$ HFPs channel. Aim of this section is to prove that, as was for the normal phase, no nematicity occurs in the AF phase, giving $u^{(d)} = 0$.

While the proof given in Sec. 2.5.2 was long and quite elaborated, this one is actually very short. Let us start from recalling the normal phase context:

$$\text{Normal phase:} \qquad \begin{aligned} u^{(s^*)} &= \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\text{BZ}}(\cos k_x + \cos k_y)f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu}) \\ u^{(d)} &= \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\text{BZ}}(\cos k_x - \cos k_y)f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu}) \end{aligned}$$

while, using $\boldsymbol{\pi}$ periodicity, in the present context we can re-write all equations doubling the integration domain,

$$\text{AF phase:} \qquad \begin{aligned} u^{(s^*)} &= -\frac{1}{2L_x L_y}\sum_{\mathbf{k}\in\text{BZ}}(\cos k_x + \cos k_y)\frac{\tilde{\epsilon}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}}\tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta,\tilde{\mu}] \\ u^{(d)} &= -\frac{1}{2L_x L_y}\sum_{\mathbf{k}\in\text{BZ}}(\cos k_x - \cos k_y)\frac{\tilde{\epsilon}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}}\tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta,\tilde{\mu}] \end{aligned}$$

Let us define the pseudo distribution

$$f'(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu}) \equiv -\frac{\tilde{\epsilon}_{\mathbf{k}}}{2\tilde{E}_{\mathbf{k}}}\tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta,\tilde{\mu}]$$

which, substituted in the self-consistency equations, makes them very similar to the normal ones, up to a $f \to f'$ exchange.

The fact that $u^{(d)} = 0$ follows from the simple consideration that in Sec. 2.5.2 (when showing the same for the normal phase) we never used the specific form of f; instead, we only made use of it being monotonically descendent as a function of $\epsilon_{\mathbf{k}}$. If f' also is a monotonically descendent function of $\epsilon_{\mathbf{k}}$, we can re-use the entire derivation of Sec. 2.5.2. Let us show that using a somewhat lighter notation,

$$\epsilon_{\mathbf{k}} \to x \qquad mU \to \Delta$$

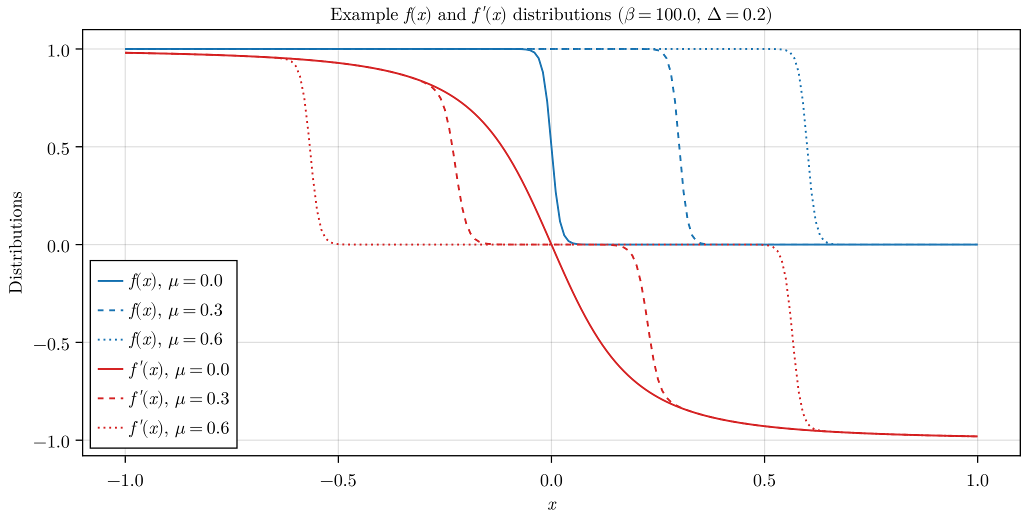


Figure 3.3 | Sketch of the Fermi-Dirac distribution $f(x)$ and the pseudo-distribution $f'(x)$ described in text for the normal phase. These example data have been plotted arbitrarily at $mU = \Delta = 0.2$ for $\mu = 0.0, 0.3, 0.6$.

and omitting parametric arguments. The function f' is given by the product

$$f'(x) = g(x)h(x)$$

with

$$\begin{aligned} g(x) &= -\frac{x}{\sqrt{x^2+\Delta^2}} \\ h(x) &= \left(\frac{1}{e^{\beta(\sqrt{x^2+\Delta^2}-\mu)}+1} - \frac{1}{e^{\beta(-\sqrt{x^2+\Delta^2}-\mu)}+1} \right) \end{aligned}$$

The following points are true:

1. The function is antisymmetric about $x = 0$. This is evident by considering that $g(x)$ is evidently antisymmetric, while $h(x)$ is symmetric.

2. $g(x)$ is monotonically descendent. Its derivative is

$$\begin{aligned} \frac{\mathrm{d}g}{\mathrm{d}x} &= -\frac{\sqrt{x^2+\Delta^2} - x^2/\sqrt{x^2+\Delta^2}}{x^2+\Delta^2} \\ &= -\frac{\Delta^2}{(x^2+\Delta^2)^{3/2}} \end{aligned}$$

   clearly always negative.

3. The product $g(x)h(x)$ is monotonically descendent at positive finite doping. To assert this, we can compute the derivative of f',

$$\frac{\mathrm{d}f'}{\mathrm{d}x} = \frac{\mathrm{d}g}{\mathrm{d}x}h(x) + g(x)\frac{\mathrm{d}h}{\mathrm{d}x}$$

   Now, evidently $h(x) > 0$ and as we said $\mathrm{d}g/\mathrm{d}x < 0$, which makes the first term negative. The second is negative too: let

$$E(x) \equiv \sqrt{x^2+\Delta^2} \qquad \Longrightarrow \qquad \frac{\mathrm{d}E}{\mathrm{d}x} = \frac{x}{E(x)} = -g(x)$$

Then:

$$\frac{\mathrm{d}h}{\mathrm{d}x} = \frac{\mathrm{d}}{\mathrm{d}x}\left[\frac{1}{e^{\beta(-E(x)-\mu)}+1} - \frac{1}{e^{\beta(E(x)-\mu)}+1}\right]$$
$$= \beta\frac{\mathrm{d}E}{\mathrm{d}x}\underbrace{\left[\frac{e^{\beta(-E(x)-\mu)}}{\left[e^{\beta(-E(x)-\mu)}+1\right]^2} + \frac{e^{\beta(E(x)-\mu)}}{\left[e^{\beta(E(x)-\mu)}+1\right]^2}\right]}_{k(x)}$$

The term in square brackets is obviously positive, $k(x) > 0$. Taking all together, we have

$$g(x)\frac{\mathrm{d}h}{\mathrm{d}x} = -\beta g^2(x)k(x) < 0 \qquad \Longrightarrow \qquad \frac{\mathrm{d}f'}{\mathrm{d}x} < 0$$

making f' descendent monotonicity proven.

These points end the proof: as anticipated, f' is a monotonically descendent function of $\epsilon_{\mathbf{k}}$, a feature that allows us to re-use the entire derivation of Sec. 2.5.2 without hesitation, and conclude that also for the AF phase

$$u^{(d)} = 0$$

while $u^{(s^*)}$ remains constrained in the limits of Eq. (3.43). Starting from a rather complicated scenario, we were able to resolve all the AF HFPs down to the pair

$$m, u^{(s^*)}$$

Perhaps not-so-excitingly, the real net effect of the non-local extension of the HM is, once again, the shrinking effect on bands. Now it is a matter of numeric analysis to assess if Mott localization takes place.

3.8 Discussion of numeric results

In this section we present numeric results. For the AF phase, the HFPs vector, as we thoroughly showed, has two components

$$\mathbf{v} = \begin{bmatrix} m \\ u^{(s^*)} \end{bmatrix} \qquad \mathbf{v} = \mathbb{R}^2$$

The resulting model RMPs are, as before:

$$\tilde{t}_{ij} \equiv \left(t - \frac{u^{(s^*)}V}{2}\right)\varphi_{ij}^{(s^*)}$$
$$\tilde{\mu} \equiv \mu + n(2zV - U)$$

In the end, the computational scenario is not so different from the normal phase. The only real difference is our choice of parametric boundaries for simulations:

$$U = 0,\ 0.05,\ 0.1,,\ \ldots,\ 8.0 \qquad V = 0,\ 0.05,\ 0.1,,\ \ldots,\ 8.0 \qquad \delta = 0.0$$

We set $\delta = 0.0$. As we anticipated in Sec. 3.5, any finite doping implies immediate disruption of commensurate antiferromagnetism, a consideration that makes any simulation at $\delta > 0$ of low physical significance. Following this consideration, we omit data at finite $\delta \neq 0$.

3.8.1 Magnetization boost

The convergence values for m computed in (U, V) plane are shown in the panels of Fig. 3.4, along with a level curves plot to identify isomagnetic curves (right panel). Note that, to increase graphic clarity in heatmaps bottom left sector, from the plots are omitted the points at $U \geq 4$ and $V \geq 6$. Tolerances on density and HFPs have been set to

$$\Delta n = 10^{-2} \qquad \Delta\mathbf{v} = \begin{bmatrix} m\colon 5\times 10^{-4} \\ u^{(s^*)}\colon 5\times 10^{-4} \end{bmatrix}$$

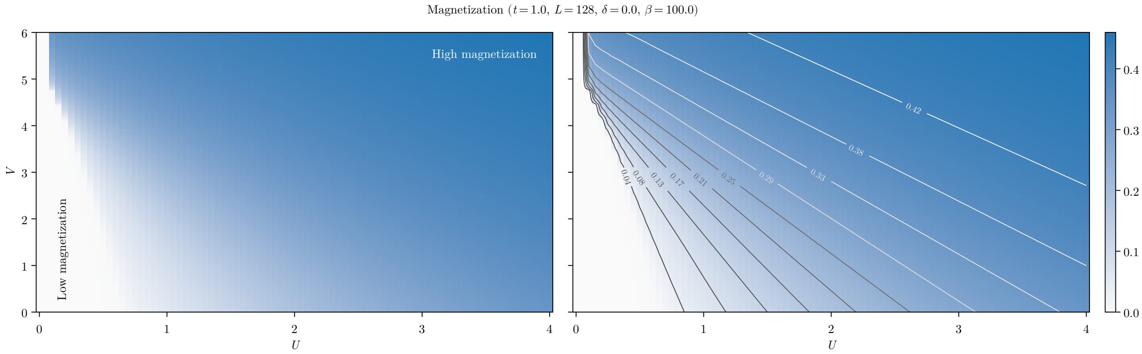
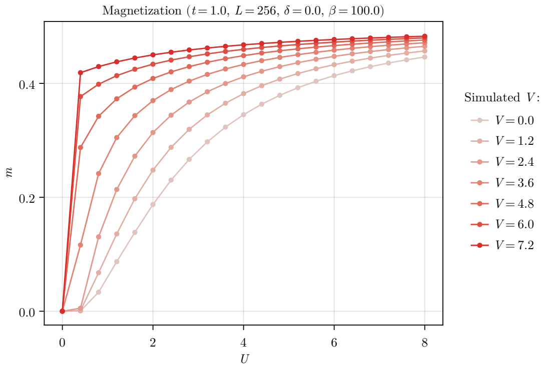
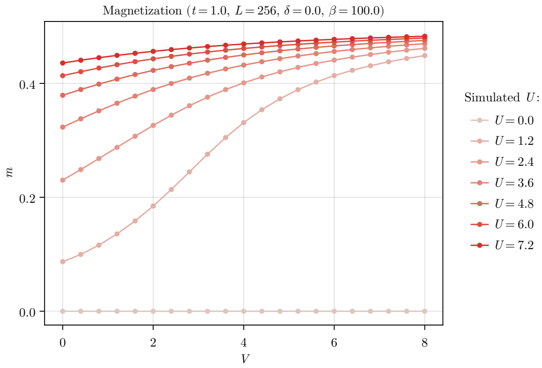


Figure 3.4 | Plot of the self-consistent convergence values for m (left panel) and isomagnetic curves computed on the same data (right panel), along which magnetization is constant. For the sake of graphic clarity, only the points at $0 \leq U \leq 4$ and $0 \leq V \leq 6$ are plotted. [Something broke at $U = 0.05$, $V \geq 5$, change with new data.]



a | U-dependency of m varying V.



b | V-dependency of m varying U.

Figure 3.5 | Plot of the self-consistent convergence values for the HFPs m and $u^{(s^*)}$ in the AF phase.

The most interesting evident visible directly from heatmaps is that $\hat{H}_V$ actually boosts magnetization. This phenomenon is intuitive and easy to explain: since the only observable net effect of the non-local interaction is to lower t by shrinking bands, at any fixed point (t, U, V) the model can be effectively mapped to

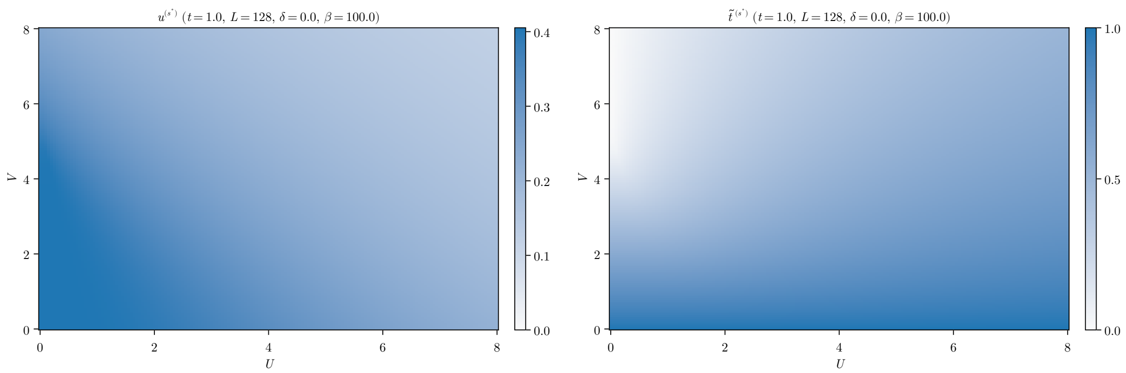
$$\text{EHM: } (t, U, V) \quad \rightarrow \quad \text{HM: } (\tilde{t}, U, 0)$$

being $\tilde{t}$ dependent on V. For the pure HM, to magnetize the system what matters is just U/t. To lower t is physically the same as to increase U, and from this point comes the magnetization precession visible in Fig. 3.4. Isomagnetic lines on the right panel show, for U significantly greater than 0, interesting linear trends whose angular coefficient depend on m itself. The low-U limit of isomagnetic lines presents some oscillations that are evidently imputable to numeric discretization of parametric space. [Why isomagnetic curves are lines? Something about $U/(V - V_\mathrm{c})$ ratio being constant?]

Data shown in Fig. 3.5 confirm clearly the aforementioned magnetization "boost" given by $\hat{H}_V$. The effect is more enhanced at low values of U. Interestingly enough, as is visible from Fig. 3.5b, magnetization does not always saturate the same way. For example, considering the curve at $U = 1.2$, magnetization varies following a somewhat tilted sigmoid shape, making boost most effective around $V \approx 2 \div 4$. Because of m saturation, this behaviour is lost at higher Us.

3.8.2 Bands shrinking

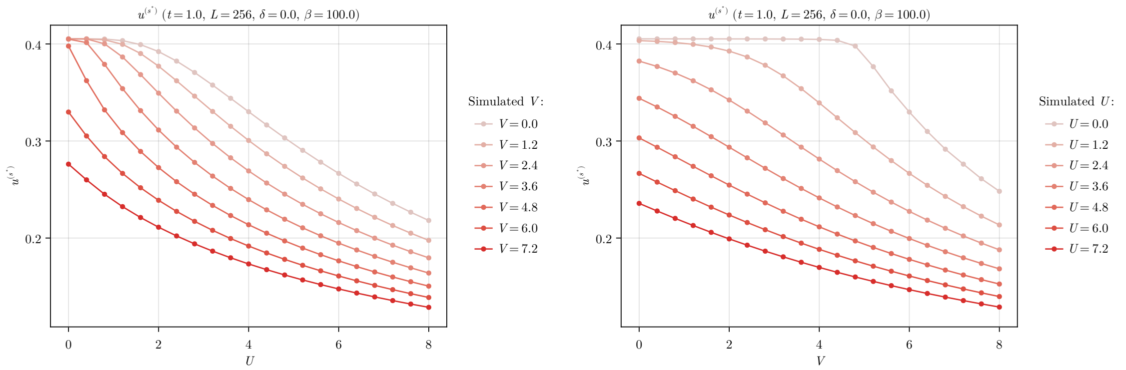
Bands shrinking effect was discussed in Sec. 3.7.1 within C_4 symmetry, and then confirmed in Sec. 3.7.4 when trying to extend to nematic structures. As was for the normal phase, we have



a | Heatmap of $u^{(s^*)}$ in UV plane.

b | Heatmap of $t^{(s^*)}$ in UV plane.

Figure 3.6 | Data for the HFP $u^{(s^*)}$ and $\tilde{t}^{(s^*)}$ varying both U and V.



a | U-dependency of $u^{(s^*)}$ varying V.

b | V-dependency of $u^{(s^*)}$ varying U.

Figure 3.7 | Plot of the self-consistent convergence values for the HFPs m and $u^{(s^*)}$ in the AF phase.

proven that $\hat{H}_V$ net effect on the hamiltonian kinetic part is to shrink bands,

$$t \quad \rightarrow \quad \tilde{t} < t$$

For the normal phase, in Sec. 2.6.1 we presented numeric results showing a Mott-localized region appears in (δ, V) plane, whose $\delta \rightarrow 0$ limit was approximately $V = 4.93t$, as shown in Fig. 2.7b.

The convergence values for the HFP $u^{(s^*)}$ are shown in Fig. 3.6a, and more in details in the two panels of Fig. 3.7. In general, $u^{(s^*)}$ decreases monotonically both in U and in V. As we already discussed in normal phase, the $U = 0$ data of Fig. 3.7b present a plateau up to a Mott-critical value. This effect is absent at positive Us. Both interactions disrupt the HFP; however, the two limits

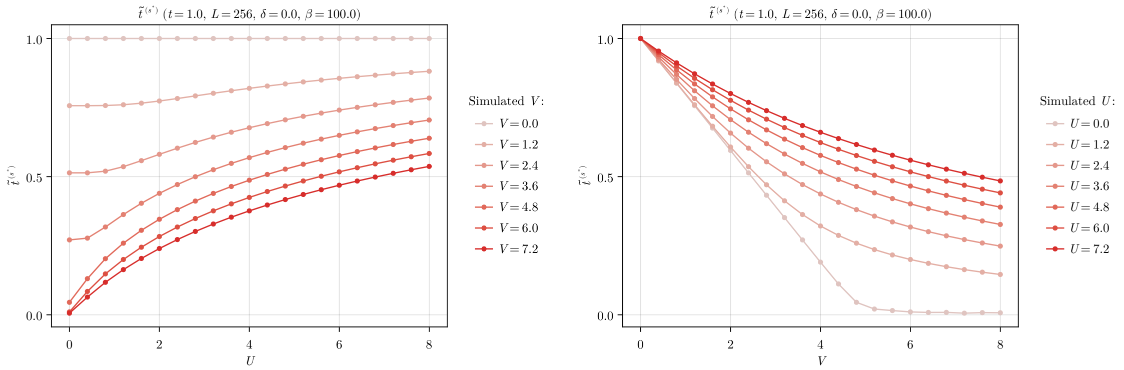
$$U \gg 1 \quad \text{and} \quad V \gg 1$$

are much different. By computing the value of $\tilde{t}^{(s^*)}$,

$$\tilde{t}^{(s^*)} = 2t - u^{(s^*)}V$$

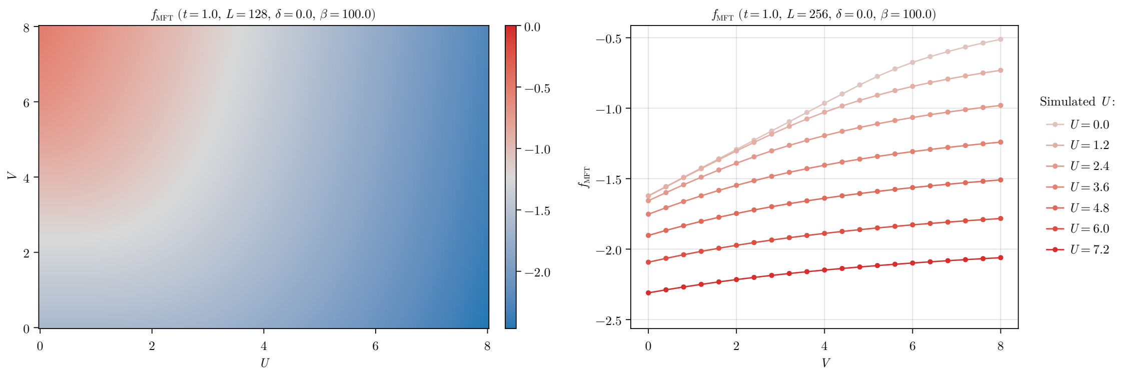
we obtain the data of Fig. 3.6b. The limit $U \gg 1$ taken at fixed V, also shown in Fig. 3.8a, provides "U-induced antiferromagnetism", with bands shrinking effect being almost irrelevant as V lowers; on the contrary, the opposite limit $V \gg 1$ taken at fixed U, also shown in Fig. 3.8b, shrinks bands strongly providing a "Mott-induced antiferromagnetism".

Interestingly, in (U, V) space there is not Mott-localized region of the kind we had in (δ, V) space for the normal phase (see Fig. 2.7b). At half filling, complete localization only occurs at $U = 0$, as can be seen in the bottom left sector of Fig. 3.8a. Any finite value of U disrupts bands flattening, thus – being the local repulsion an essential ingredient in Hubbard antiferromagnetism



a | U-dependency of $t^{(s^*)}$ varying V.   b | V-dependency of $t^{(s^*)}$ varying U.

Figure 3.8 | Plot of the self-consistent convergence values for the HFPs m and $u^{(s^*)}$ in the AF phase.



a | Heatmap of f_{MFT} in δV plane.   b | V-dependency of f_{MFT} varying δ.

Figure 3.9 | Free energy density of the AF phase.

– we can safely sustain, based on numeric evidence, that bands flattening is incompatible with a commensurate AF phase on the half-filled EHM.

3.8.3 Solution free energy

In Fig. 3.9 the solution free energy is plotted. The right panel, Fig. 3.9b, makes it becomes evident how the energetically most stable state is realized at low V, high U. The opposite limit of weak local repulsion, strong non-local attraction (top-left corner of Fig. 3.9a) instead increases f_{MFT} a lot, correctly reaching zero energy at $U = 0$, $V > 4.93t$ – the critical value discussed in Sec. 2.6.1 and computed in Eq. (2.34).

It is then evident how, even if the macroscopic magnetic properties are the same along one isomagnetic line of those plotted in Fig. 3.4, right panel, and thus the realized state exhibits an identical value of m, much different is the energetic content of the state itself. While in the $U \gg 1$, fixed V limit free energy gain comes from gap amplification – which moves down the Bogoliubov bands $\tilde{E}_{\mathbf{k}}$ –, in the $V \gg 1$, fixed U limit energies of the lower Bogoliubov band (full at low temperature) are shifted upwards via the Mott localization effect.

The AF phase is always energetically favoured at half-filling, with respect to the normal phase. Considering the energy difference,

$$\Delta f_{\mathrm{MFT}} \equiv f^{(\mathrm{N})}_{\mathrm{MFT}} - f^{(\mathrm{AF})}_{\mathrm{MFT}}$$

shown for a slightly more coarse-grained dataset in Fig. 3.10, we see immediately how $\Delta f_{\mathrm{MFT}} \geq 0$ in all (U, V) plane. Note that, in order to compare the two acquisitions, being the normal phase data of Sec. 2.6 having been simulated at $U = 0$ due to their intrinsic U independence, we obtained

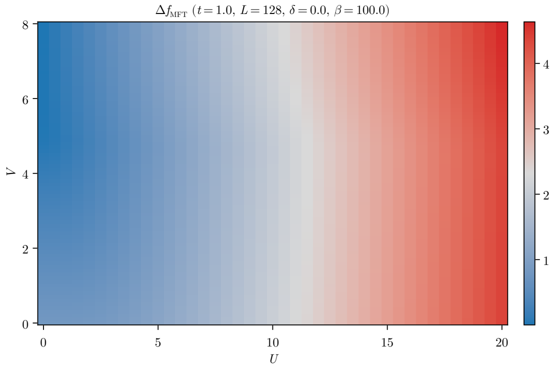


Figure 3.10 | Difference of free energies for the normal phase minus the AF phase.

a comparable dataset for the normal phase by just concatenating the filtered data for $U = 0$ at all values of U we are interested in. As we expect, in the $V \gg 1$ and low U limit the energy difference is null: in this limit the two phases tend to coincide, since at $U = 0.0$ no magnetization is physically possible. Moreover, the $U \gg 1$ limit makes the AF phase undoubtedly energetically favoured. Overall, at half filling the interesting free fermions Mott-localization effect is, at least when comparing these phases, existent but surpassed by the high energetic gain of establishing an AF structure at infinitesimal U. The situation could change at finite doping, a regime we do not investigate limitedly to AF Physics.

At low temperature and half filling, it is not worth plotting entropy density. Half-filled commensurate antiferromagnets are insulators, with two gapped Bogoliubov bands. At low temperature, electronic distribution fills completely the lower band and empties the upper one. Because of this, insulating gapped systems inherently are low entropic.

3.8.4 Fixed-point interpretation of HF solution

As a final remark discussing the mathematical structure of the algorithm, we anticipated in Sec. **??** how the reject-mix-iterate scheme upon which our HF algorithm is built can be seen as an effective search for the singular fixed point of a nonlinear map. A physical solution is represented as a HFP vector $\mathbf{v}$ such that the algorithm is a local identity,

$$\mathrm{A}(\mathbf{v}) = \mathbf{v}$$

To predict for a given initializer $\mathbf{v}_0$ the path it will follow through HFPs space is, in general, highly non-trivial due to the intrinsic nonlinearity of the algorithmic map defined by self-consistency equations for the various HFPs. An example of such nonlinearity is given in the derivation of Secs. 3.7.2, 3.7.3.

In Fig. 3.11 we sketch this concept in a practical situation. In figure are plotted various algorithmic paths through AF HFPs space, with operational setup at

$$U = 3 \qquad V = 6 \qquad L = 128 \qquad \beta = 100 \qquad \delta = 0$$

At this point, the self-consistent HF "fixed-point" solution is given by

$$m \approx 0.449 \qquad u^{(s^*)} \approx 0.212$$

indicated in panels as the black dot. The three panels show different map evolutions varying the mixing parameter within three choices, $g \in \{0.25, 0.1, 0.05\}$. We chose various initializers in the set:

$$\mathbf{v}_0 = \begin{bmatrix} m_0 \\ u_0^{(s^*)} \end{bmatrix} \quad : \quad m_0 \in \{0.2,\ 0.3,\ ...,\ 0.6\} \quad \text{and} \quad u_0^{(s^*)} \in \{0.0,\ 0.1,\ ...,\ 0.4\}$$

In figures, different colors indicate different evolutions, while the tip size indicates the magnitude of the algorithmic step in passing from one iteration to the following.

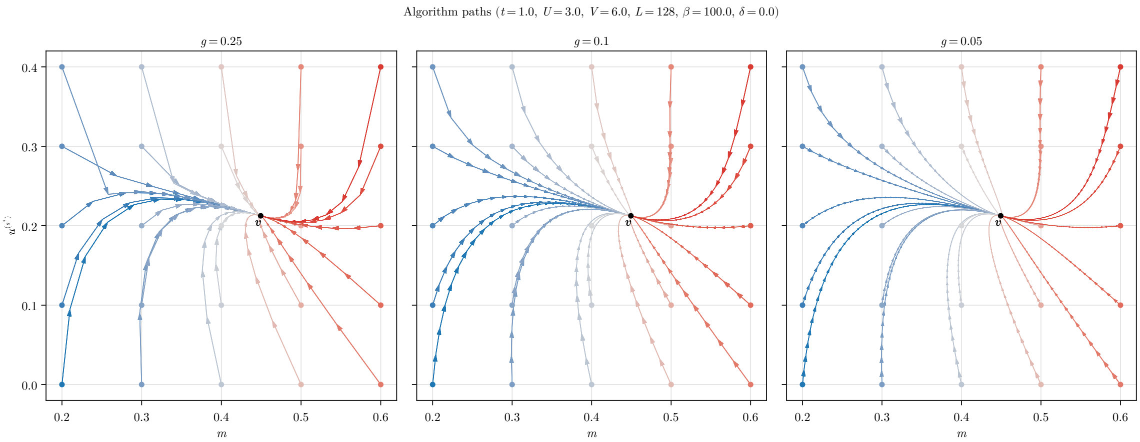


Figure 3.11 Quiver plot of algorithmic evolution in HFPs space at fixed model parametrization. The black dot $\mathbf{v}$ represents the nonlinear map fixed point, while the colored paths the sequences of HFPs coordinates simulated via iterative HF procedure, starting from different HFPs initializers (colored dots).

All paths are able to converge to $\mathbf{v}$, and the quantity of steps necessary to reach the fixed point, a sort of attractor, increases as we lower g. To select the correct g turns out to be the real algorithm fine-tuning knob: as we saw for the normal phase, some model parametrization could lead to unavoidable algorithmic instabilities. The problem is that we do not know beforehand where the instabilities will occur, and how. The correct g is the compromise between keeping low the number of steps (larger mixing) and moving slow enough to not diverge when encountering unstable points (smaller mixing). By recursively fine-tuning g repeating non-converged simulation with more care, we are essentially making the nonlinear map "softer" and curvier, as is in the right panel of Fig. 3.11. It's not always resource convenient to do so, so for most simulations we stick to a rougher map (like the left panel one) by setting $g = 0.5$.

Appendix A

Mean-Field Theory in Hubbard lattices

In this Appendix the MFT solutions to the conventional Hubbard hamiltonian of Eq. (**??**),

$$\hat{H} = -t \sum_{\langle ij \rangle} \sum_{\sigma} \left[\hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} + \text{h.c.} \right] + U \sum_{i} \hat{n}_{i\uparrow} \hat{n}_{i\downarrow} \qquad t, U > 0 \tag{A.1}$$

are described. The discussion is limited to the two-dimensional square lattice. The results of ths appendix are relevant and used throughout the text as the groundwork to be extended by the means of MFT applied to the non-local part of the EHM.

A.1 Antiferromagnetic phase

We work in a commensurate AF phase ordering, which is, we explicitly break symmetry by realizing a SDW phase with spin density unevenly distributed on sites with periodicity of two sites, each direction. Recalling the discussion of Sec. **??**, AF long-range ordering we deal with here is described by the composite SSB

$$\text{SU}(2) \overset{\text{SSB}}{\rightarrow} \text{U}^{z}(1)$$
$$\text{T}^{(x)}_{1} \otimes \text{T}^{(y)}_{1} \overset{\text{SSB}}{\rightarrow} \text{T}^{(x)}_{2} \otimes \text{T}^{(y)}_{2}$$

Recalling Tab. **??**, for such a physical phase the only Wick contraction channel available is the Hartree one. Let us proceed in such sense. As in Eqns. (3.3), (3.4), it is useful to define the density and AF fluctuation operators:

$$\hat{n}_{\mathbf{k}\sigma} \equiv \hat{c}^{\dagger}_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} \tag{A.2}$$
$$\hat{\psi}_{\mathbf{k}\sigma} \equiv \hat{c}^{\dagger}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \hat{c}_{\mathbf{k}\sigma} \tag{A.3}$$

whose EVs are relevant order parameters for the normal and AF phases.

A.1.1 MFT treatment of the Hartree channel

In terms of the physical magnetization m, the MFT Ansatz is given by

$$\langle \hat{n}_{\mathbf{r}\sigma} \rangle = n - (-1)^{x+y+\delta_{\sigma=\uparrow}} m \tag{A.4}$$

Decoupling the HM hamiltonian and substituting,

$$\begin{aligned} \hat{H} &\simeq -t \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \left[\hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} + \text{h.c.} \right] + U \sum_{\mathbf{r}} \left[\hat{n}_{\mathbf{r}\uparrow} \langle \hat{n}_{\mathbf{r}\downarrow} \rangle + \langle \hat{n}_{\mathbf{r}\uparrow} \rangle \hat{n}_{\mathbf{r}\downarrow} - \langle \hat{n}_{\mathbf{r}\uparrow} \rangle \langle \hat{n}_{\mathbf{r}\downarrow} \rangle \right] \\ &= -t \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \left[\hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} + \text{h.c.} \right] + nU \sum_{\mathbf{r}} \left[\hat{n}_{\mathbf{r}\uparrow} + \hat{n}_{\mathbf{r}\downarrow} \right] - mU \sum_{\mathbf{r}} (-1)^{x+y} \left[\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow} \right] - E^{(\text{AF})}_{\text{H}/U} \end{aligned} \tag{A.5}$$

with $E^{(\mathrm{AF})}_{\mathrm{H}/U}$ the HM Hartree contraction energy,

$$
\begin{aligned}
E^{(\mathrm{AF})}_{\mathrm{H}/U} &\equiv U \sum_{\mathbf{r}} \langle \hat{n}_{\mathbf{r}\uparrow} \rangle \langle \hat{n}_{\mathbf{r}\downarrow} \rangle \\
&= U \sum_{\mathbf{r}} \left(n + (-1)^{x+y} m\right) \left(n - (-1)^{x+y} m\right) \\
&= L_x L_y \times U \left(n^2 - m^2\right) \qquad \text{(A.6)}
\end{aligned}
$$

By a Fourier transform, the phase factor of Eq. (A.5) can be absorbed in the destruction operator inside of $\hat{n}_{\mathbf{r}\sigma}$:

$$
\begin{aligned}
\sum_{\mathbf{r}} (-1)^{x+y} \hat{n}_{\mathbf{r}\sigma} &= \sum_{\mathbf{r}} (-1)^{x+y} \hat{c}^{\dagger}_{\mathbf{r}\sigma} \hat{c}_{\mathbf{r}\sigma} \\
&\overset{\mathcal{F}}{=} \sum_{\mathbf{r}} e^{i\boldsymbol{\pi}\cdot\mathbf{r}} \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k}\in \mathrm{BZ}} e^{i\mathbf{k}\cdot\mathbf{r}} \hat{c}^{\dagger}_{\mathbf{k}\sigma} \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k}'\in \mathrm{BZ}} e^{-i\mathbf{k}'\cdot\mathbf{r}} \hat{c}_{\mathbf{k}'\sigma} \\
&= \sum_{\mathbf{k}\in \mathrm{BZ}} \sum_{\mathbf{k}'\in \mathrm{BZ}} \hat{c}^{\dagger}_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}'\sigma} \frac{1}{L_x L_y} \sum_{\mathbf{r}} e^{-i[\mathbf{k}' - (\mathbf{k}+\boldsymbol{\pi})]\cdot\mathbf{r}} \\
&= \sum_{\mathbf{k}\in \mathrm{BZ}} \hat{c}^{\dagger}_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \qquad \text{(A.7)}
\end{aligned}
$$

where $\boldsymbol{\pi} = (\pi, \pi)$. It follows:

$$
mU \sum_{\mathbf{r}} (-1)^{x+y} \left[\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}\right] = mU \sum_{\mathbf{k}\in \mathrm{BZ}} \left[\hat{c}^{\dagger}_{\mathbf{k}\uparrow} \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\uparrow} - \hat{c}^{\dagger}_{\mathbf{k}\downarrow} \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\downarrow}\right]
$$

Note that the Hartree channel renormalizes chemical potential,

$$
nU \sum_{\mathbf{r}} \left[\hat{n}_{\mathbf{r}\uparrow} + \hat{n}_{\mathbf{r}\downarrow}\right] = nU \times \hat{N} \qquad \Longrightarrow \qquad \mu \to \tilde{\mu} \equiv \mu - nU
$$

an effect that is present also in the EHM. The kinetic part of the hamiltonian transforms as in Eq. (1.28),

$$
-t \sum_{\langle \mathbf{r}\mathbf{r}'\rangle} \sum_{\sigma} \left[\hat{c}^{\dagger}_{\mathbf{r}\sigma} \hat{c}_{\mathbf{r}'\sigma} + \mathrm{h.c.}\right] \overset{\mathcal{F}}{=} \sum_{\mathbf{k}\in \mathrm{BZ}} \sum_{\sigma} \epsilon_{\mathbf{k}} \hat{c}^{\dagger}_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma}
$$

being $\epsilon_{\mathbf{k}}$ the bare bands of Eq. (1.29).

Take everything together. Using the property:

$$
\begin{aligned}
\sum_{\mathbf{k}\in \mathrm{BZ}} \hat{c}^{\dagger}_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} &= \sum_{\mathbf{k}\in \mathrm{MBZ}} \left[\hat{c}^{\dagger}_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} + \hat{c}^{\dagger}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \hat{c}_{\mathbf{k}+2\boldsymbol{\pi}\sigma}\right] \\
&= \sum_{\mathbf{k}\in \mathrm{MBZ}} \left[\hat{c}^{\dagger}_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} + \hat{c}^{\dagger}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \hat{c}_{\mathbf{k}\sigma}\right] \qquad \text{(A.8)}
\end{aligned}
$$

and reduce Eq. (A.5) to its reciprocal form

$$
\hat{H} \simeq \sum_{\mathbf{k}\in \mathrm{MBZ}} \sum_{\sigma} \epsilon_{\mathbf{k}} \left[\hat{n}_{\mathbf{k}\sigma} + \hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma}\right] - mU \sum_{\mathbf{k}\in \mathrm{MBZ}} \sum_{\sigma} (-1)^{\delta_{\sigma=\downarrow}} \left[\hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}^{\dagger}_{\mathbf{k}\sigma}\right] + nU\hat{N} - E^{(\mathrm{AF})}_{\mathrm{H}/U}
$$

A.1.2 More on AF Ansatz

Before moving on, let us give a little detail on the AF Ansatz. The decoupling in Hartree sector of the density-density interaction can be treated with more care: denoting by d.c. the double counting terms we ignore,

$$
\begin{aligned}
\hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}\downarrow} - \mathrm{d.c.} &= \hat{n}_{\mathbf{r}\uparrow} \langle \hat{n}_{\mathbf{r}\downarrow} \rangle + \langle \hat{n}_{\mathbf{r}\uparrow} \rangle \hat{n}_{\mathbf{r}\downarrow} \\
&= \hat{n}_{\mathbf{r}\uparrow} \frac{\langle \hat{n}_{\mathbf{r}\downarrow} \rangle + \langle \hat{n}_{\mathbf{r}\uparrow} \rangle}{2} + \hat{n}_{\mathbf{r}\uparrow} \frac{\langle \hat{n}_{\mathbf{r}\downarrow} \rangle - \langle \hat{n}_{\mathbf{r}\uparrow} \rangle}{2} + \frac{\langle \hat{n}_{\mathbf{r}\uparrow} \rangle + \langle \hat{n}_{\mathbf{r}\uparrow} \rangle}{2} \hat{n}_{\mathbf{r}\downarrow} + \frac{\langle \hat{n}_{\mathbf{r}\uparrow} \rangle - \langle \hat{n}_{\mathbf{r}\uparrow} \rangle}{2} \hat{n}_{\mathbf{r}\downarrow} \\
&= \frac{\langle \hat{n}_{\mathbf{r}\uparrow} \rangle + \langle \hat{n}_{\mathbf{r}\downarrow} \rangle}{2} \left(\hat{n}_{\mathbf{r}\uparrow} + \hat{n}_{\mathbf{r}\downarrow}\right) - \frac{\langle \hat{n}_{\mathbf{r}\uparrow} \rangle - \langle \hat{n}_{\mathbf{r}\downarrow} \rangle}{2} \left(\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}\right)
\end{aligned}
$$

Our commensurate AF Ansatz essentially states that each site has the same spin-averaged density,

$$n \equiv \frac{\langle \hat{n}_{\mathbf{r}\uparrow} \rangle + \langle \hat{n}_{\mathbf{r}\downarrow} \rangle}{2} \qquad \text{is } \mathbf{r}\text{-independent}$$

while the spin averaged unbalance has a two-sites periodicity

$$m \equiv (-1)^{x+y} \frac{\langle \hat{n}_{\mathbf{r}\uparrow} \rangle - \langle \hat{n}_{\mathbf{r}\downarrow} \rangle}{2} \qquad \text{is } \mathbf{r}\text{-independent}$$

This way we reduce translational invariance. Of course, if the two quantities above are $\mathbf{r}$-independent indeed, it must hold

$$n = \frac{1}{L_x L_y} \sum_{\mathbf{r}} \frac{\langle \hat{n}_{\mathbf{r}\uparrow} \rangle + \langle \hat{n}_{\mathbf{r}\downarrow} \rangle}{2} \tag{A.9}$$

$$m = \frac{1}{L_x L_y} \sum_{\mathbf{r}} (-1)^{x+y} \frac{\langle \hat{n}_{\mathbf{r}\uparrow} \rangle - \langle \hat{n}_{\mathbf{r}\downarrow} \rangle}{2} \tag{A.10}$$

While from the first equation we gain an operational method to compute density, from the second we get a magnetization self-consistency equation.

A.1.3 Nambu representation of the AF phase and diagonalization

We now map the reciprocal version of the HM onto a simpler problem. Throughout this section we use the notation $\epsilon_{\mathbf{k}} \to \epsilon_{\mathbf{k}\sigma}$ with identical meaning. Let the Nambu spinors:

$$\hat{\Psi}_{\mathbf{k}\sigma} \equiv \begin{bmatrix} \hat{c}_{\mathbf{k}\sigma} \\ \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \end{bmatrix}$$

Let also the gap function:

$$\Delta_{\mathbf{k}\sigma} = (-1)^{\delta_{\sigma=\downarrow}} mU$$

which gives the matricial decomposition

$$\hat{H} \simeq \sum_{\mathbf{k}\in\mathrm{MBZ}} \sum_{\sigma} \hat{\Psi}^{\dagger}_{\mathbf{k}\sigma} h_{\mathbf{k}\sigma} \hat{\Psi}_{\mathbf{k}\sigma} - E^{(\mathrm{AF})}_{\mathrm{H}/U} \quad \text{being} \quad h_{\mathbf{k}\sigma} \equiv \begin{bmatrix} \epsilon_{\mathbf{k}\sigma} & -\Delta^*_{\mathbf{k}\sigma} \\ -\Delta_{\mathbf{k}\sigma} & -\epsilon_{\mathbf{k}\sigma} \end{bmatrix} \tag{A.11}$$

Notice: the Nambu hamiltonian is a 2×2 matrix over the MBZ – which is half the full BZ, coherently with a solution which essentially bipartites the lattice giving back a double sized unit cell.

The system ground-state is obtained by the means of a Bogoliubov rotation. The hamiltonian maps onto the simple one of a spin in a magnetic field (the gap is purely real here),

$$h_{\mathbf{k}\sigma} = \epsilon_{\mathbf{k}\sigma} \tau^z - \Delta_{\mathbf{k}\sigma} \tau^x$$

being τ^α the Pauli matrices. Then, defining

$$\hat{s}^{\alpha}_{\mathbf{k}\sigma} \equiv \hat{\Psi}^{\dagger}_{\mathbf{k}\sigma} \tau^{\alpha} \hat{\Psi}_{\mathbf{k}\sigma} \quad \text{and} \quad \mathbf{b}_{\mathbf{k}\sigma} \equiv \begin{bmatrix} -\Delta_{\mathbf{k}\sigma} \\ 0 \\ \epsilon_{\mathbf{k}\sigma} \end{bmatrix}$$

one gets:

$$\hat{H} \simeq \sum_{\mathbf{k}\in\mathrm{MBZ}} \sum_{\sigma} \mathbf{b}_{\mathbf{k}\sigma} \cdot \hat{\mathbf{s}}_{\mathbf{k}\sigma} - E^{(\mathrm{AF})}_{\mathrm{H}/U} \quad \text{where} \quad \hat{\mathbf{s}}_{\mathbf{k}\sigma} = \begin{bmatrix} \hat{s}^x_{\mathbf{k}\sigma} \\ \hat{s}^y_{\mathbf{k}\sigma} \\ \hat{s}^z_{\mathbf{k}\sigma} \end{bmatrix} \tag{A.12}$$

The hamiltonian represents a system of spins subject to local magnetic fields, each tilted by an angle $\tan 2\theta_{\mathbf{k}\sigma} = \Delta_{\mathbf{k}\sigma}/\epsilon_{\mathbf{k}\sigma}$, as sketched in Fig. A.1. Evidently the following relations, also reported in Tab. 3.2, hold:

$$\hat{s}^x_{\mathbf{k}\sigma} = \hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}^{\dagger}_{\mathbf{k}\sigma} \tag{A.13}$$

$$\hat{s}^y_{\mathbf{k}\sigma} = i \left(\hat{\psi}_{\mathbf{k}\sigma} - \hat{\psi}^{\dagger}_{\mathbf{k}\sigma} \right) \tag{A.14}$$

$$\hat{s}^z_{\mathbf{k}\sigma} = \hat{n}_{\mathbf{k}\sigma} - \hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \tag{A.15}$$

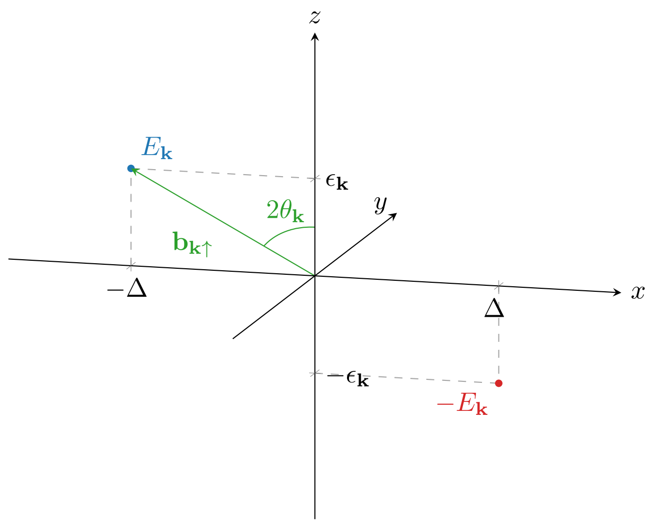


Figure A.1 | Pseudo-magnetic field originating from mean-field treatment of the square Hubbard hamiltonian. Here, only the $\sigma = \uparrow$ situation is plotted.

Let also:

$$\begin{aligned} \hat{\mathbb{I}}_{\mathbf{k}\sigma} &\equiv \hat{\Psi}^{\dagger}_{\mathbf{k}\sigma} \mathbb{I}_{2\times 2} \hat{\Psi}_{\mathbf{k}\sigma} \\ &= \hat{n}_{\mathbf{k}\sigma} + \hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \end{aligned} \tag{A.16}$$

Diagonalization of each $h_{\mathbf{k}\sigma}$ is obtained trivially by a simple rotation around the y axis:

$$d_{\mathbf{k}\sigma} = W_{\mathbf{k}\sigma} h_{\mathbf{k}\sigma} W^{\dagger}_{\mathbf{k}\sigma} \quad \text{with} \quad d_{\mathbf{k}\sigma} \equiv \begin{bmatrix} E_{\mathbf{k}\sigma} & \\ & -E_{\mathbf{k}\sigma} \end{bmatrix}$$

and where the diagonalization unitary matrix is simply a rotation by an angle $2\theta_{\mathbf{k}\sigma}$ around the y axis, counter-clockwise

$$\begin{aligned} W_{\mathbf{k}\sigma} &\equiv e^{-i\theta_{\mathbf{k}\sigma}\tau^y} \\ &= \cos\theta_{\mathbf{k}\sigma} - i\sin\theta_{\mathbf{k}\sigma}\tau^y \\ &= \begin{bmatrix} \cos\theta_{\mathbf{k}\sigma} & -\sin\theta_{\mathbf{k}\sigma} \\ \sin\theta_{\mathbf{k}\sigma} & \cos\theta_{\mathbf{k}\sigma} \end{bmatrix} \end{aligned} \tag{A.17}$$

Eigenvalues are:

$$E_{\mathbf{k}\sigma} \equiv \sqrt{\epsilon^2_{\mathbf{k}\sigma} + |\Delta_{\mathbf{k}\sigma}|^2} \tag{A.18}$$

At any finite temperature, the ground-state of such a system is well-known: a thermal superposition of "down" spins with "up" spins. In the pseudo-spin picture, at each "site" ($\mathbf{k}\sigma$), the lowest-energy state is the one anti-aligned with the pseudo-magnetic field. It is useful now to define the thermal factors,

$$\mathrm{Th}^{(\pm)}_{\mathbf{k}\sigma}[\beta,\mu] \equiv f\left(-E_{\mathbf{k}\sigma};\beta,\mu\right) \pm f\left(E_{\mathbf{k}\sigma};\beta,\mu\right) \tag{A.19}$$

that will appear quite often. Then it's immediate to derive the various expectation values.

$$\langle \hat{s}^x_{\mathbf{k}\sigma} \rangle = \frac{\mathrm{Re}\{\Delta_{\mathbf{k}\sigma}\}}{E_{\mathbf{k}\sigma}} \mathrm{Th}^{(-)}_{\mathbf{k}\sigma}[\beta,\mu] \tag{A.20}$$

$$\langle \hat{s}^y_{\mathbf{k}\sigma} \rangle = \frac{\mathrm{Im}\{\Delta_{\mathbf{k}\sigma}\}}{E_{\mathbf{k}\sigma}} \mathrm{Th}^{(-)}_{\mathbf{k}\sigma}[\beta,\mu] \tag{A.21}$$

$$\langle \hat{s}^z_{\mathbf{k}\sigma} \rangle = -\frac{\epsilon_{\mathbf{k}\sigma}}{E_{\mathbf{k}\sigma}} \mathrm{Th}^{(-)}_{\mathbf{k}\sigma}[\beta,\mu] \tag{A.22}$$

Note that considering the gran canonical hamiltonian $\hat{K} \equiv \hat{H} - \mu\hat{N}$ and using Eq. (A.16) the diagonalizing matrix remains the same, being

$$-\mu\hat{N} = \sum_{\mathbf{k}\in\text{MBZ}} \begin{bmatrix} -\mu & \\ & -\mu \end{bmatrix} \quad \Longrightarrow \quad W_{\mathbf{k}\sigma}\left[h_{\mathbf{k}\sigma} - \mu\mathbb{I}_{2\times 2}\right] W^{\dagger}_{\mathbf{k}\sigma} = d_{\mathbf{k}\sigma} - \mu\mathbb{I}_{2\times 2} \tag{A.23}$$

thus for non-null chemical potential (i.e. finite doping) the eigenvalues are just shifted, $\pm E_{\mathbf{k}} - \mu$. In other words, the new bands retain the same chemical potential of the bare bands. Notice also that the presence of an absolute value makes the eigenvalues independent of the σ index. Eigenvector fermion operators are obtained simply as:

$$\hat{\Gamma}_{\mathbf{k}\sigma} = W_{\mathbf{k}\sigma}\hat{\Psi}_{\mathbf{k}\sigma} \tag{A.24}$$

The $\hat{\Gamma}$ spinor operators are effectively free fermionic spinor fields and as such must be treated. Explicitly,

$$\begin{aligned}\hat{\Gamma}_{\mathbf{k}\sigma} &= \begin{bmatrix} \cos\theta_{\mathbf{k}\sigma} & -\sin\theta_{\mathbf{k}\sigma} \\ \sin\theta_{\mathbf{k}\sigma} & \cos\theta_{\mathbf{k}\sigma} \end{bmatrix} \begin{bmatrix} \hat{c}_{\mathbf{k}\sigma} \\ \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \end{bmatrix} \\ &= \begin{bmatrix} \cos\theta_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} - \sin\theta_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \\ \sin\theta_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} + \cos\theta_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \end{bmatrix} \equiv \begin{bmatrix} \hat{\gamma}^{(+)}_{\mathbf{k}\sigma} \\ \hat{\gamma}^{(-)}_{\mathbf{k}\sigma} \end{bmatrix}\end{aligned} \tag{A.25}$$

The consequence of Eq. (A.23) is interesting by itself: within this HF solution, the gap opens always at $E = 0$. In other words, the bands sketched in Fig. A.2b retain the same geometric features regardless of the doping, what changes is the bands width and the gap between them. The very key point here is that, being the two bands symmetric (thus containing each half of the electronic states) and defined over the MBZ, any finite doping $\delta > 0$ will produce a metallic behaviour, making chemical potential crossing the upper band. It is widely observed and confirmed by more refined numerical procedures, however, that AF establishes an insulating phase – here only reproduced at half-filling, where chemical potential lies within the gap. As will be detailed in A.1.8, the commensurate nature of AF SSB at wavevector $\boldsymbol{\pi}$ of this self-consistent solution is highly unstable and atypical at finite doping [13].

A.1.4 Explicit zero-temperature AF ground state at half-filling

The AF ground state can be expressed easily at $n = 1/2$ and $\beta \to +\infty$. Let $|\Uparrow_{\mathbf{k}\sigma}\rangle$ and $|\Downarrow_{\mathbf{k}\sigma}\rangle$ be respectively the "up" and "down" $\mathbf{k}$-th pseudo-spin states. Let $\left|\overline{\Downarrow}_{\mathbf{k}\sigma}\right\rangle$ be the "down" state in the tilted frame. The ground state is given by

$$|\Psi\rangle \equiv \bigotimes_{\sigma} \bigotimes_{\mathbf{k}\in\text{MBZ}} \left|\overline{\Downarrow}_{\mathbf{k}\sigma}\right\rangle \tag{A.26}$$

We can link the "down" states via an inverse rotation,

$$\begin{aligned}\left|\overline{\Downarrow}_{\mathbf{k}\sigma}\right\rangle &= e^{-i\theta_{\mathbf{k}\sigma}\hat{s}^{y}_{\mathbf{k}\sigma}} |\Downarrow_{\mathbf{k}\sigma}\rangle \\ &= \cos\theta_{\mathbf{k}\sigma} |\Downarrow_{\mathbf{k}\sigma}\rangle - i\sin\theta_{\mathbf{k}\sigma}\hat{s}^{y}_{\mathbf{k}\sigma} |\Downarrow_{\mathbf{k}\sigma}\rangle \\ &= \cos\theta_{\mathbf{k}\sigma} |\Downarrow_{\mathbf{k}\sigma}\rangle - \sin\theta_{\mathbf{k}\sigma}(\hat{s}^{+}_{\mathbf{k}\sigma} - \hat{s}^{-}_{\mathbf{k}\sigma}) |\Downarrow_{\mathbf{k}\sigma}\rangle \\ &= \cos\theta_{\mathbf{k}\sigma} |\Downarrow_{\mathbf{k}\sigma}\rangle - \sin\theta_{\mathbf{k}\sigma} |\Uparrow_{\mathbf{k}\sigma}\rangle\end{aligned} \tag{A.27}$$

having we used

$$\hat{s}^{\pm}_{\mathbf{k}\sigma} \equiv \frac{\hat{s}^{x}_{\mathbf{k}\sigma} \pm i\hat{s}^{y}_{\mathbf{k}\sigma}}{2}$$

Use now Eqns. (A.16) and (A.15),

$$\frac{\hat{\mathbb{I}}_{\mathbf{k}\sigma} + \hat{s}^{z}_{\mathbf{k}\sigma}}{2} = \hat{n}_{\mathbf{k}\sigma} \qquad \frac{\hat{\mathbb{I}}_{\mathbf{k}\sigma} - \hat{s}^{z}_{\mathbf{k}\sigma}}{2} = \hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma}$$

Evaluating the above operators on the "up" and "down" states, we get immediately

$$\begin{aligned} \frac{1}{2}\langle \Uparrow_{\mathbf{k}\sigma}|\hat{\mathbb{I}}_{\mathbf{k}\sigma} + \hat{s}^z_{\mathbf{k}\sigma}|\Uparrow_{\mathbf{k}\sigma}\rangle &= 1 \qquad & \frac{1}{2}\langle \Uparrow_{\mathbf{k}\sigma}|\hat{\mathbb{I}}_{\mathbf{k}\sigma} - \hat{s}^z_{\mathbf{k}\sigma}|\Uparrow_{\mathbf{k}\sigma}\rangle &= 0 \\ \frac{1}{2}\langle \Downarrow_{\mathbf{k}\sigma}|\hat{\mathbb{I}}_{\mathbf{k}\sigma} + \hat{s}^z_{\mathbf{k}\sigma}|\Downarrow_{\mathbf{k}\sigma}\rangle &= 0 \qquad & \frac{1}{2}\langle \Downarrow_{\mathbf{k}\sigma}|\hat{\mathbb{I}}_{\mathbf{k}\sigma} - \hat{s}^z_{\mathbf{k}\sigma}|\Downarrow_{\mathbf{k}\sigma}\rangle &= 1 \end{aligned}$$

Thus, up to a phase

$$|\Uparrow_{\mathbf{k}\sigma}\rangle \sim \hat{c}^\dagger_{\mathbf{k}\sigma}|\Omega_{\mathbf{k}\sigma}\rangle \qquad |\Downarrow_{\mathbf{k}\sigma}\rangle \sim \hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma}|\Omega_{\mathbf{k}\sigma}\rangle$$

being $|\Omega_{\mathbf{k}\sigma}\rangle$ the $(\mathbf{k},\sigma)$ Hilbert subspace vacuum.

It follows that composing Eqns. (A.26) and (A.27) the ground state $|\Psi\rangle$ can be written as:

$$|\Psi\rangle = \bigotimes_{\sigma}\bigotimes_{\mathbf{k}\in\mathrm{MBZ}}\left[\sin\theta_{\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{k}\sigma} + e^{2i\zeta_{\mathbf{k}\sigma}}\cos\theta_{\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma}\right]|\Omega_{\mathbf{k}\sigma}\rangle \tag{A.28}$$

having we absorbed in the relative phase $e^{2i\zeta_{\mathbf{k}\sigma}}$ the two states phases differences as well as the original sign difference of Eq. (A.27). Now, inserting the explicit form of the ground state from equation above:

$$\begin{aligned} \langle \overline{\Downarrow}_{\mathbf{k}\sigma}|\hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}^\dagger_{\mathbf{k}\sigma}|\overline{\Downarrow}_{\mathbf{k}\sigma}\rangle &= \left(e^{2i\zeta_{\mathbf{k}\sigma}} + e^{-2i\zeta_{\mathbf{k}\sigma}}\right)\sin\theta_{\mathbf{k}\sigma}\cos\theta_{\mathbf{k}\sigma} \\ &= \cos 2\zeta_{\mathbf{k}\sigma}\sin 2\theta_{\mathbf{k}\sigma} \end{aligned}$$

but from Eq. (A.13)

$$\begin{aligned} \langle \overline{\Downarrow}_{\mathbf{k}\sigma}|\hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}^\dagger_{\mathbf{k}\sigma}|\overline{\Downarrow}_{\mathbf{k}\sigma}\rangle &= \langle \overline{\Downarrow}_{\mathbf{k}\sigma}|\hat{s}^x_{\mathbf{k}\sigma}|\overline{\Downarrow}_{\mathbf{k}\sigma}\rangle \\ &= \sin 2\theta_{\mathbf{k}\sigma} \end{aligned}$$

since the pseudofield has no y component. This implies necessarily

$$\cos 2\zeta_{\mathbf{k}\sigma} \quad \Longrightarrow \quad \zeta_{\mathbf{k}\sigma} = k\pi \quad \text{with} \quad k\in\mathbb{Z}$$

and lets us conclude from Eq. (A.28)

$$|\Psi\rangle = \bigotimes_{\sigma}\bigotimes_{\mathbf{k}\in\mathrm{MBZ}}\left[\sin\theta_{\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{k}\sigma} + \cos\theta_{\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma}\right]|\Omega_{\mathbf{k}\sigma}\rangle \tag{A.29}$$

For low-temperature physics, we can expect the ground-state density matrix to be highly dominated by this BCS-like zero-temperature ground-state, which highlights the essential features of the system.

Finally, using Eq. (A.25), we check that correctly the ground-state describes a perfect degenerate $\gamma^{(-)}$ quasiparticles Fermi gas,

$$|\Psi\rangle = \bigotimes_{\sigma}\bigotimes_{\mathbf{k}\in\mathrm{MBZ}}\gamma^{(-)\dagger}_{\mathbf{k}\sigma}|\Omega_{\mathbf{k}\sigma}\rangle \tag{A.30}$$

The AF phase can be thought of as a long-range coordination between fermionic states related by a $\boldsymbol{\pi}$ shift in reciprocal space, producing a new free gas where all interactions have been "condensed" in these AF pairs.

A.1.5 Spin degeneracy removal

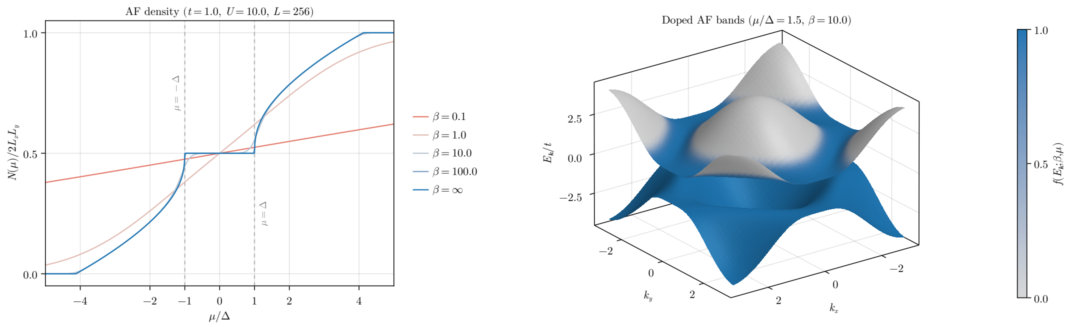
Up to now, we wrote explicitly the σ index. However,

$$\epsilon_{\mathbf{k}\uparrow} = \epsilon_{\mathbf{k}\downarrow} \qquad \Delta_{\mathbf{k}\uparrow} = -\Delta_{\mathbf{k}\downarrow}$$

thus $E_{\mathbf{k}\uparrow} = E_{\mathbf{k}\downarrow}$. Since the gap function sign does has not a measurable impact, we might as well take the spin index as a pure degeneracy and from now on work in the $\sigma = \uparrow$ sector, dropping the σ index. We can further simplify by dropping all indices from the gap function,

$$\Delta_{\mathbf{k}\uparrow} = mU \equiv \Delta$$

being the latter purely s-wave. This is a property of the pure HM: for the EHM, the gap acquires a non-trivial topological structure.



a | Electronic density.

b | Example doped bands.

Figure A.2 | Left panel: electronic density in the AF phase as a function of the chemical potential, measured in relation to the gap $\Delta = mU$, at various temperatures. Right panel: example doped bands, with $\mu/\Delta = 1.5$ set arbitrarily and temperature set to $\beta = 10$ to enhance smearing effects around FS. Single-state filling is proportional to color magnitude.

A.1.6 Chemical potential and system density

Using the fact that

$$\left\langle \hat{\Gamma}^\dagger_{\mathbf{k}\sigma} \hat{\Gamma}_{\mathbf{k}\sigma} \right\rangle = \left\langle \hat{\Psi}^\dagger_{\mathbf{k}\sigma} \hat{\Psi}_{\mathbf{k}\sigma} \right\rangle$$

due to the $W_{\mathbf{k}\sigma}$ unitarity properties, the total number of electrons is given by (also recalling Eq. (A.9))

$$\begin{aligned} N &= \sum_{\mathbf{k}\sigma} \langle \hat{n}_{\mathbf{k}\sigma} \rangle \\ &= \sum_{\sigma} \sum_{\mathbf{k}\in\mathrm{MBZ}} \langle \hat{n}_{\mathbf{k}\sigma} + \hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \rangle \\ &= \sum_{\sigma} \sum_{\mathbf{k}\in\mathrm{MBZ}} \left\langle \hat{\Gamma}^\dagger_{\mathbf{k}\sigma} \hat{\Gamma}_{\mathbf{k}\sigma} \right\rangle \\ &= 2 \sum_{\mathbf{k}\in\mathrm{MBZ}} \mathrm{Th}^{(+)}_{\mathbf{k}}[\beta, \mu] \end{aligned} \tag{A.31}$$

being f be the Fermi-Dirac distribution at inverse temperature β and chemical potential μ,

$$f(\epsilon; \beta, \mu) = \frac{1}{e^{\beta(\epsilon-\mu)} + 1}$$

The 2 prefactor appeared because of spin degeneracy. Since the gapped bands refer to the smaller MBZ, thus exhibit the periodicity

$$E_{\mathbf{k}} = E_{\mathbf{k}+\boldsymbol{\pi}}$$

it holds equivalently

$$N = \sum_{\mathbf{k}\in\mathrm{BZ}} \mathrm{Th}^{(+)}_{\mathbf{k}}[\beta, \mu] \tag{A.32}$$

as is obvious. The 2 prefactor was absorbed in doubling the integration region, MBZ $\to$ BZ. Next sections are devoted to simplify the above theoretical results in order to obtain an algorithmic estimation for the magnetization m, which is just the AF instability order parameter.

In Fig. A.2a the density $n(\mu)$ is plotted as a function of μ. As is evident, at nearly zero temperature ($\beta \to \infty$) the system is half filled for $-\Delta < \mu < \Delta$. In order to obtain a finite doping, it must be $|\mu| > \Delta$; which means, the chemical potential needs to cross the electronic bands. This HF solution leads invariably to a rather unusual metallic AF at any finite doping.

A.1.7 Self-consistency gap equation and magnetization

A convergence algorithm can be designed to find the Hartree-Fock solution to the model. Ultimately, we aim to extract m self-consistently. Combining Eqns. (A.7), (A.8) and the definition of Eq. (A.3), we get

$$\frac{1}{2L_xL_y}\sum_{\mathbf{r}}(-1)^{x+y}\hat{n}_{\mathbf{r}\sigma} = \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\left[\hat{\psi}_{\mathbf{k}\sigma}+\hat{\psi}^{\dagger}_{\mathbf{k}\sigma}\right]$$

thus employing Eq. (A.10)

$$\begin{aligned} m &= \frac{1}{2L_xL_y}\sum_{\mathbf{r}}(-1)^{x+y}\langle\hat{n}_{\mathbf{r}\uparrow}-\hat{n}_{\mathbf{r}\downarrow}\rangle \\ &= \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\left\langle\left(\hat{\psi}_{\mathbf{k}\uparrow}+\hat{\psi}^{\dagger}_{\mathbf{k}\uparrow}\right)-\left(\hat{\psi}_{\mathbf{k}\downarrow}+\hat{\psi}^{\dagger}_{\mathbf{k}\downarrow}\right)\right\rangle \end{aligned}$$

Using now Eq. (A.13) we get the self-consistency equation for the HFP m,

$$\begin{aligned} m &= \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\left[\langle\hat{s}^x_{\mathbf{k}\uparrow}\rangle-\langle\hat{s}^x_{\mathbf{k}\downarrow}\rangle\right] \\ &= \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\frac{\mathrm{Re}\{\Delta_{\mathbf{k}\uparrow}\}-\mathrm{Re}\{\Delta_{\mathbf{k}\downarrow}\}}{E_{\mathbf{k}}}\mathrm{Th}^{(-)}_{\mathbf{k}}[\beta,\mu] \\ &= \frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\frac{\Delta}{E_{\mathbf{k}}}\mathrm{Th}^{(-)}_{\mathbf{k}}[\beta,\mu] \end{aligned} \tag{A.33}$$

The magnetization m is the physical order parameter of the phase transition from the normal phase to the antiferromagnetic one. Within such phase transition, in fact, it is the expectation value

$$\langle\hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma}\rangle$$

that goes from zero (normal phase, free Fermi gas) to a non-zero value (AF). As is evident from equations above, m is just its (only) s-wave component.

Consider specifically the half-filled situation and multiply Eq. (A.33) on both sides by U. Since due to symmetry $\mu = 0$, we have the finite-temperature self-consistent equation

$$\Delta = \frac{U}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\frac{\Delta}{\sqrt{\epsilon^2_{\mathbf{k}}+\Delta^2}}\tanh\left(\frac{\beta}{2}\sqrt{\epsilon^2_{\mathbf{k}}+\Delta^2}\right)$$

where we used the relation

$$\mathrm{Th}^{(-)}_{\mathbf{k}}[\beta,0] = \tanh\left(\frac{\beta E_{\mathbf{k}}}{2}\right)$$

In thermodynamic limit $L\to\infty$

$$\frac{1}{U} = \frac{1}{2}\int_{\mathrm{BZ}}\frac{d\mathbf{k}}{(2\pi)^2}\frac{1}{\sqrt{\epsilon^2_{\mathbf{k}}+\Delta^2}}\tanh\left(\frac{\beta}{2}\sqrt{\epsilon^2_{\mathbf{k}}+\Delta^2}\right)$$

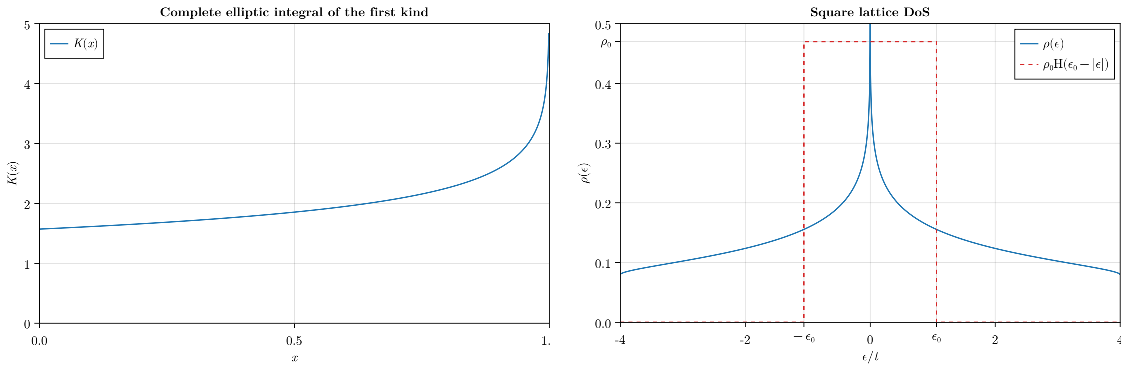
Substitute the momentum integral with an integral over energies:

$$\frac{2}{U} = \int_{-4t}^{4t}\frac{d\epsilon\rho(\epsilon)}{\sqrt{\epsilon^2+\Delta^2}}\tanh\left(\frac{\beta}{2}\sqrt{\epsilon^2+\Delta^2}\right)$$

being $\rho(\epsilon)$ the 2D square lattice DoS. Its explicit form is given by [26]

$$\rho(\epsilon) = \frac{1}{2\pi^2}K\left(\sqrt{1-\left(\frac{\epsilon}{4t}\right)^2}\right) \qquad K(x)\equiv\int_0^{\pi/2}\frac{d\theta}{\sqrt{1-x^2\sin^2\theta}}$$

$K(x)$ is the complete elliptic integral of the first kind, shown in Fig. A.3a.



a | Plot of $K(x)$.

b | DoS and its approximation.

Figure A.3 | Plots of the complete elliptic integral of the first kind $K(x)$ and the density of states for the planar square lattice.

As is plotted in Fig. A.3b, the resulting DoS is logarithmically peaked at $\epsilon = 0$, on the MBZ. An extremely rough but common approximation is to substitute the actual DoS with a step function,

$$\rho(\epsilon) \simeq \rho_0 \mathrm{H}\left(\epsilon_0 - |\epsilon|\right) \quad \text{with} \quad 2\rho_0\epsilon_0 = 1$$

Here, the peak parameters ϵ_0 and ρ_0 are chosen in order to maintain a correct DoS integral, considering also the 2 pre-factor of spin degeneracy. Within this extremely rough approximation, we have

$$\frac{2}{U} \simeq \rho_0 \int_{-\epsilon_0}^{\epsilon_0} \frac{\mathrm{d}\epsilon}{\sqrt{\epsilon^2 + \Delta^2}} \tanh\left(\frac{\beta}{2}\sqrt{\epsilon^2 + \Delta^2}\right)$$

Suppose the system is magnetized, $\Delta > 0$. At low temperature, $\beta \gg 1$, as a first approximation we can substitute the hyperbolic tangent with unity, obtaining

$$\begin{aligned}
\frac{2}{\rho_0 U} &\simeq \int_{-\epsilon_0}^{\epsilon_0} \frac{\mathrm{d}\epsilon}{\sqrt{\epsilon^2 + \Delta^2}} \\
&= \int_{-\epsilon_0/\Delta}^{\epsilon_0/\Delta} \frac{\mathrm{d}x}{\sqrt{1 + x^2}} \\
&= \sinh^{-1}\left(\frac{\epsilon_0}{\Delta}\right) - \sinh^{-1}\left(-\frac{\epsilon_0}{\Delta}\right) \\
&= 2\sinh^{-1}\left(\frac{\epsilon_0}{\Delta}\right)
\end{aligned}$$

thus

$$m \simeq \frac{\epsilon_0}{U \sinh(1/\rho_0 U)}$$

having we used $\Delta = mU$. Now: at large U,

$$\lim_{U \to +\infty} m \simeq \epsilon_0 \rho_0 = \frac{1}{2}$$

which gives the correct saturation limit at high local repulsion. In the opposite limit,

$$\begin{aligned}
\lim_{U \to 0} m &\simeq \lim_{U \to 0} \frac{2\epsilon_0/U}{\exp(1/\rho_0 U) - \exp(-1/\rho_0 U)} \\
&\simeq \lim_{U \to 0} \frac{2\epsilon_0}{U} \exp\left(-\frac{1}{\rho_0 U}\right)
\end{aligned}$$

giving a near-0 exponential suppression of magnetization, typical of MFT self-consistent solutions.

A.1.8 Instability of the commensurate AF solution and atypical metallic bands

A key point to be discussed is the known instability of the AF HF solution of the EHM. As is thoroughly discussed in [13], the solution we presented above is only stable at half-filling, becoming hardly unstable at infinitesimal doping. This is due to the commensurate nature of the established AF phase: we impose SSB to take place at wavevector $\boldsymbol{\pi}$, shrinking the BZ to exactly half its size and giving back on the newly defined MBZ two bands $\pm E_{\mathbf{k}}$. This procedure actually works well for the half-filled case: magnetization essentially depends on the scattering rate between points on the FS connected by a $\boldsymbol{\pi}$ vector. The half filled case, taking advantage of the natural nesting property of the bare bands,

$$\forall \mathbf{k} \in \partial\mathrm{MBZ} : \ \epsilon_{\mathbf{k}} = \epsilon_{\mathbf{k}+\boldsymbol{\pi}}$$

(being ∂MBZ the boundary of the MBZ) exhibits a magnetization at infinitesimal interaction. For a many body state resulting from continuous SSB, to establish a given phase means to show the presence of a stable gapless Goldstone mode associated to the broken symmetry. Such a Goldstone mode is essentially a sharp peak at zero frequency for some dressed response function. For the AF phase, it follows from the reduced U^z(1) rotational symmetry and is related to the proper transverse pseudo-spin to pseudo-spin response function, 2×2 matrix equation,

$$\tilde{\chi}_{\hat{\mathbf{S}}^-\hat{\mathbf{S}}^+}(\mathbf{k},\omega) = \frac{\chi_{\hat{\mathbf{S}}^-\hat{\mathbf{S}}^+}(\mathbf{k},\omega)}{1 - U\chi_{\hat{\mathbf{S}}^-\hat{\mathbf{S}}^+}(\mathbf{k},\omega)} \tag{A.34}$$

being χ the free transverse response. The pseudo-spin hereby considered is the one of Eq. (A.12). As is shown in [13], for the commensurate AF SSB at wavevector $\boldsymbol{\pi}$ the condition for the presence of a stable gapless mode reduces to a 2×2 matrix equation,

$$\text{Stability condition:} \qquad \mathbb{I}_{2\times 2} - U\left[\chi_{\hat{\mathbf{S}}^-\hat{\mathbf{S}}^+}(\boldsymbol{\pi},0)\right] \geq 0 \tag{A.35}$$

with $\left[\chi_{\hat{\mathbf{S}}^-\hat{\mathbf{S}}^+}(\boldsymbol{\pi},\omega)\right]$ the response function matrix obtained by considering a matrix response function for a two-sites unit cell. The above equation is to be interpreted as a matricial equation [Explain better..]. Considering the eigenvalue of largest amplitude of the matrix λ_δ at filling δ, the one most prone to instabilities, it is found that

$$\begin{aligned}
\lambda_0(\mathbf{k}\to\boldsymbol{\pi},0) &= \frac{1}{U} && \text{(half filling)}\\
\lambda_\delta(\mathbf{k}\to\boldsymbol{\pi},0) &= \frac{1}{U} + \frac{2\epsilon_{\mathrm{F}}}{\Delta}N(\epsilon_{\mathrm{F}}) && \text{(finite doping)}
\end{aligned}$$

The reason for the distinction above is the intrinsic metallic nature of the system for infinitesimal doping. At half-filling the system is an insulator, therefore no Fermi energy is defined, as well as no FS. Evidently this implies

$$\begin{aligned}
1 - U\lambda_0(\mathbf{k}\to\boldsymbol{\pi},0) = 0 \quad &\text{(half filling)} &&\Longrightarrow \quad \text{stable Goldstone mode}\\
1 - U\lambda_\delta(\mathbf{k}\to\boldsymbol{\pi},0) < 0 \quad &\text{(finite doping)} &&\Longrightarrow \quad \text{unstable Goldstone mode}
\end{aligned}$$

For any finite doping, the commensurate AF phase becomes unstable and is destroyed by transverse spin fluctuations. Nevertheless, for the sake of completeness and because we noticed too late, we will study the metallic AF also at finite doping.

A.1.9 Hartree-Fock algorithm results in reciprocal space

The above sections offer a self-consistency equation for the only HFP m; W is determined by $\Delta = mU$ indeed. In an iterative HF fashion, Eq. (A.33) reads:

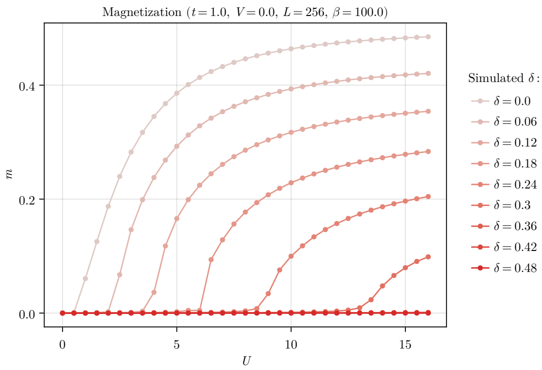
$$m_{i+1} = \frac{U}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{MBZ}} \frac{m_i}{2E_{\mathbf{k}}[\Delta_i]}\left[f(-E_{\mathbf{k}}[m_i];\beta,\mu) - f(E_{\mathbf{k}}[m_i];\beta,\mu)\right] \tag{A.36}$$

Then a self-consistent algorithm to search for a self consistent estimate of the magnetization may be sketched, following the general idea of Sec. **??**:

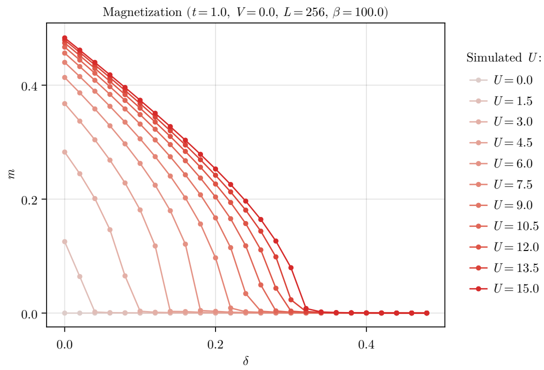
1. Algorithm setup: initialize a counter $i = 1$ and choose:
   - the local repulsion U/t or equivalently the hopping amplitude t/U;
   - the coarse-graining of the BZ, which is, fix L_x and L_y. Then
   $$k_x = 2\pi n_x/L_x \qquad k_y = 2\pi n_y/L_y$$
   where $-L_j/2 \leq n_j \leq L_j/2$, $n_j \in \mathbb{Z}$;
   - the density n or equivalently the doping $\delta \equiv n - 0.5$;
   - the inverse temperature β;
   - the maximum number of iterations $p \in \mathbb{N}$;
   - the mixing parameter $g \in [0, 1]$;
   - the tolerance parameters $\delta_m, \delta_n \in \mathbb{R}$ (respectively for the HFP m and density);
2. Select a random starting value $m_i > 0$ for $i = 1$;
3. Find the optimal chemical potential μ as follows:
   a) Compute $\pm E_{\mathbf{k}}[\Delta_i, U]$;
   b) Define the function of the chemical potential μ,
   $$\delta n(\mu; \beta, m_i, U) \equiv \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathrm{MBZ}} \left[f\left(-E_{\mathbf{k}}[U, m_i]; \beta, \mu\right) + f\left(E_{\mathbf{k}}[U, m_i]; \beta, \mu\right)\right] - n$$
   c) Find the root of δn,
   $$\delta n(\mu_i) = 0$$
   Then μ_i is the chemical potential which, at step i with HFP m_i, realizes the best approximation of density n;
4. Compute m_{i+1} using Eq. (A.36) with chemical potential μ_i and update the counter, $i \to i+1$;
5. Check if $|m_{i+1} - m_i| \leq \delta_m$:
   - If yes, halt;
   - If not: check if $i > p$
     - If yes, halt and consider choosing better tolerance and model parameters;
     - If not, define
     $$m_{i+1} : g m_{i+1} + (1 - g) m_i$$
     (logical assignment notation used) and repeat from step 2.

For the sake of completeness, we decided to run our algorithm also at finite doping, knowing from the previous discussion of Sec. A.1.8 that the results at $\delta > 0$ are unstable. The algorithm sketched hereby was ran with the following setup:

```
UU = [U for U in 0.5:0.5:20.0]        # Local repulsions
LL = [256]                            # Lattice sizes
dd = [d for d in 0.0:0.05:0.45]       # Dopings
bb = [100.0, Inf]                     # Inverse temperatures
p = 100                               # Max number of iterations
dm = 1e-4                             # Tolerance on magnetization
dn = 1e-2                             # Tolerance on density
g = 0.5                               # Mixing parameter
```



a | Local repulsion U/t depencence.



b | Doping $\delta = n - 0.5$ dependence.

Figure A.4 | Plots of the zero-temperature AF instability order parameter m as a function of both the local repulsion U/t (Fig. A.4a) and the doping $\delta = n - 1/2$ (Fig. A.4b).

In Fig. A.4 plots at $\beta = 100$ of the magnetization m is reported. As is evident in Fig. A.4a, doping disrupts AF ordering; the horizontal asymptote lowers as δ gets bigger, a mere consequence of the fact that the free space at each single-particle state in k-space shrinks as density increases; also, for $\delta > 0$, a finite magnetization m exists only for $U \geq U_c[\delta]$ (a critical value parametrically dependent on doping). Fig. A.4b shows the dependence of magnetization on doping at various fixed local repulsions U: once again, to dope the material disrupt AF ordering. Identical analysis have been performed for different temperatures, leading to analogous results as long as $\beta \gtrsim 10$ and to $m \simeq 0$ for higher temperatures (predictably).

A.2 (Impossibility of a stable) superconducting phase

Let us now move to the discussion of superconductivity in the HM. As for the AF phase, we use the results and analyses of this section as groundwork for the more sophisticated case of SC phase in the EHM. We anticipate here that BCS-like superconductivity cannot develop in the pure HM, due to the inherently repulsive nature of $\hat{H}_U$. BCS theory admits, for an half filled model, finite Cooper fluctuations for an infinitesimal attractive interaction. However, as will be discussed, by the means of MFT it is perfectly possible and coherent to derive a self-consistent superconducting solution to the HM. The point here is that such self-consistent solution sets to zero Cooper fluctuations.

As done for the AF phase in Sec. A.1 and following the introductory discussion of Sec. [Missing], let us start from the discussion of the phase symmetries. We aim to describe a uniform-density superconductor,

$$\langle \hat{n}_{i\sigma} \rangle = n$$

taking into account Cooper RWCs,

$$\langle \hat{c}^\dagger_{i\sigma} \hat{c}^\dagger_{i\bar{\sigma}} \rangle$$

which, having we broken $U^{\mathrm{c}}(1)$ charge symmetry, we allow to fluctuate to non-zero values. For the pure HM, we know from Tab. **??** that both Hartree and Cooper pairing channels contribute to the RWCs set. We handle them separately.

A.2.1 MFT treatment of the Hartree channel

For the Hartree channel, knowing the deal with a uniform-density phase, we can re-use the derivation of Sec. A.1.1 simply by setting $m = 0$. We immediately get from Eq. (A.5):

$$\begin{aligned}
\hat{H}_U &\simeq U \sum_{\mathbf{r}} \left[\hat{n}_{\mathbf{r}\uparrow} \langle \hat{n}_{\mathbf{r}\downarrow} \rangle + \langle \hat{n}_{\mathbf{r}\uparrow} \rangle \hat{n}_{\mathbf{r}\downarrow} - \langle \hat{n}_{\mathbf{r}\uparrow} \rangle \langle \hat{n}_{\mathbf{r}\downarrow} \rangle \right] + (\text{Cooper channel}) \\
&= nU \times \hat{N} - E^{(\mathrm{SC})}_{\mathrm{H}/U} + (\text{Cooper channel})
\end{aligned} \tag{A.37}$$

thus chemical potential renormalization $\tilde{\mu} \equiv \mu - nU$ is present also for the SC phase. The contraction energy comes immediately from Eq. (A.6),

$$E^{(\mathrm{SC})}_{\mathrm{H}/U} = L_x L_y \times n^2 U \tag{A.38}$$

We now move to Cooper channel.

A.2.2 MFT treatment of the Cooper channel

Recall the result of Eq. (??) which gave the Fourier-transformed version of the local repulsion, and perform a Cooper class Wick's contraction on it,

$$\begin{aligned}\hat{H}_U \simeq (\text{Hartree channel}) + \frac{U}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \Big[\big\langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \big\rangle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \\ + \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \big\langle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \big\rangle - \big\langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \big\rangle \big\langle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \big\rangle \Big]\end{aligned} \tag{A.39}$$

We are not breaking translational invariance, thus only Cooper fluctuations with net zero total momentum are allowed. This means only $\mathbf{K} = \mathbf{0}$ contributes. Define as in Eq. (??) the pairing operator

$$\hat{\phi}_{\mathbf{k}} \equiv \hat{c}_{-\mathbf{k}\downarrow} \hat{c}_{\mathbf{k}\uparrow} \tag{A.40}$$

and define $w_{\mathbf{q}}$ as in Eq. (??),

$$w_{\mathbf{q}} \equiv \langle \phi^\dagger_{\mathbf{q}} \rangle$$

Its s-wave part is given by:

$$w^{(s)} \equiv \frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{BZ}} w_{\mathbf{q}} \tag{A.41}$$

Let $\hat{n}_{\mathbf{k}\sigma}$ as in Eq. (A.2). Eq. (A.37) reduces to

$$\hat{H}_U \simeq (\text{Hartree channel}) + \frac{U}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \Big[\langle \hat{\phi}^\dagger_{\mathbf{q}} \rangle \hat{\phi}_{\mathbf{q}'} + \hat{\phi}^\dagger_{\mathbf{q}} \langle \hat{\phi}_{\mathbf{q}'} \rangle \Big] - E^{(\mathrm{SC})}_{\mathrm{C}/U} \tag{A.42}$$

being $E^{(\mathrm{SC})}_{\mathrm{C}/U}$ the HM Cooper contraction energy,

$$\begin{aligned} E^{(\mathrm{SC})}_{\mathrm{C}/U} &\equiv \frac{U}{L_x L_y} \sum_{\mathbf{q}\mathbf{q}'} \langle \hat{\phi}^\dagger_{\mathbf{q}} \rangle \langle \hat{\phi}_{\mathbf{q}'} \rangle \\ &= L_x L_y \times U \big| w^{(s)} \big|^2 \end{aligned} \tag{A.43}$$

Let now

$$\Delta^{(s)} \equiv -U w^{(s)}$$

We finally reduce $\hat{H}_U$ to

$$\hat{H}_U \simeq (\text{Hartree channel}) - \sum_{\mathbf{q}\in\mathrm{BZ}} \Big[\Delta^{(s)} \hat{\phi}_{\mathbf{q}} + \Delta^{(s)*} \hat{\phi}^\dagger_{\mathbf{q}} \Big] - E^{(\mathrm{SC})}_{\mathrm{C}/U} \tag{A.44}$$

which can be easily mapped on a pseudo-spin model.

Composing Eqns. (A.37) and (A.44), we get the full MFT hamiltonian

$$\begin{aligned}\hat{H} \simeq \sum_{\mathbf{k}\in\mathrm{BZ}} \sum_{\sigma} \epsilon_{\mathbf{k}\sigma} \Big[\hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} - \hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma} \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \Big] \\ - \sum_{\mathbf{q}\in\mathrm{BZ}} \Big[\Delta^{(s)} \hat{\phi}_{\mathbf{q}} + \Delta^{(s)*} \hat{\phi}^\dagger_{\mathbf{q}} \Big] + nU \times \hat{N} - \Big[E^{(\mathrm{SC})}_{\mathrm{H}/U} + E^{(\mathrm{SC})}_{\mathrm{C}/U} \Big]\end{aligned} \tag{A.45}$$

A.2.3 Nambu representation of the SC phase and diagonalization

We now proceed precisely as in Sec. A.1.3. Define the SC Nambu spinor

$$\hat{\Phi}_{\mathbf{k}} \equiv \begin{bmatrix} \hat{c}_{\mathbf{k}\uparrow} \\ \hat{c}^{\dagger}_{-\mathbf{k}\downarrow} \end{bmatrix}$$

Define the pseudo-spin operator:

$$\hat{s}^{\alpha}_{\mathbf{k}} \equiv \hat{\Phi}^{\dagger}_{\mathbf{k}} \tau^{\alpha} \hat{\Phi}_{\mathbf{k}}$$

and the shifted bare bands,

$$\xi_{\mathbf{k}} \equiv \epsilon_{\mathbf{k}} - \tilde{\mu}$$

Notice we are using the chemical potential RMP. We employ from the start the spin-independency of the bare bands, as well as their pure s^*-wave structure, since no Fock interaction in the present context chan change that. The following chain holds:

$$\begin{aligned} \hat{n}_{\mathbf{k}\uparrow} + \hat{n}_{-\mathbf{k}\downarrow} &= \hat{c}^{\dagger}_{\mathbf{k}\uparrow}\hat{c}_{\mathbf{k}\uparrow} + \hat{c}^{\dagger}_{-\mathbf{k}\downarrow}\hat{c}_{-\mathbf{k}\downarrow} \\ &= \hat{c}^{\dagger}_{\mathbf{k}\uparrow}\hat{c}_{\mathbf{k}\uparrow} - \hat{c}_{-\mathbf{k}\downarrow}\hat{c}^{\dagger}_{-\mathbf{k}\downarrow} + \hat{\mathbb{I}}_{\mathbf{k}} \\ &= \hat{\Phi}^{\dagger}_{\mathbf{k}}\tau^{z}\hat{\Phi}_{\mathbf{k}} + \hat{\mathbb{I}}_{\mathbf{k}} \\ &= \hat{s}^{z}_{\mathbf{k}} + \hat{\mathbb{I}}_{\mathbf{k}} \end{aligned} \tag{A.46}$$

Then, considering the kinetic part of the gran-canonical hamiltonian,

$$\begin{aligned} \hat{H}_t - \tilde{\mu}\hat{N} &= \sum_{\sigma} \sum_{\mathbf{k}\in \mathrm{BZ}} \left(\epsilon_{\mathbf{k}\sigma} - \tilde{\mu}\right) \hat{n}_{\mathbf{k}\sigma} \\ &= \sum_{\mathbf{k}\in \mathrm{BZ}} \left[\xi_{\mathbf{k}}\hat{n}_{\mathbf{k}\uparrow} + \xi_{-\mathbf{k}}\hat{n}_{-\mathbf{k}\downarrow}\right] \\ &= \sum_{\mathbf{k}\in \mathrm{BZ}} \xi_{\mathbf{k}}\hat{s}^{z}_{\mathbf{k}} + E^{(\mathrm{SC})}_{\mathrm{a}} \end{aligned}$$

being $E^{(\mathrm{SC})}_{\mathrm{a}}$ the "anticommutation energy" coming from the kinetic term,

$$E^{(\mathrm{SC})}_{\mathrm{a}} \equiv \sum_{\mathbf{k}\in \mathrm{BZ}} \xi_{\mathbf{k}} \tag{A.47}$$

The scalar term in the above equation is essential for estimating correctly the phase ground state free energy density. The full MFT gran-canonical hamiltonian in Nambu representation is obtained including Eq. (A.42),

$$\hat{H} - \mu\hat{N} \simeq \sum_{\mathbf{k}\in \mathrm{BZ}} \sum_{\sigma} \hat{\Phi}^{\dagger}_{\mathbf{k}} h_{\mathbf{k}} \hat{\Phi}_{\mathbf{k}} + E^{(\mathrm{SC})}_{\mathrm{a}} - \left[E^{(\mathrm{SC})}_{\mathrm{H}/U} + E^{(\mathrm{SC})}_{\mathrm{C}/U}\right] \tag{A.48}$$

(note the non-renormalized chemical potential on the left) being

$$h_{\mathbf{k}} \equiv \begin{bmatrix} \xi_{\mathbf{k}} & -\Delta^{*}_{\mathbf{k}} \\ -\Delta_{\mathbf{k}} & -\xi_{\mathbf{k}} \end{bmatrix}$$

while the off-diagonal gap is a constant,

$$\Delta_{\mathbf{k}} \equiv \Delta^{(s)}$$

This hamiltonian behaves as an ensemble of spins in local magnetic pseudofields

$$\mathbf{b}_{\mathbf{k}} \equiv \begin{bmatrix} -\operatorname{Re}\{\Delta_{\mathbf{k}}\} \\ -\operatorname{Im}\{\Delta_{\mathbf{k}}\} \\ \xi_{\mathbf{k}} \end{bmatrix} \qquad \tan 2\theta_{\mathbf{k}} \equiv \frac{\xi_{\mathbf{k}}}{\sqrt{\xi^2_{\mathbf{k}} + |\Delta_{\mathbf{k}}|^2}} \qquad \tan 2\zeta_{\mathbf{k}} \equiv \frac{\operatorname{Im}\{\Delta_{\mathbf{k}}\}}{\operatorname{Re}\{\Delta_{\mathbf{k}}\}} \tag{A.49}$$

precisely as was for the AF phase, with the essential difference that now the chemical potential RMP lies inside the z component of the pseudofield itself. This is a consequence of the Nambu spinors definition. We have the spin hamiltonian

$$\hat{H} - \mu\hat{N} \simeq \sum_{\mathbf{k}\in\mathrm{BZ}} \mathbf{b}_{\mathbf{k}} \cdot \hat{\mathbf{s}}_{\mathbf{k}} + E_{\mathrm{a}}^{(\mathrm{SC})} - \left[E_{\mathrm{H}/U}^{(\mathrm{SC})} + E_{\mathrm{C}/U}^{(\mathrm{SC})}\right] \quad \text{where} \quad \hat{\mathbf{s}}_{\mathbf{k}} = \begin{bmatrix} \hat{s}_{\mathbf{k}}^x \\ \hat{s}_{\mathbf{k}}^y \\ \hat{s}_{\mathbf{k}}^z \end{bmatrix} \tag{A.50}$$

As was for the AF phase, to diagonalize the hamiltonian means essentially to align the spin z axis with the direction of the pseudofield. Then, we diagonalize the hamiltonian via a set of rotations similarly to Eq. (A.17). The main difference is that in general the pseudofield can have a y component, thus the diagonalizing matrix is made of a sequence of two rotations, both counter-clockwise, the first around the z axis by an angle $2\zeta_{\mathbf{k}}$ to place the pseudofield in the xz plane, the second around the new y axis by an angle $2\theta_{\mathbf{k}}$ to align the z axis with the pseudofield,

$$\begin{aligned} W_{\mathbf{k}} &\equiv e^{-i\theta_{\mathbf{k}}\tau^y} e^{-i\zeta_{\mathbf{k}}\tau^z} \\ &= \begin{bmatrix} \cos\theta_{\mathbf{k}} & -\sin\theta_{\mathbf{k}} \\ \sin\theta_{\mathbf{k}} & \cos\theta_{\mathbf{k}} \end{bmatrix} \begin{bmatrix} e^{-i\zeta_{\mathbf{k}}} & \\ & e^{i\zeta_{\mathbf{k}}} \end{bmatrix} \\ &= \begin{bmatrix} \cos\theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} & -\sin\theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \\ \sin\theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} & \cos\theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \end{bmatrix} \end{aligned} \tag{A.51}$$

obtaining the diagonal matrix

$$d_{\mathbf{k}} \equiv W_{\mathbf{k}} h_{\mathbf{k}} W_{\mathbf{k}}^\dagger \quad \text{where} \quad d_{\mathbf{k}} \equiv \begin{bmatrix} E_{\mathbf{k}} & \\ & -E_{\mathbf{k}} \end{bmatrix}$$

and the bands

$$E_{\mathbf{k}} \equiv \sqrt{\xi_{\mathbf{k}}^2 + |\Delta_{\mathbf{k}}|^2}$$

Finally, the Nambu spinor for Bogoliubov quasiparticles is given similarly to Eq. (A.24)

$$\hat{\Gamma}_{\mathbf{k}} = W_{\mathbf{k}} \hat{\Phi}_{\mathbf{k}} \tag{A.52}$$

Explicitly,

$$\begin{aligned} \hat{\Gamma}_{\mathbf{k}} &= \begin{bmatrix} \cos\theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} & -\sin\theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \\ \sin\theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} & \cos\theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \end{bmatrix} \begin{bmatrix} \hat{c}_{\mathbf{k}\uparrow} \\ \hat{c}_{-\mathbf{k}\downarrow}^\dagger \end{bmatrix} \\ &= \begin{bmatrix} \cos\theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} \hat{c}_{\mathbf{k}\uparrow} - \sin\theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \hat{c}_{-\mathbf{k}\downarrow}^\dagger \\ \sin\theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} \hat{c}_{\mathbf{k}\uparrow} + \cos\theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \hat{c}_{-\mathbf{k}\downarrow}^\dagger \end{bmatrix} \equiv \begin{bmatrix} \hat{\gamma}_{\mathbf{k}}^{(+)} \\ \hat{\gamma}_{\mathbf{k}}^{(-)} \end{bmatrix} \end{aligned} \tag{A.53}$$

These operators act as free fermion fields on the Bogoliubov bands $\pm E_{\mathbf{k}}$.

A.2.4 Explicit zero-temperature SC ground state at half-filling

Reasoning precisely as in Sec. A.1.4, we can extract the half-filling zero temperature ground state, commonly referred to as the "BCS state". Eqns. (A.26) and (A.27) hold equivalently (ignoring the σ index and extending momentum tensor product to the entire BZ),

$$|\Psi\rangle = \bigotimes_{\mathbf{k}\in\mathrm{BZ}} \left[\cos\theta_{\mathbf{k}} |\Downarrow_{\mathbf{k}}\rangle - \sin\theta_{\mathbf{k}} |\Uparrow_{\mathbf{k}}\rangle\right]$$

Using now Eq. (A.46), we have

$$\begin{aligned} \langle\Downarrow_{\mathbf{k}}|\hat{s}_{\mathbf{k}}^z + \hat{\mathbb{I}}_{\mathbf{k}}|\Downarrow_{\mathbf{k}}\rangle &= 0 \\ \langle\Uparrow_{\mathbf{k}}|\hat{s}_{\mathbf{k}}^z + \hat{\mathbb{I}}_{\mathbf{k}}|\Uparrow_{\mathbf{k}}\rangle &= 2 \end{aligned}$$

so up to a phase

$$|\Downarrow_{\mathbf{k}}\rangle \sim |\Omega_{\mathbf{k}}\rangle \qquad |\Uparrow_{\mathbf{k}}\rangle \sim \hat{\phi}_{\mathbf{k}}^\dagger |\Omega_{\mathbf{k}}\rangle$$

Hence, we have

$$|\Psi\rangle = \bigotimes_{\mathbf{k}\in\mathrm{BZ}} \left[\cos\theta_{\mathbf{k}} + e^{2i\zeta_{\mathbf{k}}} \sin\theta_{\mathbf{k}} \hat{\phi}_{\mathbf{k}}^\dagger\right] |\Omega_{\mathbf{k}}\rangle \tag{A.54}$$

The argument $2\zeta_{\mathbf{k}}$ of the phase factor is precisely the phase of the gap function, as can be shown easily.

A.2.5 Superconducting fluctuations suppression in the conventional HM

Let us report the SC analogous of the AF Eqns. (A.13) and (A.14):

$$\hat{s}^x_{\mathbf{k}} = \hat{\phi}_{\mathbf{k}} + \hat{\phi}^\dagger_{\mathbf{k}} \tag{A.55}$$

$$\hat{s}^y_{\mathbf{k}} = i\left(\hat{\phi}_{\mathbf{k}} - \hat{\phi}^\dagger_{\mathbf{k}}\right) \tag{A.56}$$

while the analogous of Eq. (A.15) was given in Eq. (A.46). Let also the thermal factor

$$\mathrm{Th}^{(\pm)}_{\mathbf{k}}[\beta] \equiv f\left(-E_{\mathbf{k}};\beta,0\right) \pm f\left(E_{\mathbf{k}};\beta,0\right) \tag{A.57}$$

and the EVs:

$$\langle \hat{s}^x_{\mathbf{k}} \rangle = \frac{\mathrm{Re}\{\Delta_{\mathbf{k}}\}}{E_{\mathbf{k}}}\mathrm{Th}^{(-)}_{\mathbf{k}}[\beta] \tag{A.58}$$

$$\langle \hat{s}^y_{\mathbf{k}} \rangle = \frac{\mathrm{Im}\{\Delta_{\mathbf{k}}\}}{E_{\mathbf{k}}}\mathrm{Th}^{(-)}_{\mathbf{k}}[\beta] \tag{A.59}$$

$$\langle \hat{s}^z_{\mathbf{k}} \rangle = -\frac{\xi_{\mathbf{k}}}{E_{\mathbf{k}}}\mathrm{Th}^{(-)}_{\mathbf{k}}[\beta] \tag{A.60}$$

Up to now, the derivation was general. However, it is immediate to show that for the pure HM the SC phase can never establish. Recall Eq. (A.41): explicitly

$$\begin{aligned}
w^{(s)} &= \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \langle \hat{\phi}^\dagger_{\mathbf{k}} \rangle \\
&= \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \langle \hat{\Phi}^\dagger_{\mathbf{k}} \tau^+ \hat{\Phi}_{\mathbf{k}} \rangle \\
&= \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \frac{\langle \hat{s}^x_{\mathbf{k}} \rangle + i\langle \hat{s}^y_{\mathbf{k}} \rangle}{2} \\
&= \frac{1}{2L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \frac{\Delta_{\mathbf{k}}}{E_{\mathbf{k}}}\mathrm{Th}^{(-)}_{\mathbf{k}}[\beta]
\end{aligned} \tag{A.61}$$

Now: $\Delta_{\mathbf{k}} = \Delta^{(s)} = -Uw^{(s)}$, thus if we define

$$U_{\mathrm{c}}[\beta] \equiv \left[\frac{1}{2L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \frac{\mathrm{Th}^{(-)}_{\mathbf{k}}[\beta]}{E_{\mathbf{k}}}\right]^{-1} \tag{A.62}$$

(the reason for this notation will be clear in Sec. [Missing]) we end with the equation:

$$w^{(s)} = -\frac{U}{U_{\mathrm{c}}[\beta]}w^{(s)}$$

Since evidently $U_{\mathrm{c}}[\beta] > 0$ by definition (the thermal factor is negative by construction), the above equation can be solved in $\mathbb{C}$ only by $w^{(s)} = 0$. This implication terminates the argument: superconductivity cannot establish self-consistently at this level in the HM.

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